

AE 4610 Lab 1 Report

Section	Author
Introduction	Victor Hernandez
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Modeling	Keira Mapel
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AE 4610 – Rotary Servo Base Laboratory

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1 Rubric

MODELING (Group): 25 points

<i>Frequency Response Test</i>	6
a. Table B1 fully populated	1
b. Plot: Bode Plot from Table B1 values (<i>Cutoff frequency point should be labelled</i>)	3
c. DC Gain calculation and value	1
d. Time constant calculation and value	1
<i>Step Response Test</i>	4
a. Plot: Maximum load speed response	1
b. DC Gain calculation and value	1.5
c. Time constant calculation and value	1.5
<i>Nominal Values</i>	2
a. Nominal K and Tau calculation and values	2
<i>Model Validation</i>	13
a. Plot (same figure): i. Measured response ii. Simulated response with K and Tau from Frequency Response Test iii. Simulated response with K and Tau from Step Response Test iv. Simulated response with nominal K and Tau	10
b. Table B2 fully populated (<i>Note: $K_{e,v}$ and $\tau_{e,v}$ are the K and Tau whose simulated response matches closest with the measured</i>)	1
c. Question: How do K and Tau affect the DC motor response to step input?	2

CONTROL DESIGN (Individual): 25 points (Note: Refer to section C in the lab manual for the listed steps.)

Step 5	2.5
a. Plots/Screenshots of: i. Open-loop Bode plot with Phase margin marked ii. Closed-loop Bode plot with Bandwidth frequency marked iii. Root locus plot iv. Time response plot with steady-state value (or error) marked	2
b. Proportional gain value	0.5
Step 6	2.5
a. Plots/Screenshots of: i. Open-loop Bode plot with Phase margin marked ii. Closed-loop Bode plot with Bandwidth frequency marked iii. Root locus plot iv. Time response plot with steady-state value (or error) marked	2
b. Proportional and Derivative gain values	0.5
Step 9	3
a. Plots/Screenshots of: i. Open-loop Bode plot with Phase margin marked ii. Closed-loop Bode plot with Bandwidth frequency marked iii. Root locus plot iv. Time response plot with steady-state value (or error) marked	2
b. Proportional, Derivative, and Integral gain values (mention if values are same or readjusted from step 8)	1
Steps 10-13	7
a. Closed Loop Transfer Function	1
b. Poles, natural frequency, and damping of the closed-loop system	1
c. Image of PID controller Simulink block diagram (Step 12)	1
d. Plot (same figure) and compare (explain): Ramp response with PD vs. PID controller (additionally include zoomed-in figures to highlight difference)	2
e. Plot (same figure) and compare (explain): Error with PD vs. PID controller	2
Additional Questions	10
a. Describe the difference between Fig. 2 from the control design section of the lab manual and the control structure screenshot saved from Step 6.	1
b. Show a modified Simulink model that matches the control architecture from Step 6.	2
c. Plot (same figure): Step responses from both Simulink models (step 12 and modified) and Explain: why the difference between the two control structures is acceptable for step input.	2
d. Plot (same figure) for step response: i. Derivative of error signal from Step 12 model	3
ii. Derivative of error signal from modified model iii. Speed from Step 12 model iv. Speed from modified model	
e. Question: Explain how the feed-forward and feedback derivative gain influence system behavior as observed in the input voltage plot.	2

CONTROLLER IMPLEMENTATION (Group): 35 points

Step Response		12
a. Plot (same figure) for PD-controller at 0.002s sampling time with tuned gains: i. Commanded rotary position ii. Experimental rotary position iii. Digital Twin rotary position		3
b. Plot (same figure) for PD-controller at 0.04s sampling time with tuned gains: i. Commanded rotary position ii. Experimental rotary position iii. Digital Twin rotary position		3
c. Questions: i. Why is there no steady-state error in a PD controller with a step command?		3
ii. What is the effect of sampling time on closed-loop system behavior?		3
Ramp Response		12
a. Plot (same figure) for PD-controller at 0.002s sampling time: i. Commanded rotary position ii. Experimental rotary position iii. Digital Twin rotary position		3
b. Plot (same figure) for PID-controller at 0.002s sampling time: i. Commanded rotary position ii. Experimental rotary position iii. Digital Twin rotary position		3
d. Questions: i. What is the effect of I-gain in ramp response?		2
ii. Explain differences in ramp responses with PD gains between digital twin and experimental results.		2
iii. Explain difference in ramp responses with PID gains between digital twin and experimental results.		2
Closed Loop System		11
a. Questions i. Determine the bandwidth of the closed loop system with the tuned gains		4
ii. Determine the phase margin of the closed loop system with the tuned gains		4
iii. Discuss dead zone in DC motor and how it affects steady-state error.		3

2 Introduction

This project was split into three weeks, the second of which was completed at the individual level, while the remaining weeks were performed as a group during lab time. The experiment began with an open-loop characterization, moved into a closed-loop controller design, and finally controller validation using real hardware using a Rotary Servo Base Unit.

2.1 Week 1 Objective

The objective of the first week was to obtain a transfer function describing the rotary motion of the Rotary Servo Base Unit load shaft. Before we designed the controller, we reviewed how both the dynamic model and the Rotary Servo Base Unit transfer function model were derived, where the former is derived analytically from classical mechanics principles and using experimental methods, and the latter simply analytically in [Modeling Using First-Principles](#). From these derivations we can obtain values and expressions for the steady-state gain K and time constant τ . After the lab experiments, the experimental model parameters were compared with the nominal values.

2.2 Week 2 Objective

Building on the previous week, the objective of this experiment was to design a position control system for the motor to meet a given set of specifications in the frequency-domain. In other words, we individually focused on closing the open loop found in Week 1 in three progressive stages: first with a P controller, followed by a PD controller, and lastly with a PID controller, all tuned to satisfy bandwidth, phase-margin, and steady-state error specifications.

2.3 Week 3 Objective

Lastly, the controller developed over the past two weeks was evaluated using both a Quanser Digital Twin and the physical Servo Base hardware. Both models were subjected to a step and ramp response experiment, and once all of the data was collected, the system behaviors were compared to identify the discrepancies between the two models as well as to assess how well the idealized model from Weeks 1 and 2 predicts real system behavior.

2.4 Real-World Applications

Rotary servo systems play a crucial role across a diverse number of engineering fields, including aerospace, mechanical, robotics, industrial, etc. Their closed-loop systems allow for feedback on how fast a servo is moving, how fast its moving, how much torque it supplies, or how much voltage it requests from a system. All of these qualities are important

when working with robotic arms to calculate end effector positions, a car's mileage counter, or automated machines with a need for extreme precision such as a CNC machine or 3D printer.

2.5 Experimental Approach

The project was a blend of computational and experimental work. Using MATLAB as our primary programming language, our team was able to connect to real hardware in order to perform the open-loop characterization using Simulink. Simulink and MATLAB's Control System Designer were primarily used in Week 2, as these tools were critical in designing the closed loop controller. Lastly, MATLAB and Simulink were once again used to perform the controller validation for Week 3.

3 Design Steps

The Rotary Servo Base was first modeled as a first-order system relating motor voltage to load-shaft angular velocity,

$$\frac{\Omega_l(s)}{V_m(s)} = \frac{K}{\tau s + 1}. \quad (1)$$

The parameters K and τ were found using the nominal model, frequency-response testing, and a bump test. The frequency-response data were used to determine the steady-state gain and -3 dB cutoff frequency, while the bump-test parameters were found from the measured step response. The models were then compared with the measured servo response, and the final model-validation values were

$$K = 1.649, \quad \tau = 0.0274 \text{ s}. \quad (2)$$

Since the controller regulates angular position rather than angular velocity, an integrator was added to the plant model. The plant used for controller design was

$$G(s) = \frac{\Theta_l(s)}{V_m(s)} = \frac{K}{s(\tau s + 1)} = \frac{1.649}{0.0274s^2 + s}. \quad (3)$$

Controller design was performed in MATLAB Control System Designer. A proportional controller was done first, followed by a PD controller just the proportional control could not satisfy all of the design requirements. The PD controller had the form

$$C_{PD}(s) = K_p + K_d s. \quad (4)$$

Proportional and derivative gains were adjusted to meet the bandwidth and phase-margin requirements. The controller was then implemented on the digital twin and further tuned to achieve less than 5% overshoot and a peak time below 0.4 s.

The final tuned gains used for the Week 3 experiments were

$$K_p = 12 \quad K_d = 0.07 \quad (5)$$

Integral control was added for the PID ramp-response tests. The integral gain was set to 1% of the proportional gain,

$$K_i = 0.01K_p = 0.12, \quad (6)$$

giving the PID controller

$$C_{PID}(s) = 12 + 0.07s + \frac{0.12}{s}. \quad (7)$$

The final controller was implemented in Simulink and evaluated using both the Digital Twin and the physical Rotary Servo Base. Step and ramp responses were recorded using the same controller gains whenever possible, and the effect of sampling time was evaluated by comparing sampling periods of 0.002 s and 0.04 s.

4 Modeling

4.1 Frequency Response Test

4.1.1 Frequency Response Data

Table 1: Collected frequency response data.

F (Hz)	Amplitude (V)	Maximum Load Speed (rad/s)	$ G(\omega) $ (rad/s/V)	$ G(\omega) $ (dB)
0.0	2.0	3.3587	1.6794	4.503
1.0	2.0	3.3599	1.6799	4.506
2.0	2.0	3.2738	1.6369	4.281
3.0	2.0	3.0716	1.5358	3.727
4.0	2.0	2.9814	1.4907	3.468
5.0	2.0	2.7545	1.3773	2.780
6.0	2.0	2.5980	1.2990	2.272
7.0	2.0	2.4865	1.2432	1.891
8.0	2.0	2.3069	1.1535	1.240

4.1.2 Bode Magnitude Plot

The $f = 0$ Hz data point is excluded from the Bode plot because the logarithm of zero is undefined.

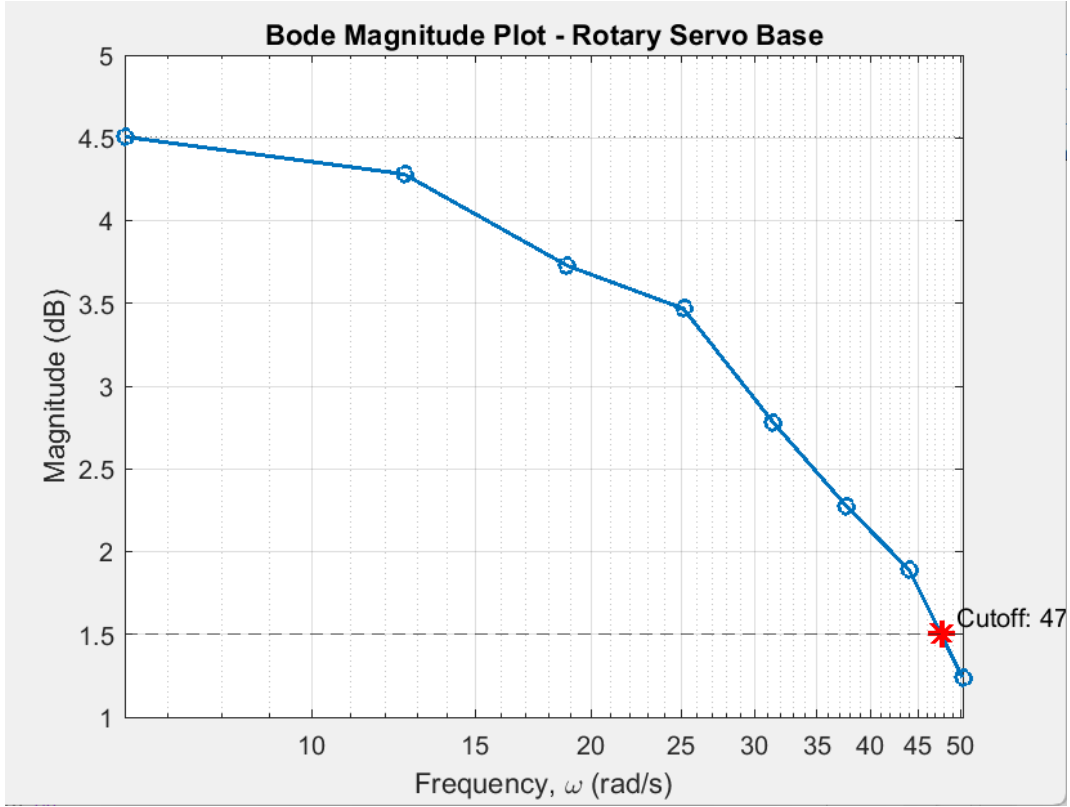


Figure 1: Bode magnitude plot generated from the frequency response data. The cutoff frequency is identified at the frequency where the gain decreases by 3 dB from the DC gain.

4.1.3 DC Gain Calculation

The DC gain represents the SS output of the servo at a constant input voltage. Here, it is the gain when the system is at zero Hertz frequency with an applied voltage of 2 V, represented by V_m . The DC was calculated with:

$$K_{DC} = \frac{\omega_l}{V_m} \quad (8)$$

The speed at zero Hz was 3.3587 rad/s, represented by ω_l . Using this, the DC gain was calculated by:

$$K_{DC} = \frac{3.3587 \text{ rad/s}}{2.0 \text{ V}} = 1.6794 \frac{\text{rad/s}}{\text{V}}. \quad (9)$$

Thus, the DC gain is:

$$\boxed{K_{DC} = 1.6794 \frac{\text{rad/s}}{\text{V}}} \quad (10)$$

To find it in decibels, the log of the gain was taken and multiplied by 20:

$$K_{DC,\text{dB}} = 20 \log_{10}(1.6794) = 4.503 \text{ dB.} \quad (11)$$

Using the gains in decibels of all the frequencies, the cutoff frequency at 3 dB from the DC gain was found and used in Table 3. It was found to be 1.503 dB.

4.1.4 Time Constant Calculation

The necessary time constant to focus on is the value found using the frequency, ω_c , at the cutoff gain, established above. The time constant of the system at 1.503 dB, or 7.6 Hz can be found by converting Hz into radians/sec by:

$$\omega_c = 2\pi(7.6) = 47.75 \text{ rad/s.} \quad (12)$$

The time constant is the inverse of the cutoff frequency in rad/s using the following equation:

$$\tau = \frac{1}{\omega_c} \quad (13)$$

Thus, the time constant of the system is:

$$\tau = \frac{1}{47.75} = 0.02094 \text{ s.} \quad (14)$$

4.2 Step Response Test

4.2.1 Maximum Load Speed Response

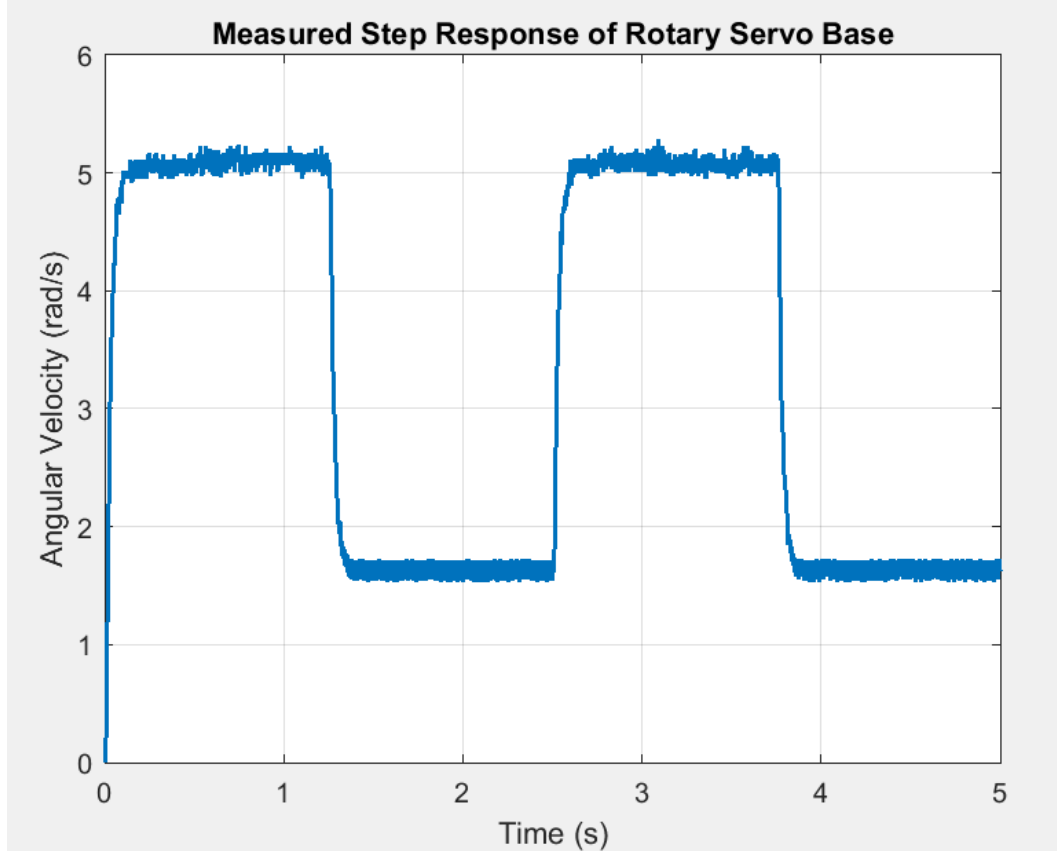


Figure 2: Maximum load speed response during the step response experiment.

Specific points were measured off the plot using the `ginput` command. To find gain and time constant, two values were measured: the steady-state value y_{ss} and the initial output signal at y_0 . These values were determined using the mean values of the high and low speeds on the plot and further explained below. The input voltage was changed from an initial value of $u_{\min} = 1$ V to a final value of $u_{\max} = 3$ V. From the plot, the initial load shaft speed was measured as $y_o = 1.6123$ rad/s while the SS load shaft speed was measured as $y_{ss} = 5.1537$ rad/s.

4.2.2 DC Gain Calculation

The DC gain was calculated using the change in output divided by the change in input:

$$K = \frac{\Delta y}{\Delta u} \quad (15)$$

In this equation, the changes were found with:

$$\Delta y = y_{ss} - y_0 \quad (16)$$

and

$$\Delta u = u_{\max} - u_{\min}. \quad (17)$$

Using the values found from the step response plot:

$$\Delta y = 5.1537 - 1.6123 = 3.5414 \text{ rad/s} \quad (18)$$

and

$$\Delta u = 3 - 1 = 2 \text{ V}. \quad (19)$$

The DC gain was found to be:

$$K = \frac{3.5414}{2} = \boxed{1.7707 \frac{\text{rad/s}}{\text{V}}}. \quad (20)$$

4.2.3 Time Constant Calculation

The time constant was found with the measured response times from the step response plot. The times used were that of the initial step, identified by the second transition from low to high. The initial time of the step was identified at 2.5 seconds. The second time was determined at the 63.2% using the following equation:

$$y(t_1) = 0.632(y_{ss} - y_0) + y_0. \quad (21)$$

Using the equations defined in the plot, described above:

$$y(t_1) = 0.632(5.1537 - 1.6123) + 1.6123 \quad (22)$$

$$y(t_1) = 3.8505 \text{ rad/s}. \quad (23)$$

To find the time that corresponded to this speed, the plot was used and the exact value was found to be 2.533 seconds. Thus, the time constant was calculated and found to be:

$$\tau = t_1 - t_0. \quad (24)$$

Therefore,

$$\tau = 2.533 - 2.500 = \boxed{0.033 \text{ s}}. \quad (25)$$

4.3 Nominal Values

In this section the electrical equation that describes the DC motor armature circuit and the mechanical equation that couples the the speed of the load shaft with respect to the applied motor torque are brought together to get an expression that represents the load shaft speed in terms of the applied motor voltage:

$$J_{eq} \frac{d\omega_l}{dt} + (B_{eq,v})\omega_l(t) = A_m V_m(t) \quad (26)$$

Taking the Laplace transform of this equation yields:

$$\mathcal{L}\{\dots\} = J_{eq} \cdot s \cdot \Omega_l(t) + B_{eq,v} \cdot \Omega_l(t) = A_m \cdot V_m(t) \quad (27)$$

Factoring $\Omega_l(t)$ from the LHS of the equation and isolating $\frac{\Omega_l(t)}{V_m(t)}$ we get:

$$\frac{\Omega_l(t)}{V_m(t)} = \frac{A_m}{J_{eq}s + B_{eq,v}} = \frac{A_m/B_{eq,v}}{J_{eq}s/B_{eq,v} + 1} \quad (28)$$

Where:

$$B_{eq,v} = \frac{k_m \eta_g K_g^2 \eta_m k_t + B_{eq} R_m}{R_m} \quad (29)$$

$$A_m = \frac{\eta_g K_g \eta_m k_t}{R_m} \quad (30)$$

The values above were provided in the lab manual's appendix. Since we know that $\frac{\Omega_\ell(s)}{V_m(s)} = \frac{K}{\tau s + 1}$, we can compare this with equation 28 and plug in the corresponding values to get our finalized nominal values of the steady-state gain and time constant:

$$K = \boxed{1.528 \frac{\text{rad}}{\text{s} \cdot \text{V}}} \quad (31)$$

and

$$\tau = \boxed{0.0217 \text{ s}}. \quad (32)$$

4.4 Model Validation

The measured response was then compared with plots from the varying experiments using their gain and tau values: frequency response, nominal, step bump test, and the original model.

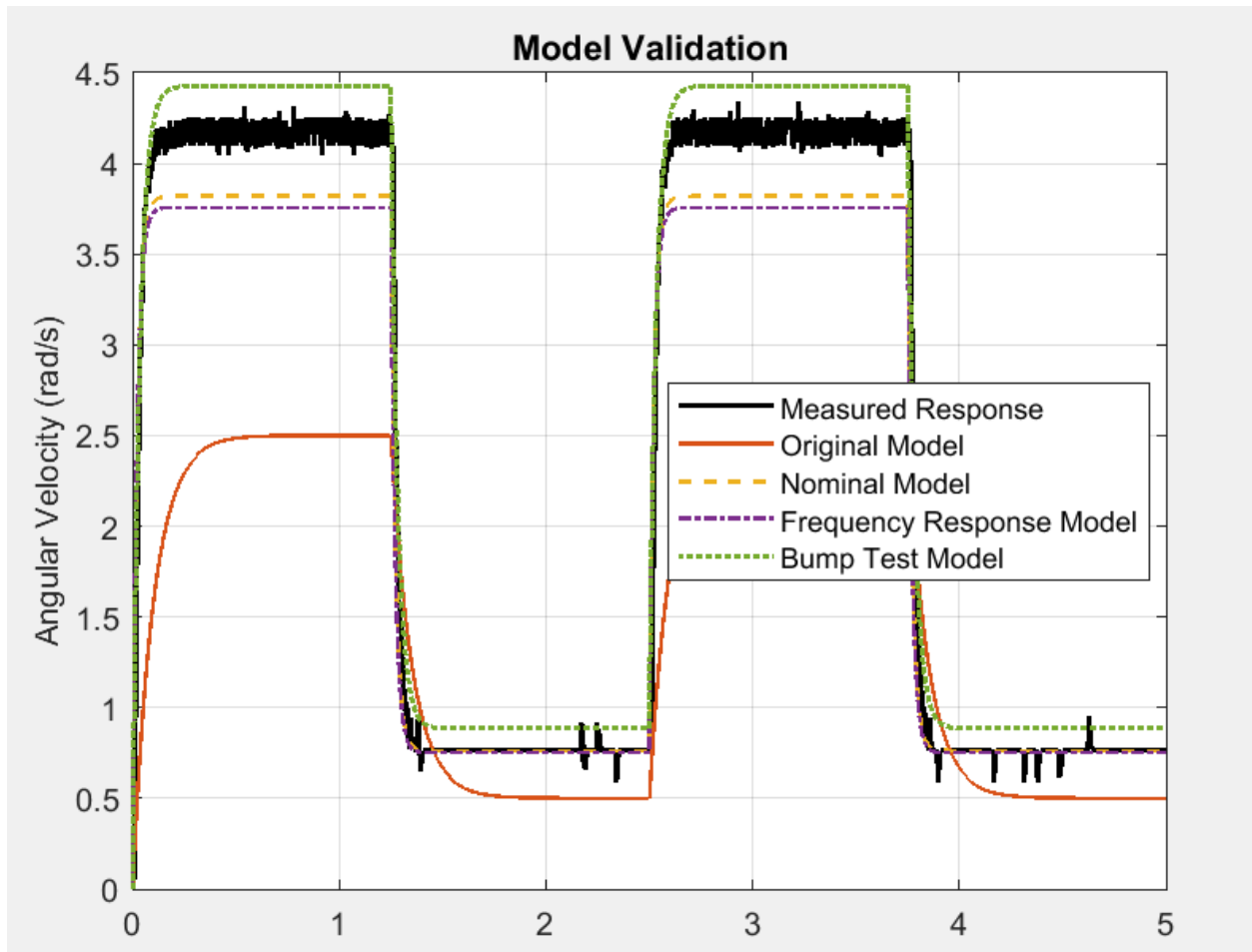


Figure 3: Model validation comparison between the measured response and simulations using the frequency response, bump response, and nominal model parameters.

4.4.1 Model Validation Results

Table 2: Summary of results for the Rotary Servo Base Unit modeling laboratory.

Section	Description	Symbol	Value	Unit
Nominal Values	Open-loop steady-state gain	K	1.528	rad/(V s)
Nominal Values	Open-loop time constant	τ	0.0217	s
Frequency Response	Open-loop steady-state gain	$K_{e,f}$	1.503	rad/(V s)
Frequency Response	Open-loop time constant	$\tau_{e,f}$	0.0209	s
Bump Test	Open-loop steady-state gain	$K_{e,b}$	1.7707	rad/(V s)
Bump Test	Open-loop time constant	$\tau_{e,b}$	0.0330	s
Model Validation	Open-loop steady-state gain	$K_{e,v}$	1.649	rad/(V s)
Model Validation	Open-loop time constant	$\tau_{e,v}$	0.0274	s

4.4.2 Effect of K and τ

Question: How do K and τ affect the DC motor response to a step input?

These two values are critical to a DC motor's response to any type of input and affect a step response in various ways. Gain assists in determining the steady state value of the motor speed, so a larger gain value will create a larger SS angular velocity in the same step response. Changing K will primarily change the magnitude of the response. On the other side, the time constant affects how quickly the motor will reach that steady state value. A smaller value will give a faster response while a larger value gives a slower response. Thus, the time constant mostly changes the speed of the response.

5 Control Design

5.1 Keira Mapel

5.1.1 Step 5

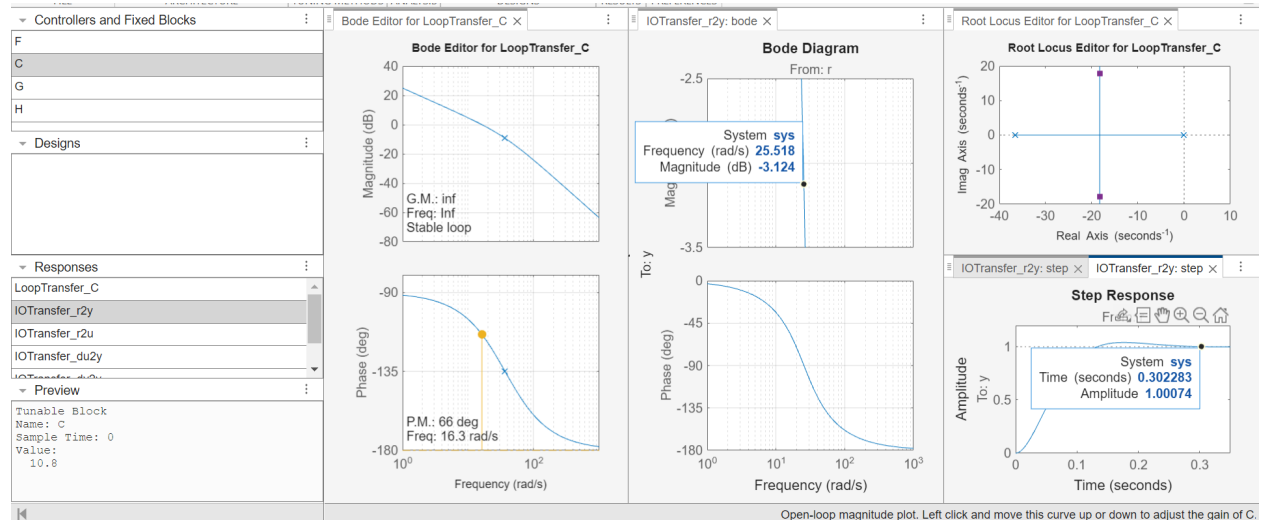


Figure 4: Keira Mapel – Step 5 Combined OL and CL Bode Plot, Root Locus Plot, and Time Response Plot.

The proportional gain for my Step 5 is

$$K_p = 10.8 \frac{\text{V}}{\text{rad/s}}. \quad (33)$$

5.1.2 Step 6

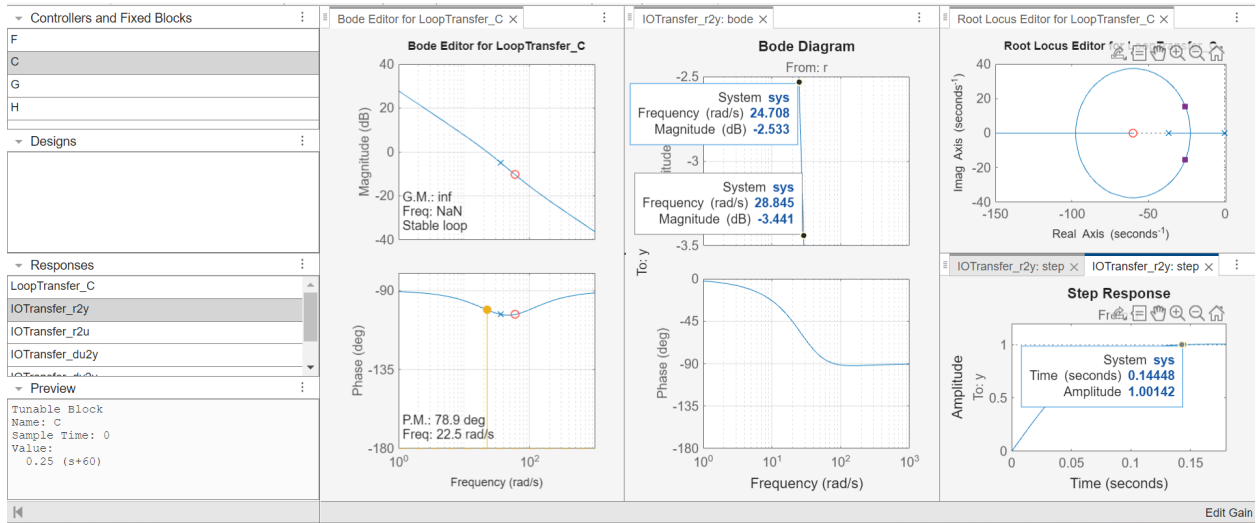


Figure 5: Keira Mapel – Step 6 Combined OL and CL Bode Plot, Root Locus Plot, and Time Response Plot.

The controller gains for Step 6 are

$$K_p = 15 \frac{\text{V}}{\text{rad/s}}, \quad K_d = 0.255 \frac{\text{V}}{\text{rad/s}}. \quad (34)$$

5.1.3 Step 9

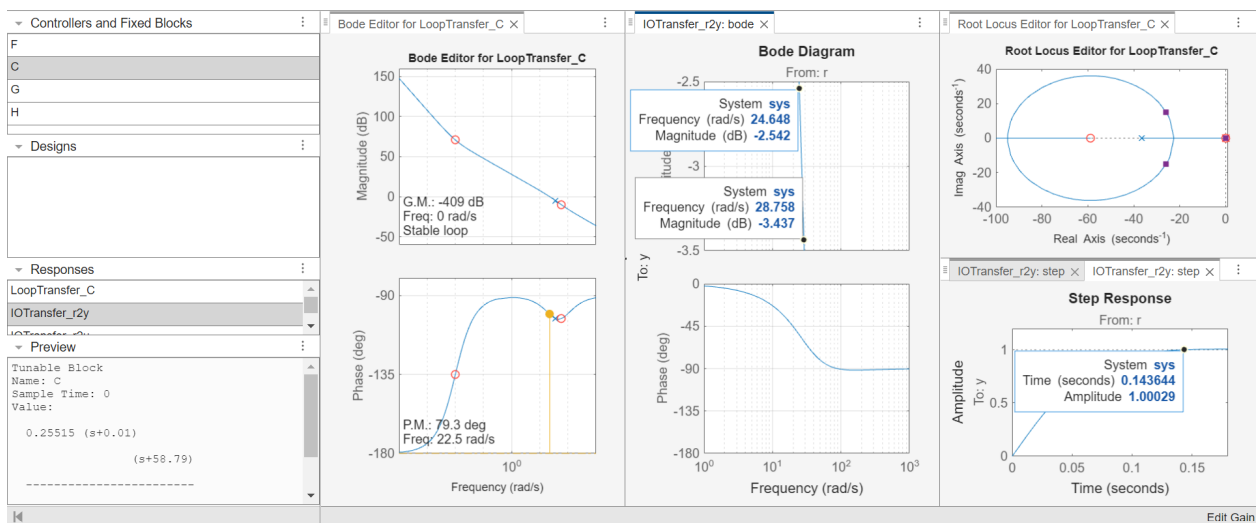


Figure 6: Keira Mapel – Step 9 Combined OL and CL Bode Plot, Root Locus Plot, and Time Response Plot.

The final PID gains are

$$K_p = \boxed{15 \frac{\text{V}}{\text{rad/s}}}, \quad K_d = \boxed{0.255 \frac{\text{V}}{\text{rad/s}}}, \quad K_i = \boxed{0.15 \frac{\text{V}}{\text{rad/s}}}. \quad (35)$$

These values are the same as the ones from step 8.

5.2 Steps 10-13

5.3 Step 10

Closed-loop transfer function obtained:

$$G(s) = \frac{15.38s^2 + 904.5s + 9.044}{s^3 + 51.94s^2 + 904.5s + 9.044}$$

5.4 Step 11

Natural frequency and damping of the dominant poles of the closed-loop system:

Table 3: System poles, damping ratios, and natural frequencies.

p	ζ	ω_n
0.0100	1.0000	$-0.0100 + 0.0000i$
30.0670	0.8636	$-25.9651 + 15.1603i$
30.0670	0.8636	$-25.9651 - 15.1603i$

5.5 Step 12

The PID controller block diagram for the system is shown below:

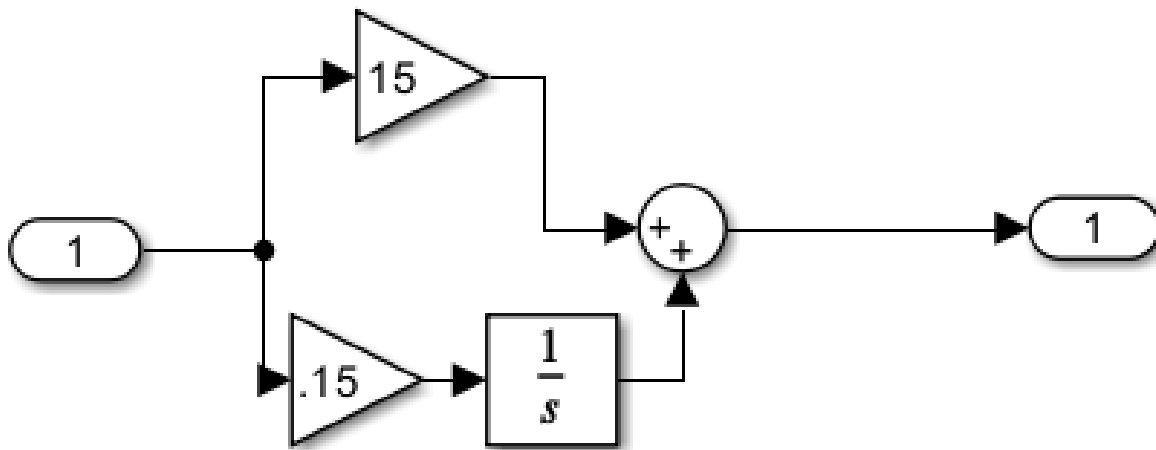
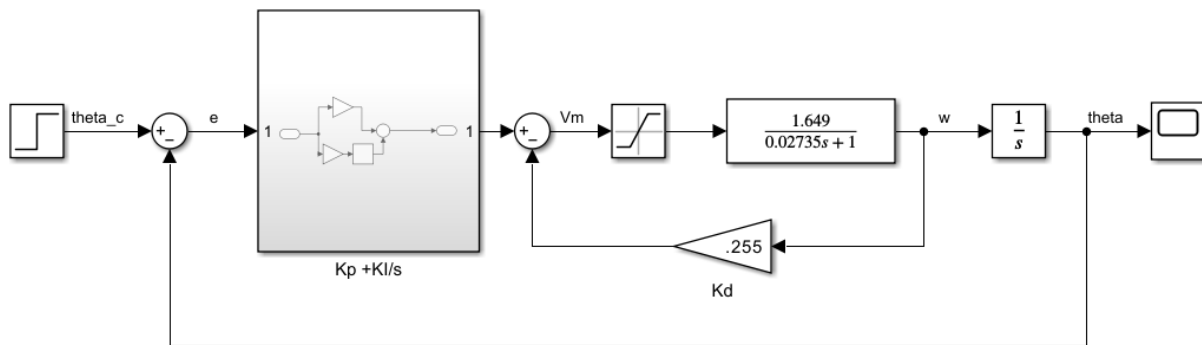


Figure 7: Keira Mapel – Closed Loop Transfer Function.

5.6 Step 13

The following two images show, on the same plot, the differences in the PD and PID controller on the response and the error. As it can be seen, in the ramp response the PD and PID controllers track the controlled ramp very closely. However, on closer view, both controllers have a small SS tracking error; the actual position is slightly below the commanded position. The PD controller produces a slighter larger tracking error than the PID, which is more close to the command. This shows how the integral term reduces SS error.

In the tracking error plot, the difference between the two errors is more prominent. The PD controller maintains a constant error of about 0.0575, but the PID controller's error

decreases from this number to about 0.0425 over the simulation. The integral term in the controller allows it to reduce the tracking error while the PD stays at a constant error. Allow the I improves performance, it does not reduce to zero over the interval.

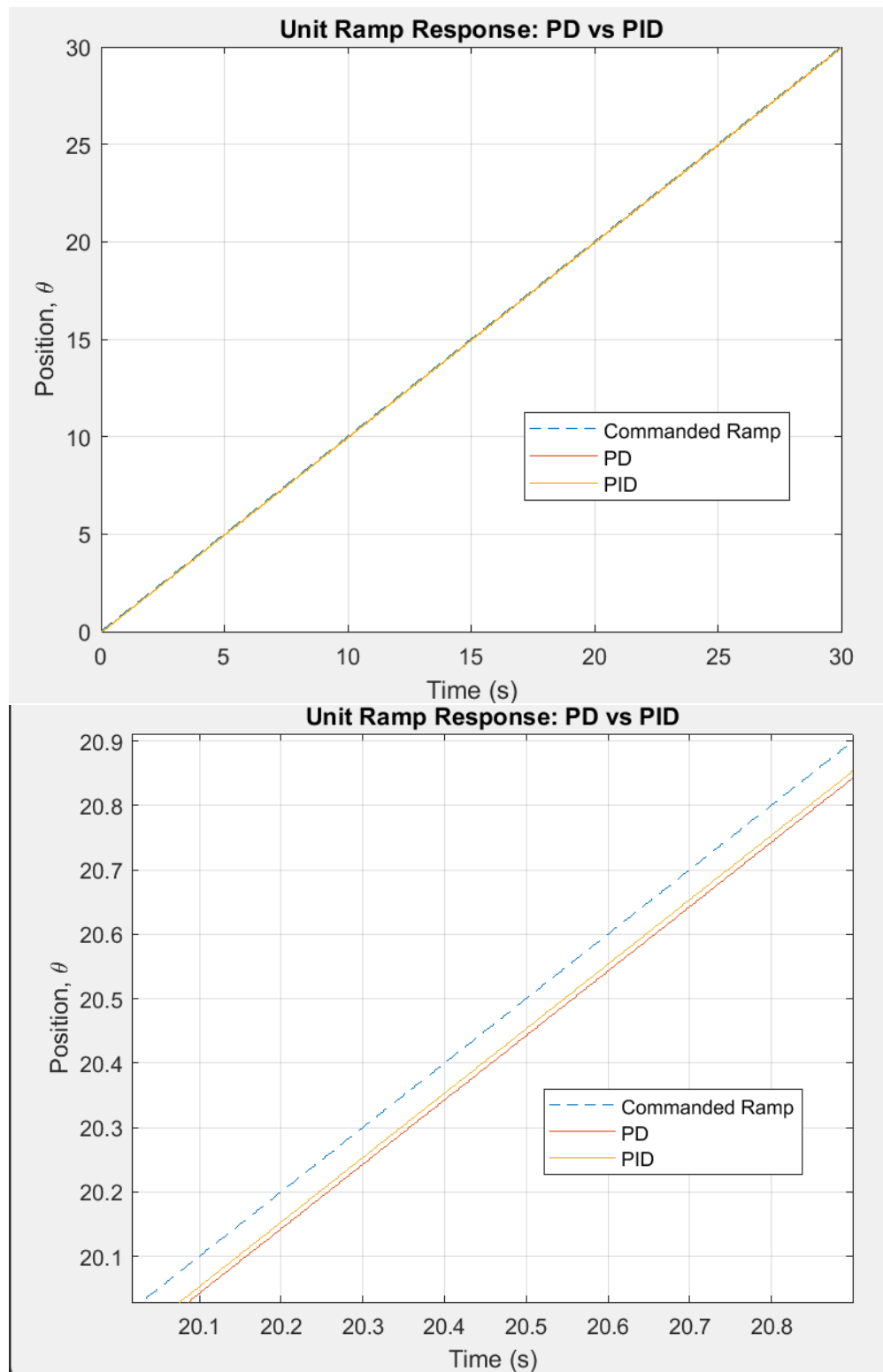


Figure 8: Keira Mapel Ramp Response with PD and PID controller. The bottom image includes a zoomed in portion to better characterize differences.

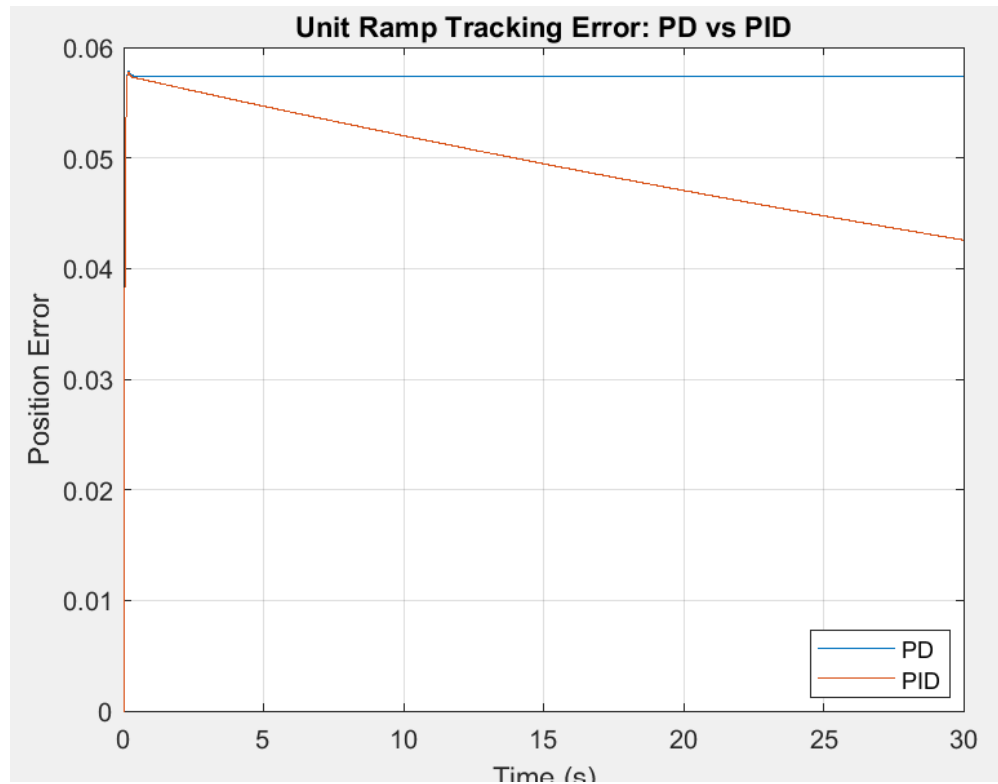


Figure 9: Error with the PD and PID controller.

5.7 Additional Questions

5.7.1 a. The difference between Fig. 2 and the Control Structure from Step 6

Figure 2 describes the control architecture from the lab manual, which uses a PD controller. The position error is formed by comparing the measured position with the commanded one. The error passes through the proportional gain. Inside the motor portion, there is an additional derivative feedback pack with the derivative gain, which feeds back the angular velocity before the motor plant. The motor output then gets integrated to obtain the position.

The second figure is the general Simulink control structure found in lab. The main difference between the figures is the location of the derivative gain and the signal it is multiplied by. In Figure 2, K_d is inside the inner feedback loop around the motor plant. It is multiplied by the angular velocity and yields a derivative feedback term that is subtracted from the proportional controller output before the motor plant. In the Simulink control structure, the controller is represented by a single C block before the plant. The derivative gain is still represented in feedback but acts directly on the error signal rather than the motor angular velocity.

5.7.2 b. A modified Simulink model matching the control architecture from step 6

The following two images show the modified Simulink models that match K_d being multiplied by the derivative of error and speed.

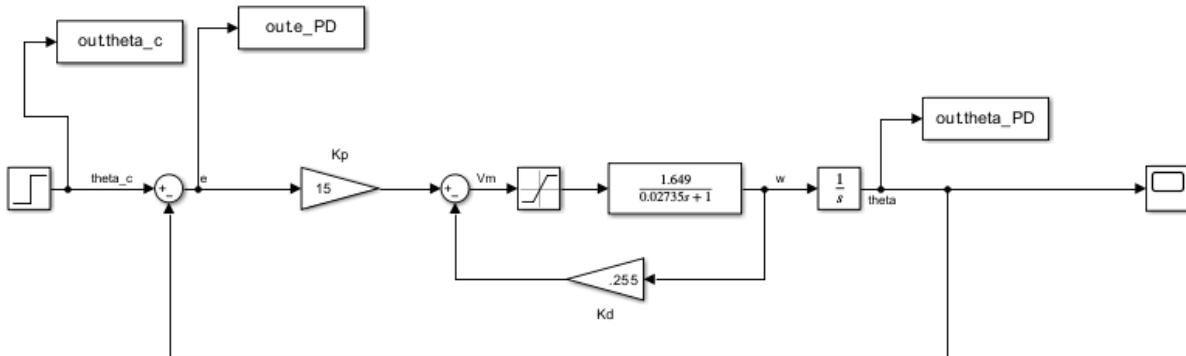


Figure 10: Simulink Control Structure of K_d multiplied by Angular Speed.

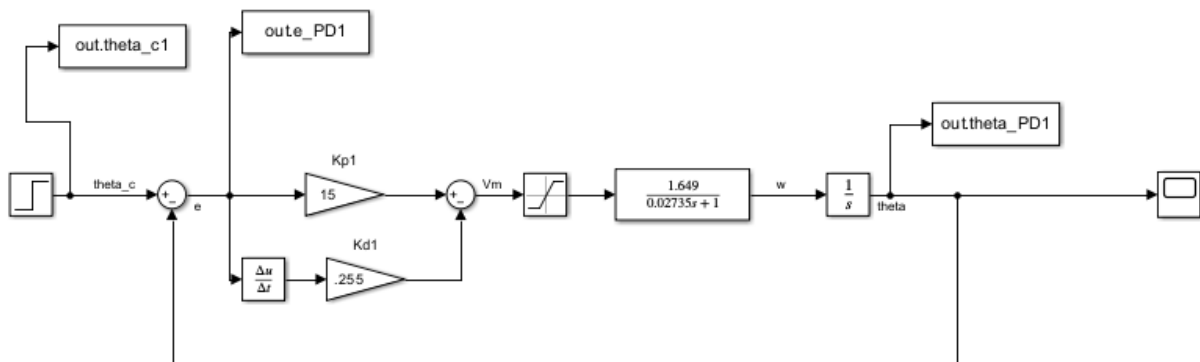


Figure 11: Simulink Control Structure of K_d multiplied by the Derivative of Error.

5.7.3 c. Plots of the Step Response from Step 12 and Modified

The following plot shows a comparison of the step responses from Step 12 and a modified version based on Step 6

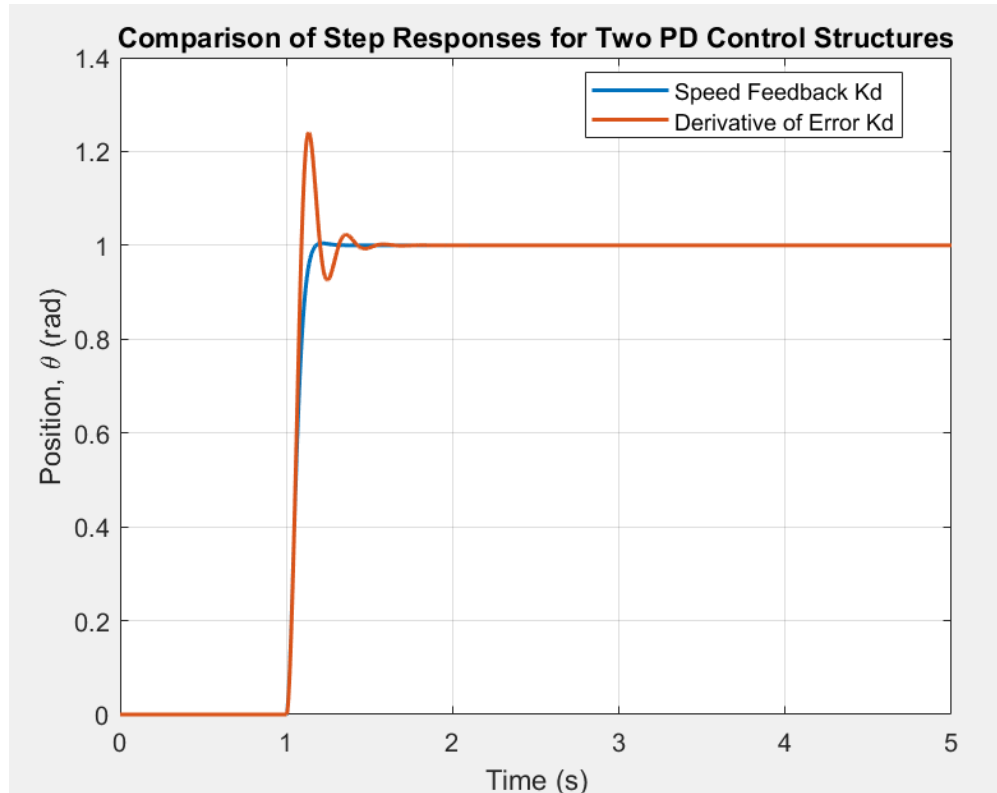


Figure 12: Plot Comparison of Step Responses of Two PD Controller Structures.

The differences between the two control structures is acceptable for the step input. This is because both systems ultimately produced the same desired response. As shown in the plot, both controllers reach the commanded position and settle at about the same value. The main difference is after the step input where the derivative of error structure creates more overshoot and oscillation because it reacts to the sudden change in the reference signal when a step is applied. Once the commanded position becomes constant, the derivative becomes zero and the term is then based on the angular velocity. Both structures behave similarly after the initial response and create the desired outcome, so the difference is acceptable.

5.7.4 d. Plots for Step Response

The following plot shows the differences in the derivative of error signal and the speed from the Step 12 model and the modified model. The modified model has a large initial spike because of how the reference position changes immediately. Other than that, the curves between controllers is very similar, and the derivative of error signal has an opposite sign from the speed signals, as expected.

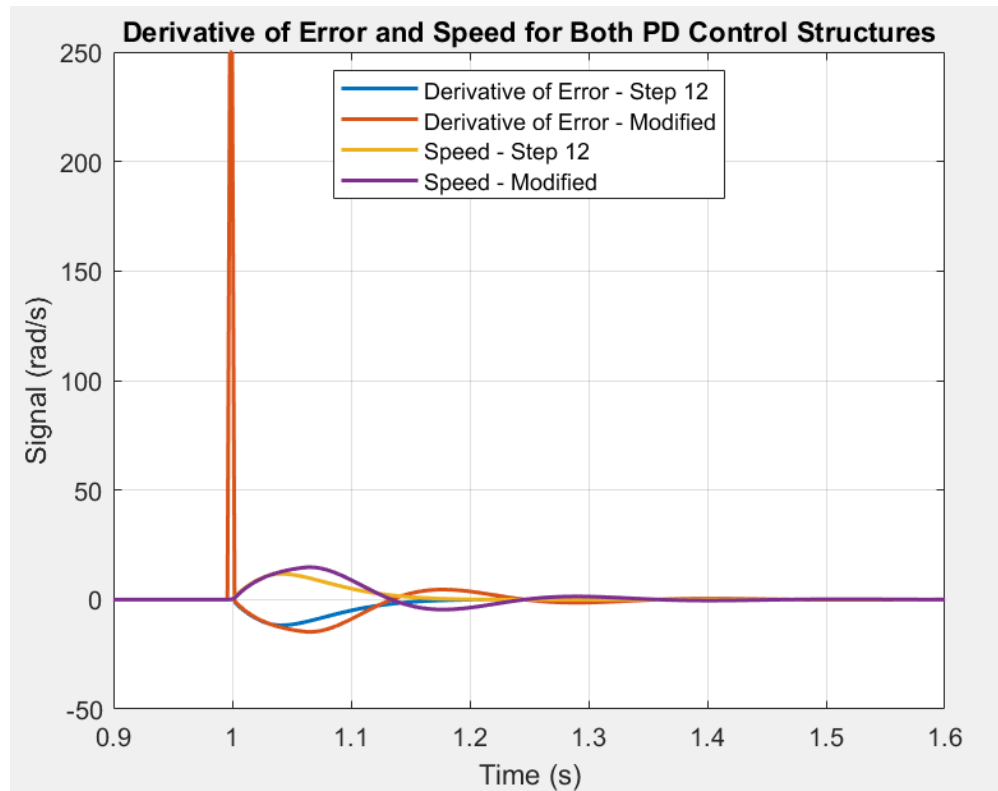


Figure 13: Plot Comparison of Step Responses of Two PD Controller Structures Using Speed and Derivative of Error Signal.

5.7.5 e. Question

Question: Explain how the feed-forward and feedback derivative gain influence system behavior as observed in the input voltage plot.

In the feed-forward structure, the derivative gain acts on the error. When the reference position changes suddenly during the step, the error term becomes very large for a short term, which will create a large change in the input voltage at the beginning of the response. In the feedback derivative, the derivative gain acts on the motor speed. Here, it will respond to how quickly the motor moves and generates a smoother change in the input voltage. When the motor reaches the commanded position, the error will change more slowly, causing motor speed to decrease. In both, the derivative contribution to the initial voltage will decrease to control oscillations and settling towards the desired position.

5.8 Victor Hernandez

5.8.1 Step 5

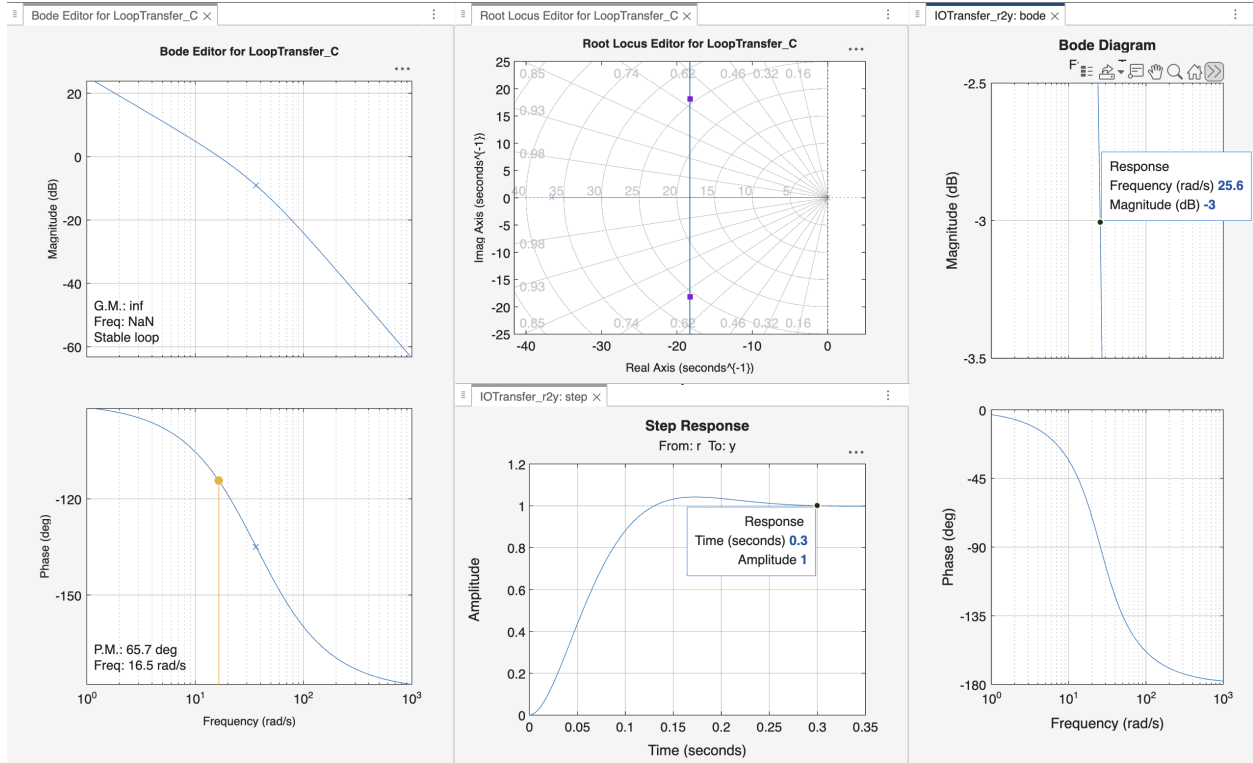


Figure 14: Step 5 Plots – Open/Closed Loop Plots, Step Response, and Root Locus.

The proportional gain K_p for Step 5 is

$$K_p = 11 \frac{\text{V}}{\text{rad/s}} \quad (36)$$

The full equation becomes:

$$C = K_p = 11. \quad (37)$$

5.8.2 Step 6

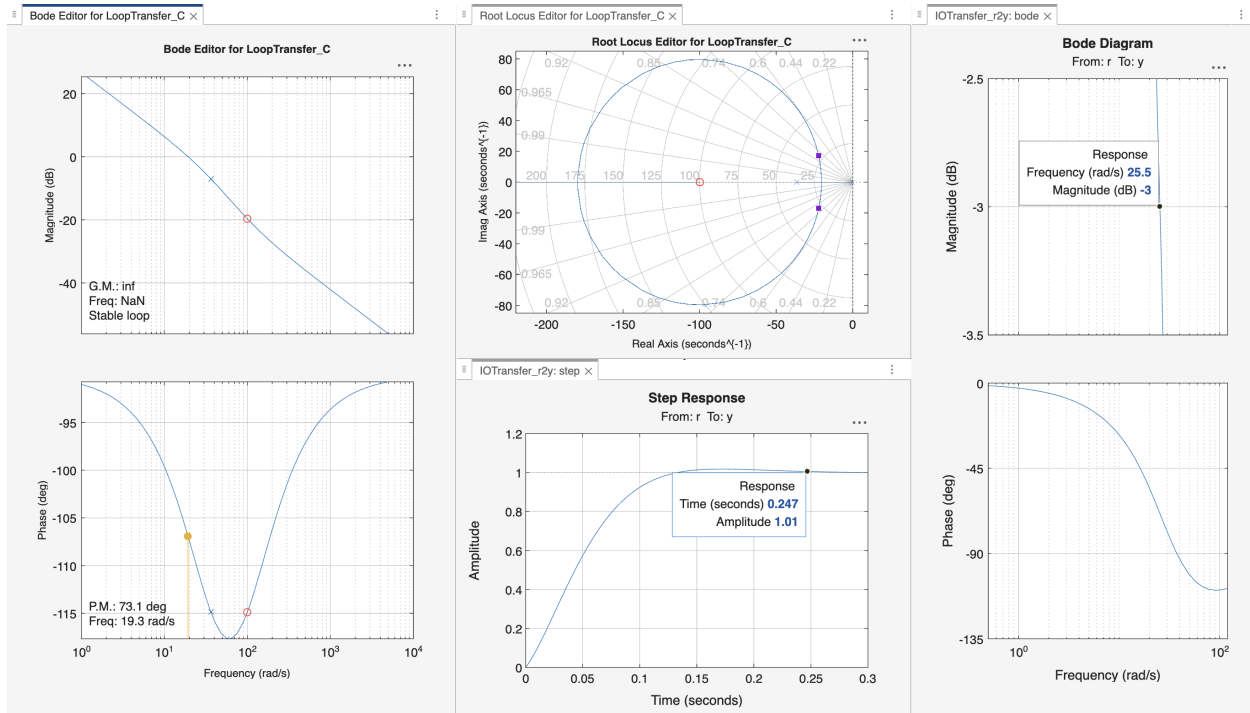


Figure 15: Step 6 Plots – Open/Closed Loop Plots, Step Response, and Root Locus.

The proportional and derivative gains K_p and K_d for Step 6 have been readjusted from step 5. They are:

$$K_p = 13 \frac{\text{V}}{\text{rad/s}}, \quad K_d = 0.13 \frac{\text{V}}{\text{rad/s}} \quad (38)$$

The full equation becomes:

$$C = K_p \left(1 + \frac{K_p}{K_d} s \right) = 13(1 + 0.01s) \quad (39)$$

5.8.3 Step 9

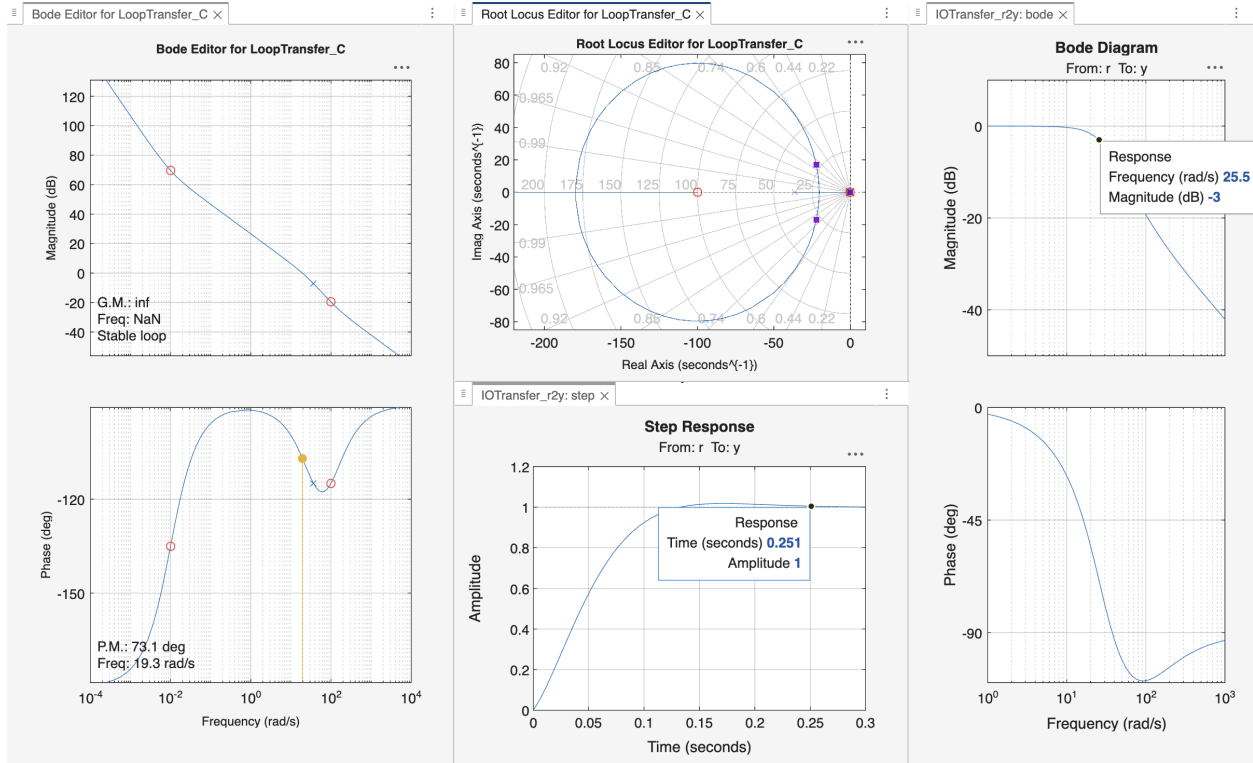


Figure 16: Step 9 Plots – Open/Closed Bode Plots, Step Response, and Root Locus.

The proportional, derivative, and integral gains K_p , K_d , and K_i for Step 9 have NOT been readjusted from step 6. They are:

$$K_p = 13 \frac{\text{V}}{\text{rad/s}}, \quad K_d = 0.13 \frac{\text{V}}{\text{rad/s}}, \quad K_i = 0.13 \frac{\text{V}}{\text{rad/s}} \quad (40)$$

The full equation becomes:

$$C = \frac{K_i \left(\frac{K_d}{K_i} s^2 + \frac{K_p}{K_i} s + 1 \right)}{s} = \frac{0.13(s^2 + 100.01s + 1)}{s}. \quad (41)$$

5.8.4 Step 10

Open-loop plant transfer function obtained:

$$G = \frac{1.649}{s(0.02735s + 1)} \quad (42)$$

Closed-loop transfer function obtained:

$$T = \frac{7.838s^2 + 783.9s + 7.838}{s^3 + 44.4s^2 + 783.9s + 7.838} \quad (43)$$

5.8.5 Step 11

Natural frequency and damping of the dominant poles of the closed-loop system:

Pole	Damping	Frequency (rad s ⁻¹)
-1.00×10^{-2}	1.00×10^0	1.00×10^{-2}
$-2.22 \times 10^1 + 1.71 \times 10^1 i$	7.93×10^{-1}	2.80×10^1
$-2.22 \times 10^1 - 1.71 \times 10^1 i$	7.93×10^{-1}	2.80×10^1

5.8.6 Step 12

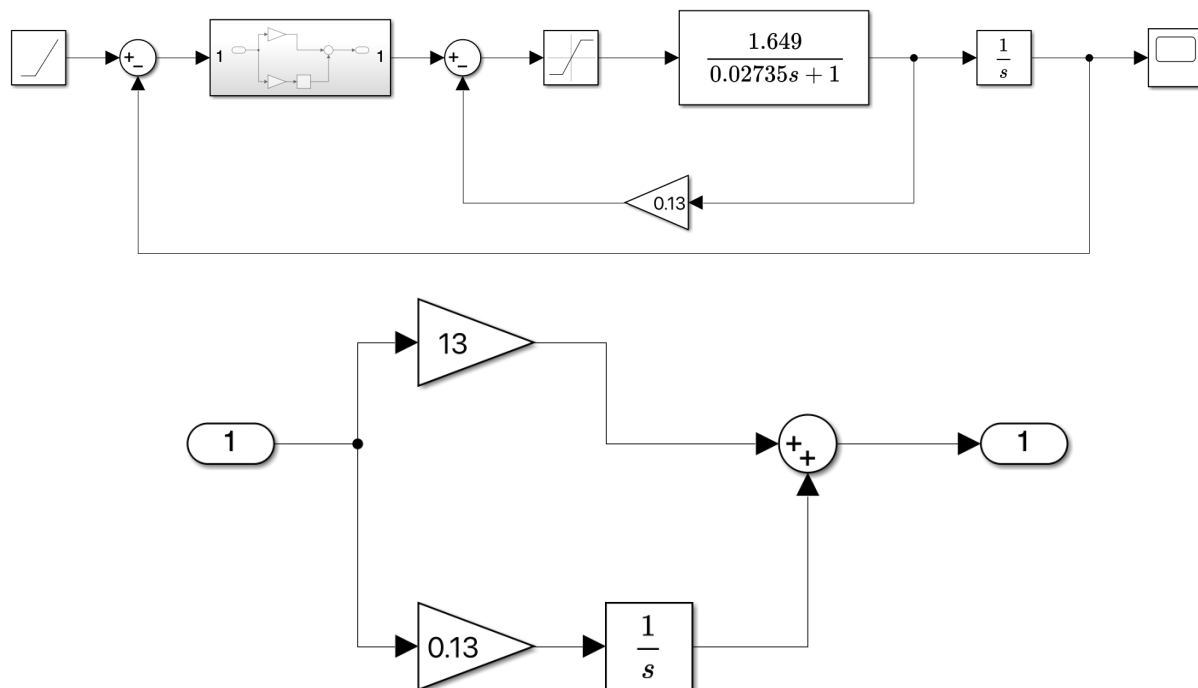


Figure 17: Step 12 Plots – Simulink diagram. The bottom figure represents the inner subsystem.

5.8.7 Step 13

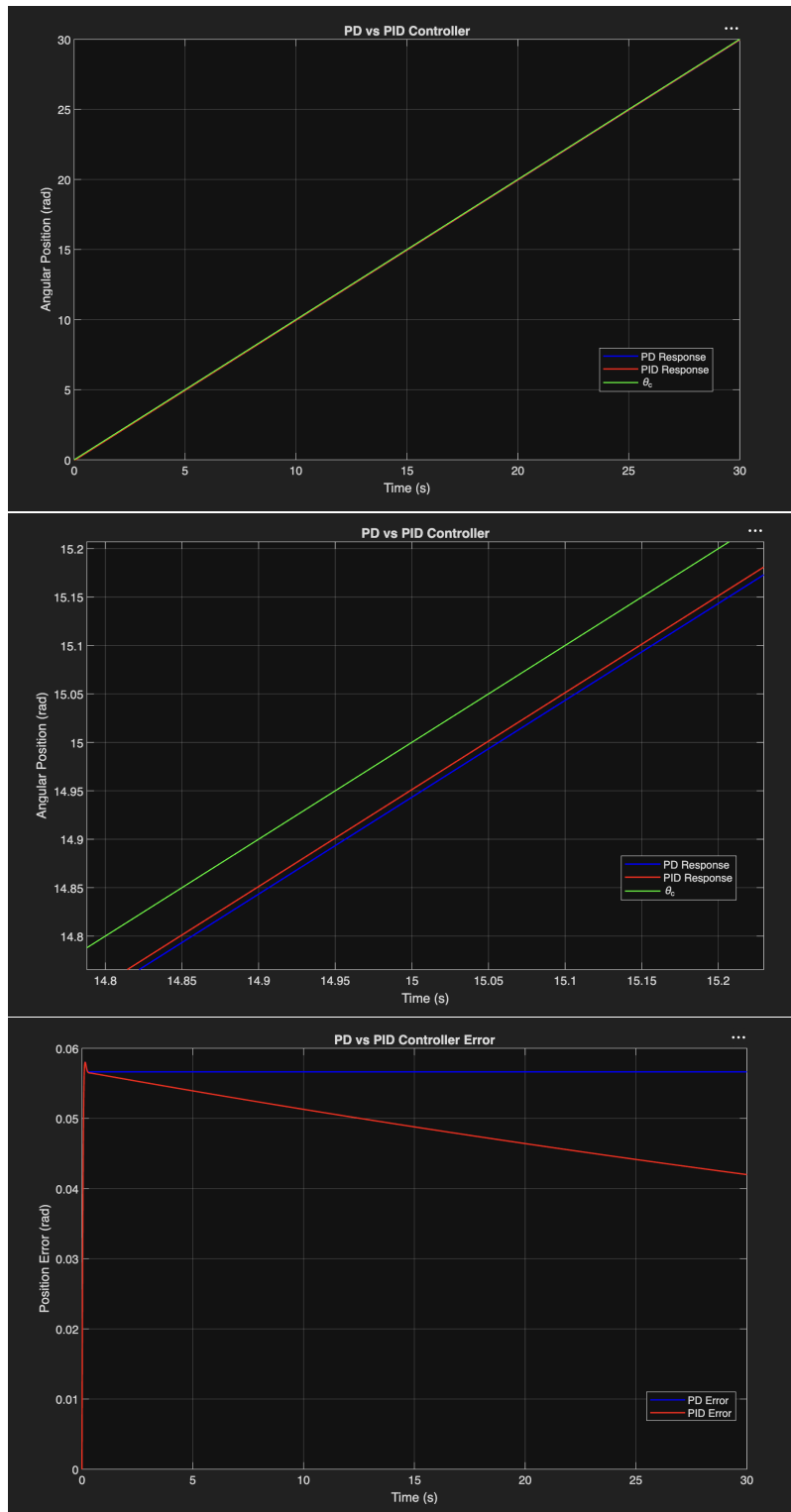
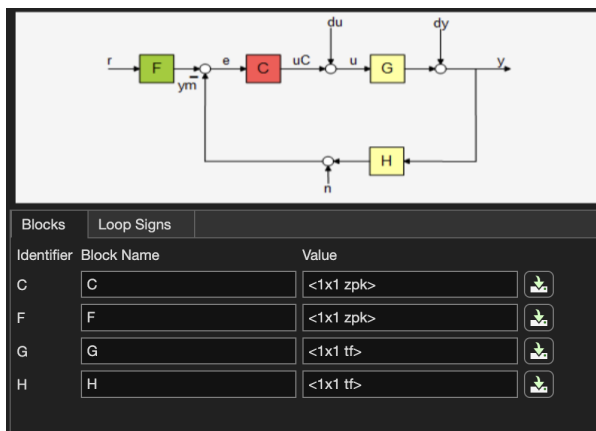


Figure 18: Step 13 Plots – Overlapped PD and PID response and error plots. The middle plot is the response zoomed in for clarity.

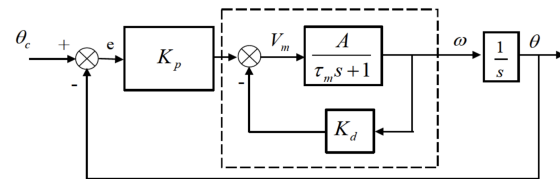
In the above plots, we can see that there is not much difference between the PD and PID controllers, as they both closely align with the ramp input. However, zooming into the controller and error plots, we can confirm that the PID controller is overall more effective as the error decreases with respect to time; in contrast with the PD controller, whose error does not decrease at all.

5.8.8 Additional Questions

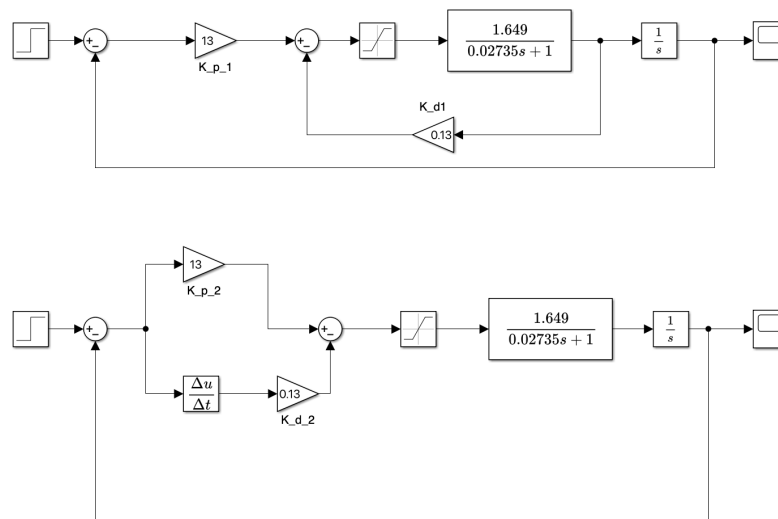
Question 1: Describe the difference between Fig. 2 from the control design section of the lab manual and the control structure screenshot saved from Step 6.



(a) Step 6 Control System Designer architecture.



(b) Closed-loop block diagram from 'Figure 2'.



(c) AQ-1 plot description.

Figure 19: (a) Step 6 architecture, (b) closed-loop block diagram, and (c) finalized Simulink diagrams. The top is from step 12 and the bottom represents the CSD architecture.

With the two diagrams side-by-side, we can see that the top figure obtained from the Control System Designer architecture is a single feedback loop, whereas the bottom figure from the lab manual has a nested subsystem. Both controllers are of the same PD type, however, the top figure obtains the K_d term by differentiating the error signal in one single forward path, while the bottom figure gets it through direct velocity feedback via an inner loop.

Question 2: Plot on the same figure the step responses from both Simulink models (Step 12 and modified), and explain why the difference between the two control structures is acceptable for a step input.

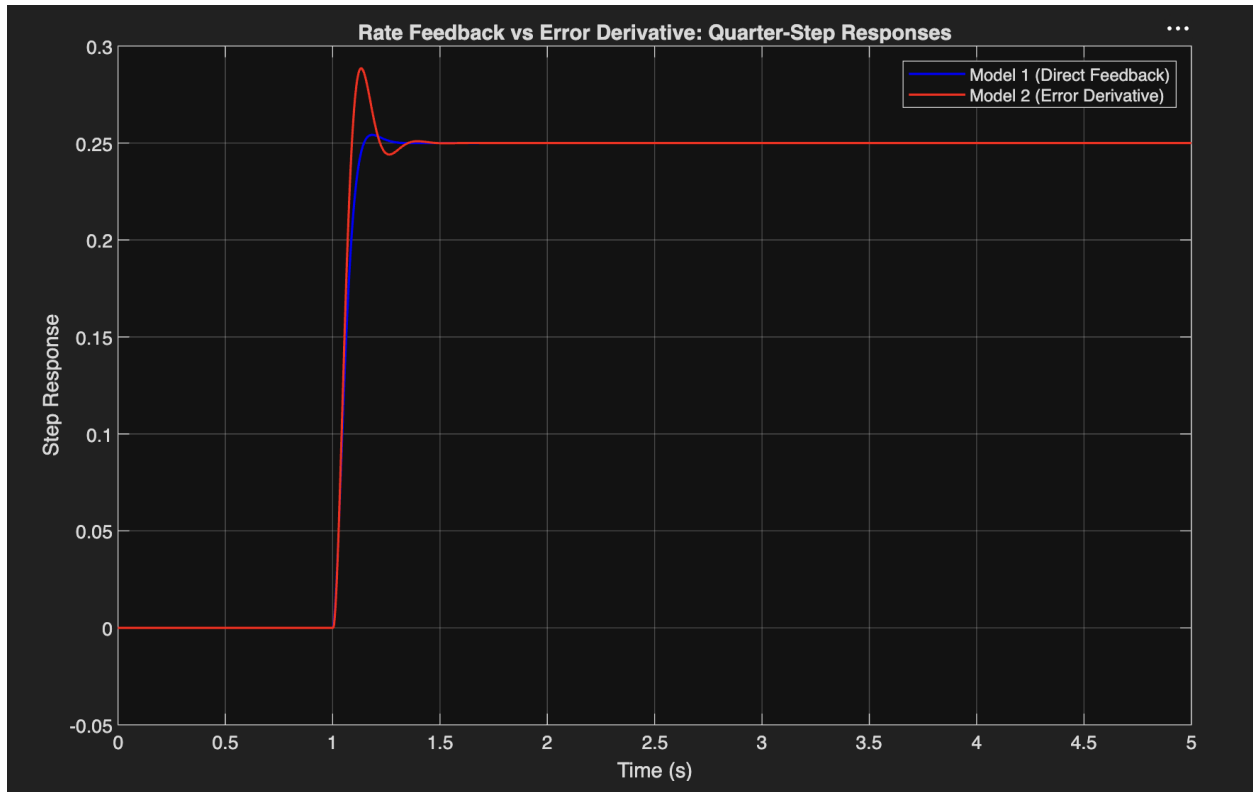


Figure 20: Quarter-step responses to two models: direct feedback and error derivative

Since both models eventually produce the same desired steady-state response, we can deem the models acceptable. However, we can see that there is more overshoot from Model 2. This is because we introduced a differentiation block after the error signal which contains a step discontinuity, and thus introduces overshoot where there otherwise was none. The overshoot is also exacerbated by the fact that we multiply this derivative by the K_d term.

Question 3: Plot on the same figure the following step response signals:

1. Derivative of error signal from the Step 12 model
2. Derivative of error signal from the modified model
3. Speed from the Step 12 model

4. Speed from the modified model

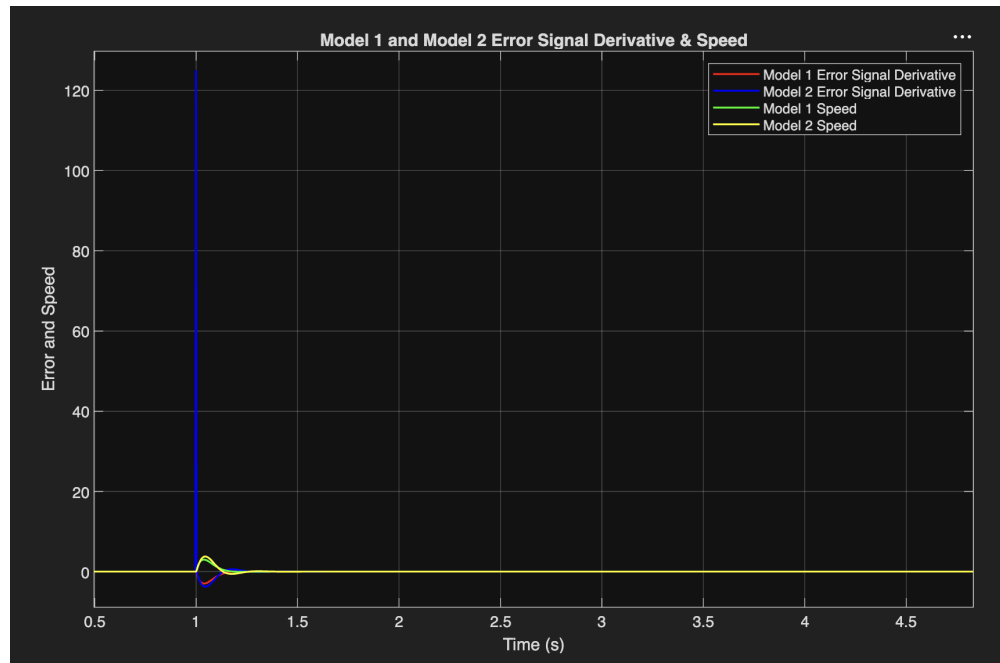


Figure 21: Model 1 and Model 2 Error Signal Derivative & Speed

Question 4: Explain how the feed-forward and feedback derivative gain influence system behavior as observed in the input voltage plot.

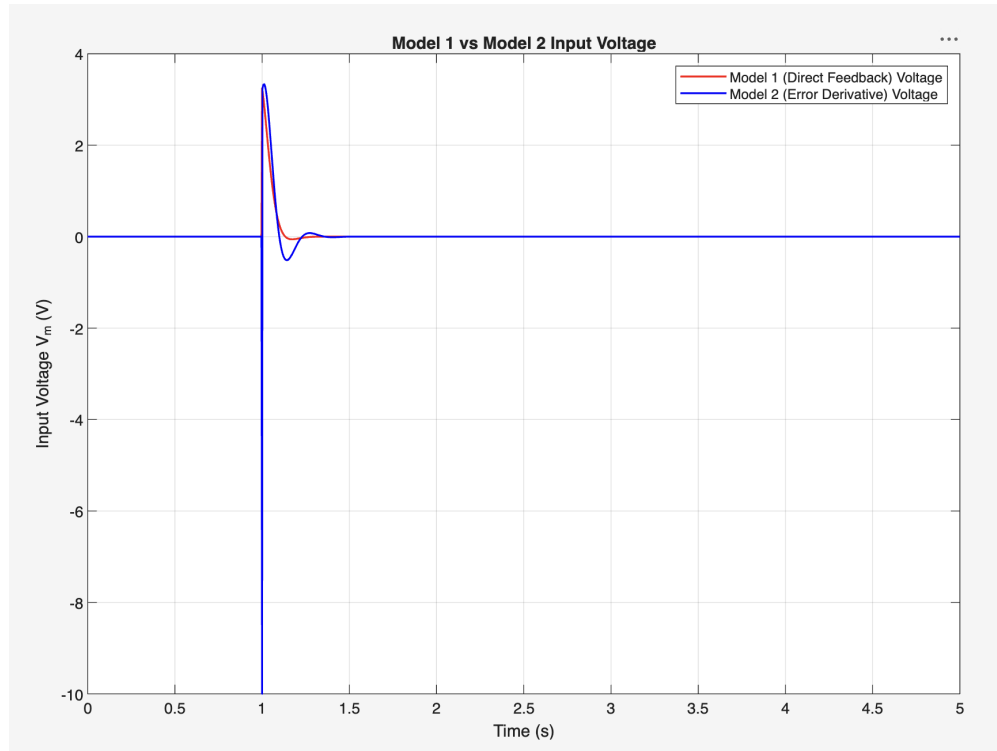


Figure 22: Model 1 and Model 2 Voltage Signal

As one can see from the above plot, Model 1 produces a smooth voltage response curve, whereas Model 2 produces a large spike, similar to the error signal derivative plot in Figure 21. This is because K_d acts on the derivative of the error – unlike the feed-forward system – which produces a large derivative spike because the error encodes the reference command θ_c . Because the feed-forward system does not rely on this reference command, it only reacts to the plant's smooth, physical motion.

5.9 Rebecca Dworkin

5.9.1 Step 5

The proportional gain used was

$$K_p = 10.7. \quad (44)$$

The resulting phase margin was approximately 66.2° , which was below the required 70° . Therefore, proportional control alone did not meet all of the design requirements.

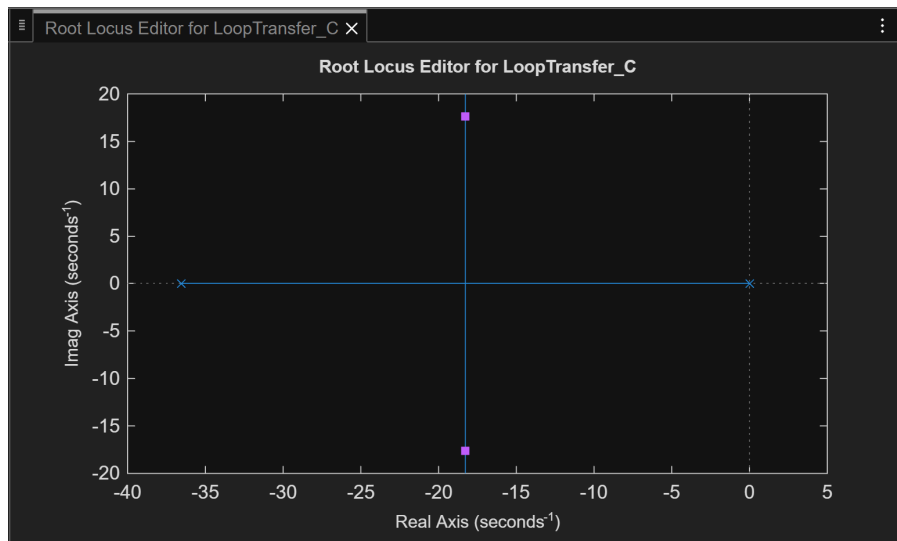


Figure 23: Step 5 root locus.

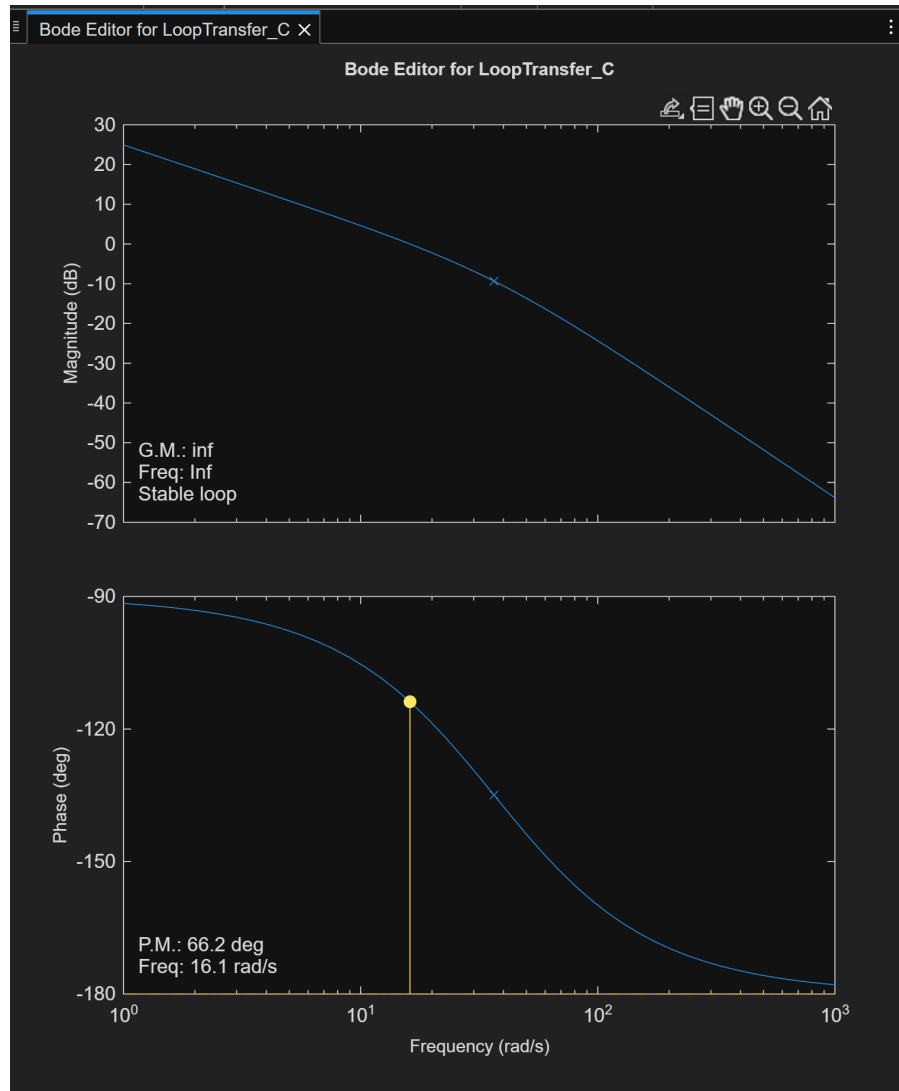


Figure 24: Step 5 open-loop Bode plot.

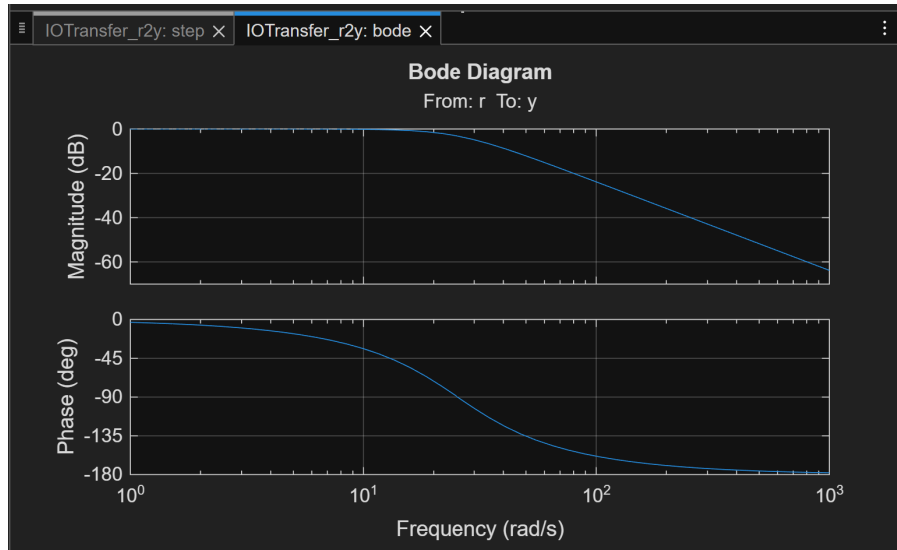


Figure 25: Step 5 closed-loop Bode plot.

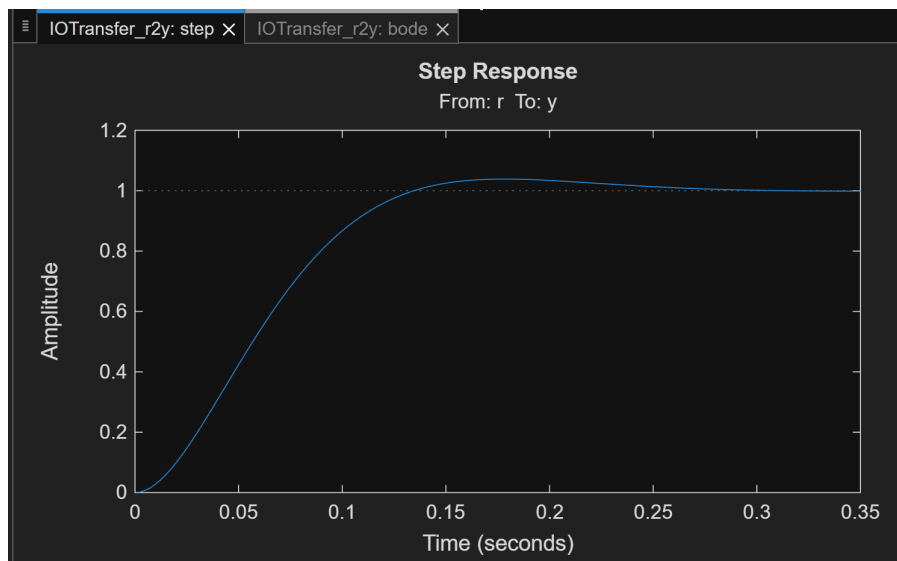


Figure 26: Step 5 step response.

5.9.2 Step 6

The below term was added to form the PD controller

$$C_{PD}(s) = K_p + K_d s. \quad (45)$$

The gains used were

$$K_p = 14.4, \quad K_d = 0.24. \quad (46)$$

The PD controller increased the phase margin to about 79.2° and gave a closed-loop bandwidth of about 26.3 rad/s, satisfying the design requirements.

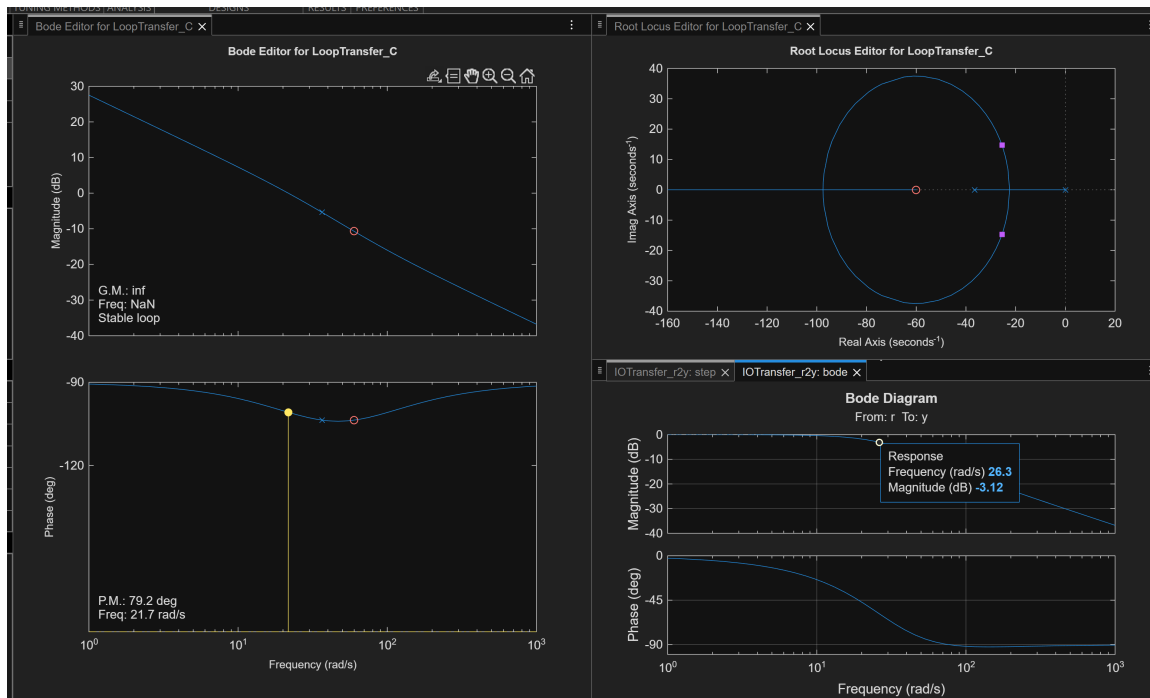


Figure 27: Step 6 controller design plots.

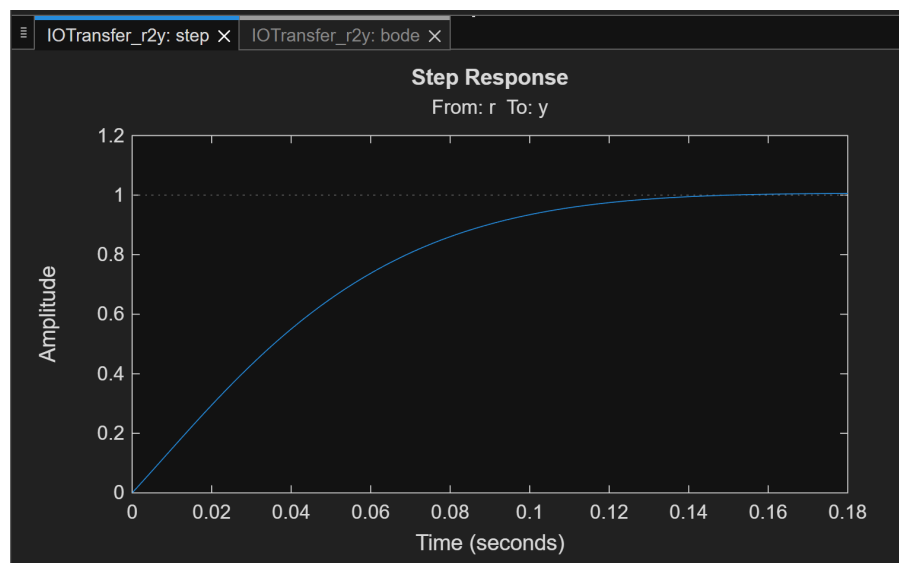


Figure 28: Step 6 step response.

5.9.3 Step 9

Integral control was added to form a PID controller. The integral gain was set to 1% of the proportional gain,

$$K_i = 0.144. \quad (47)$$

The final PID gains were

$$K_p = 14.4, \quad K_d = 0.24, \quad K_i = 0.144. \quad (48)$$

The resulting controller was

$$C_{PID}(s) = 14.4 + 0.24s + \frac{0.144}{s}. \quad (49)$$

The PID controller maintained a phase margin of about 79.2° and a closed-loop bandwidth of about 26.2 rad/s.

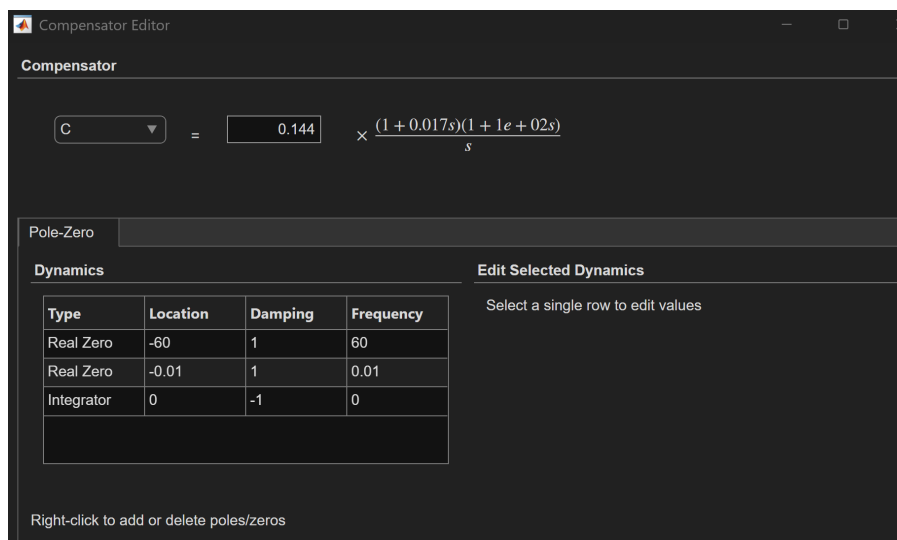


Figure 29: PID compensator.

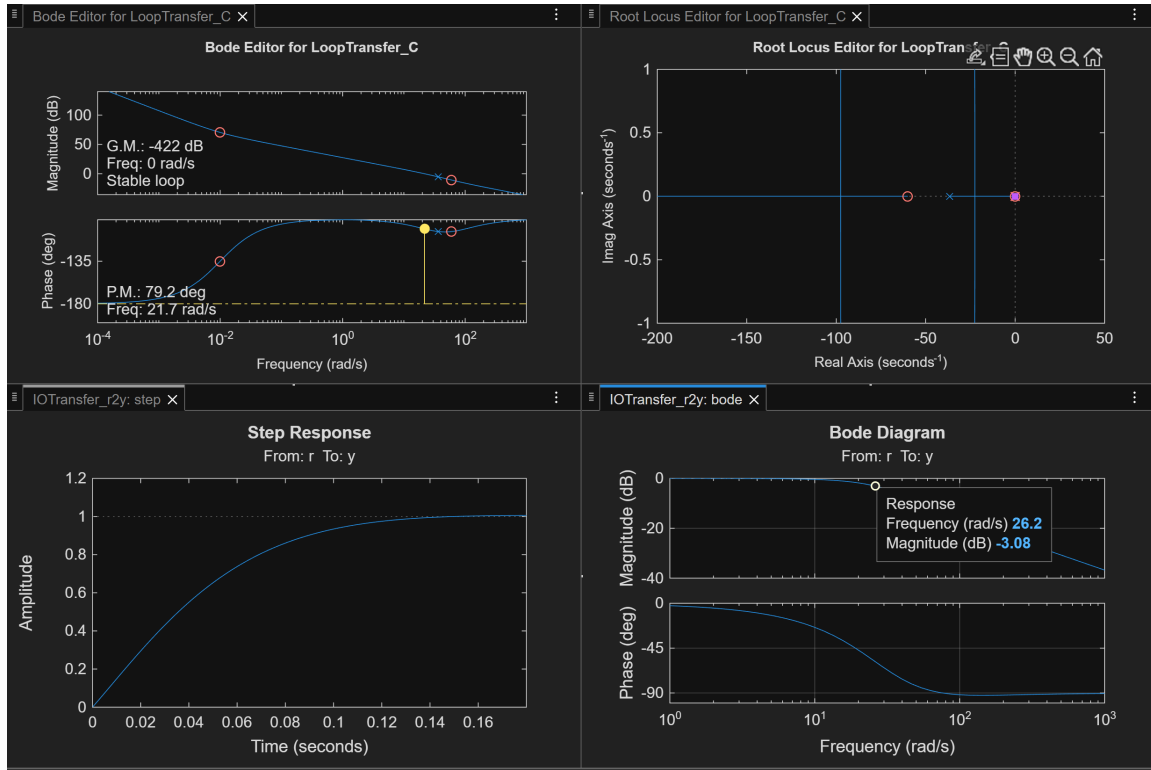


Figure 30: PID controller design plots.

5.9.4 Steps 10–13

Using the final PID controller, the closed-loop transfer function was

$$T(s) = \frac{0.3958s^2 + 23.75s + 0.2375}{0.0274s^3 + 1.396s^2 + 23.75s + 0.2375}. \quad (50)$$

The closed-loop poles were

$$p_1 = -0.0100, \quad p_{2,3} = -25.4651 \pm 14.7529j. \quad (51)$$

The complex pole pair had a natural frequency and damping ratio of

$$\omega_n = 29.4299 \text{ rad/s}, \quad \zeta = 0.8653. \quad (52)$$

The controller was then implemented in Simulink with proportional, integral, and derivative paths and a ± 10 V saturation limit.

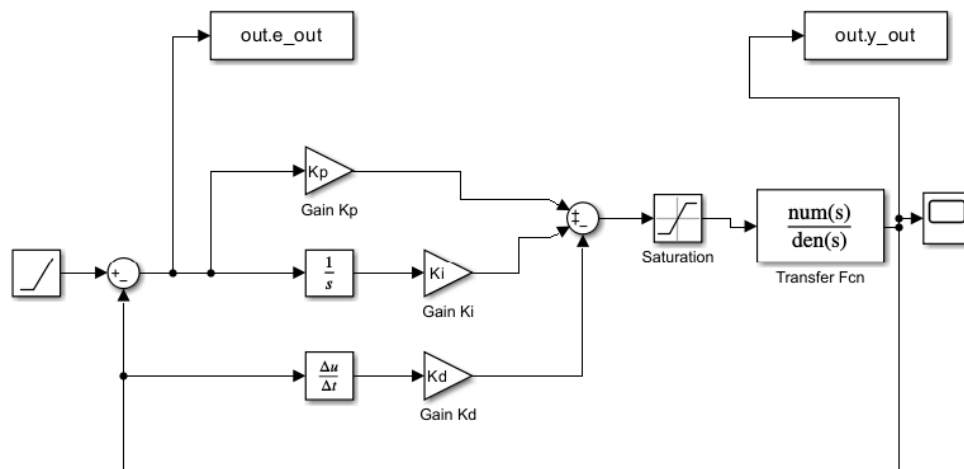


Figure 31: Simulink implementation of the controller.

The PD and PID controllers were compared using a ramp input. The overall position responses were similar, but the difference was a little clearer when comparing the tracking error.

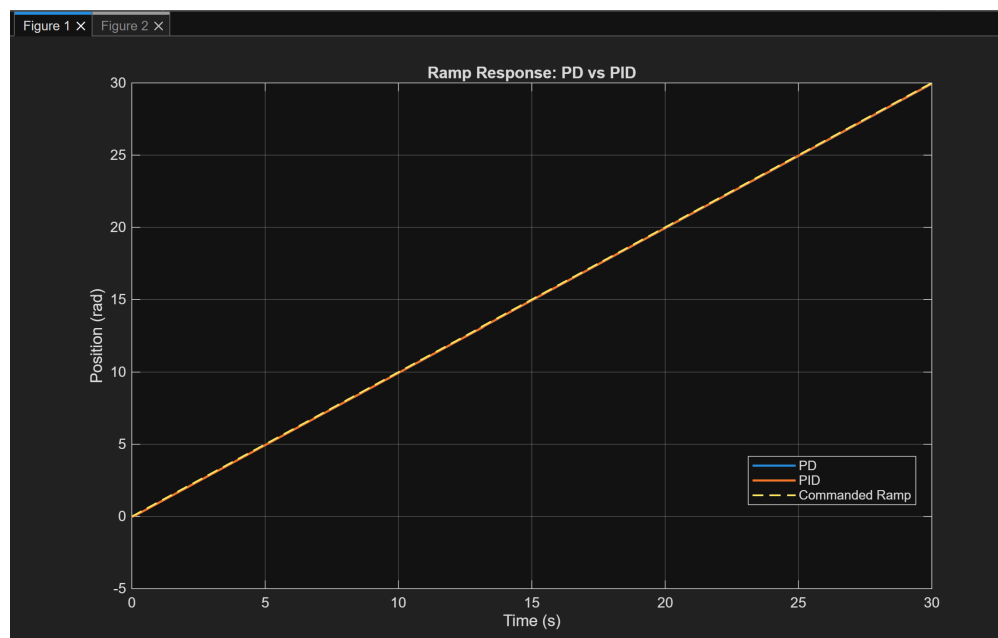


Figure 32: PD and PID ramp-response comparison.

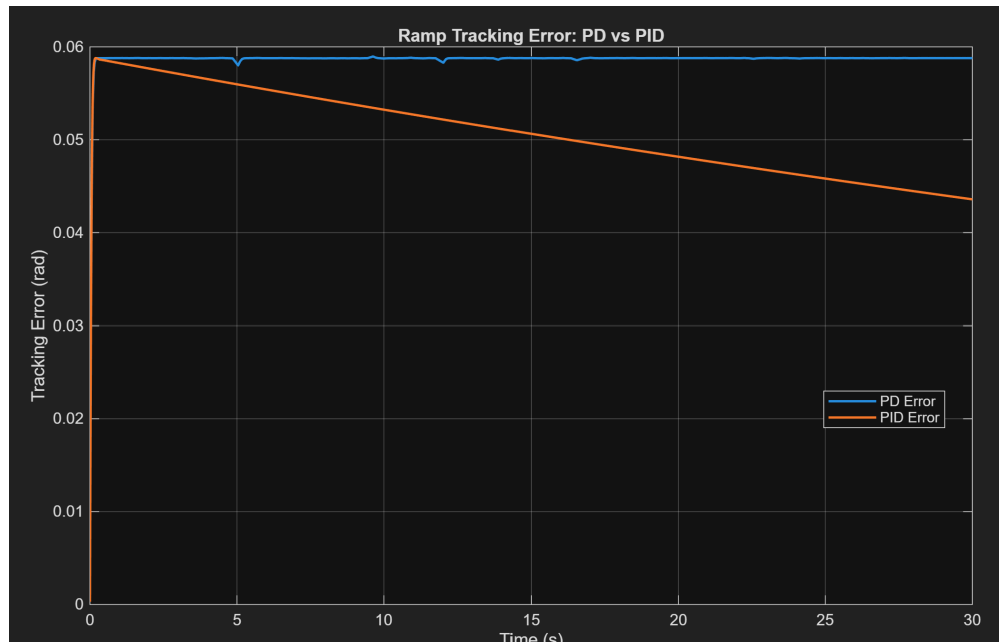


Figure 33: PD and PID ramp tracking-error comparison.

The PD controller maintained a nonzero ramp tracking error of about 0.0588 rad. Adding integral control caused the PID tracking error to decrease toward zero over time.

5.9.5 Additional Questions

1. Difference Between the Two Control Structures

Figure 2 from the Control Design lab manual applies K_d to the measured motor speed and feeds it back negatively. In the Step 6 control structure, K_d is applied to the derivative of the position error instead.

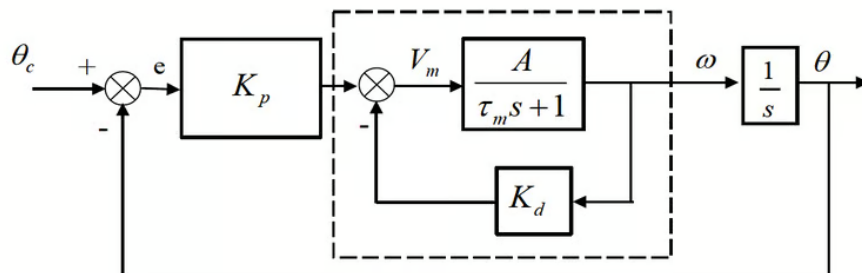


Figure 2: Closed-loop block diagram for the position control of a DC motor using a proportional-plus-derivative (PD) controller.

Figure 34: Closed-loop PD control structure from Figure 2 of the Control Design lab manual.

The Step 12 model was modified to match the Step 6 control structure by applying K_d to the derivative of the error. The Step 6 gains were used,

$$K_p = 14.4, \quad K_d = 0.24, \quad K_i = 0. \quad (53)$$

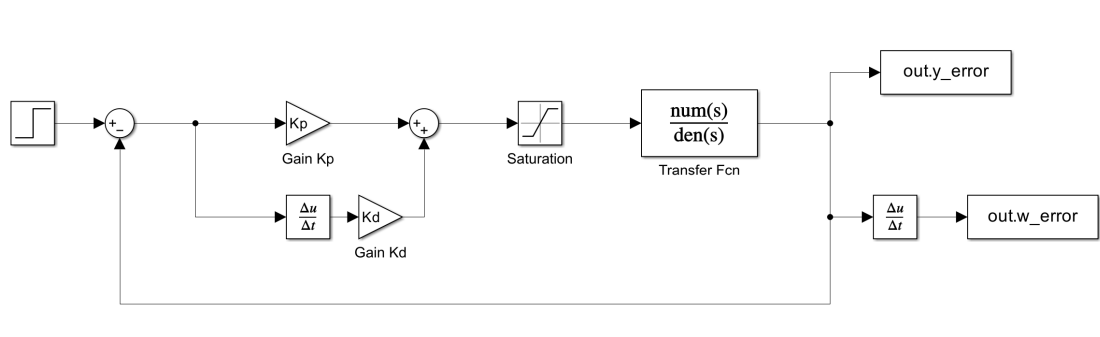


Figure 35: Modified Simulink model with K_d applied to the derivative of the error.

2. Why the Difference is Acceptable for a Step Input

Both models were run for 5 seconds using the same quarter-step input. The two control structures produced nearly identical position responses and settled to the same commanded position.

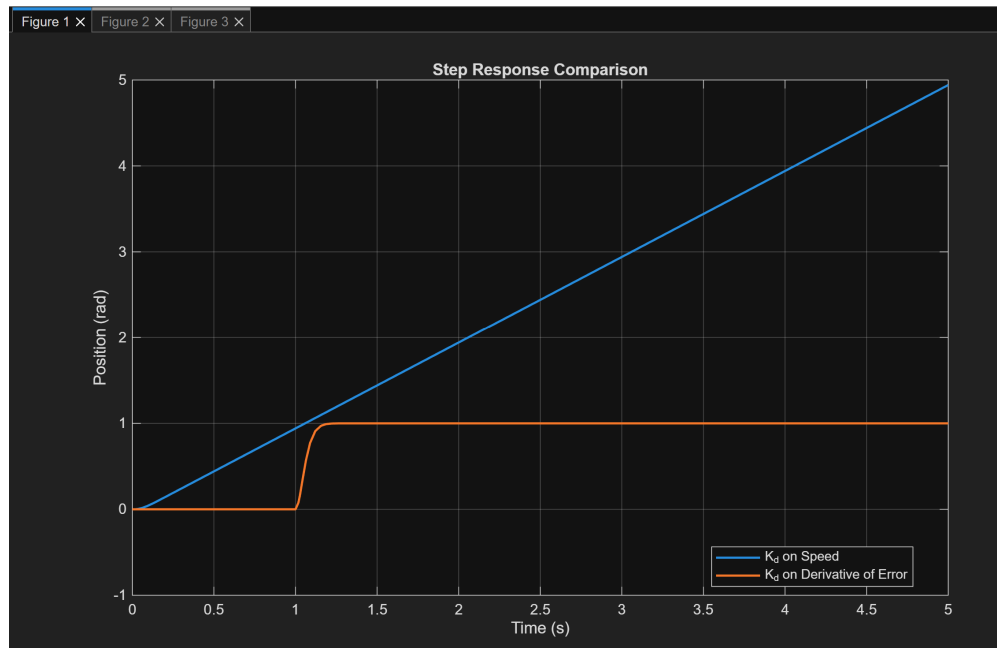


Figure 36: Step-response comparison of the two PD control structures.

After the step occurs, the commanded position is constant, so its derivative is zero. Therefore, the derivative of the error becomes the negative of the motor speed,

$$\dot{e} = -\omega. \quad (54)$$

This causes the two derivative terms to behave similarly after the initial change in the command.

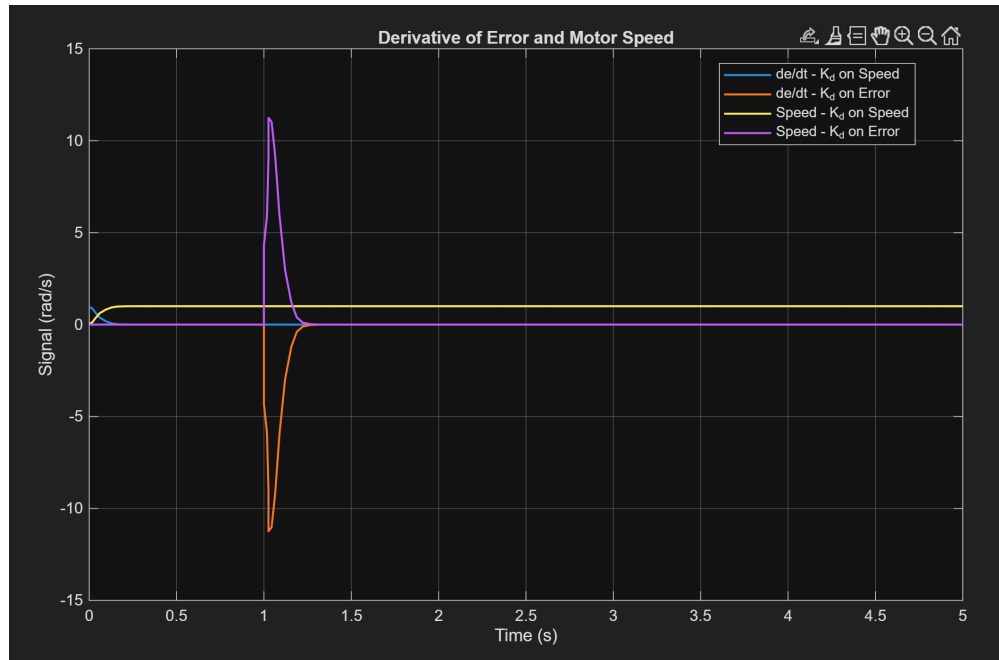


Figure 37: Comparison of the derivative-of-error and motor-speed signals for both control structures.

The largest difference occurs when the step is first applied. The derivative-of-error structure reacts directly to the sudden change in the command, which creates a large initial spike. The speed-feedback structure only responds to the actual motor motion, so the response is smoother.

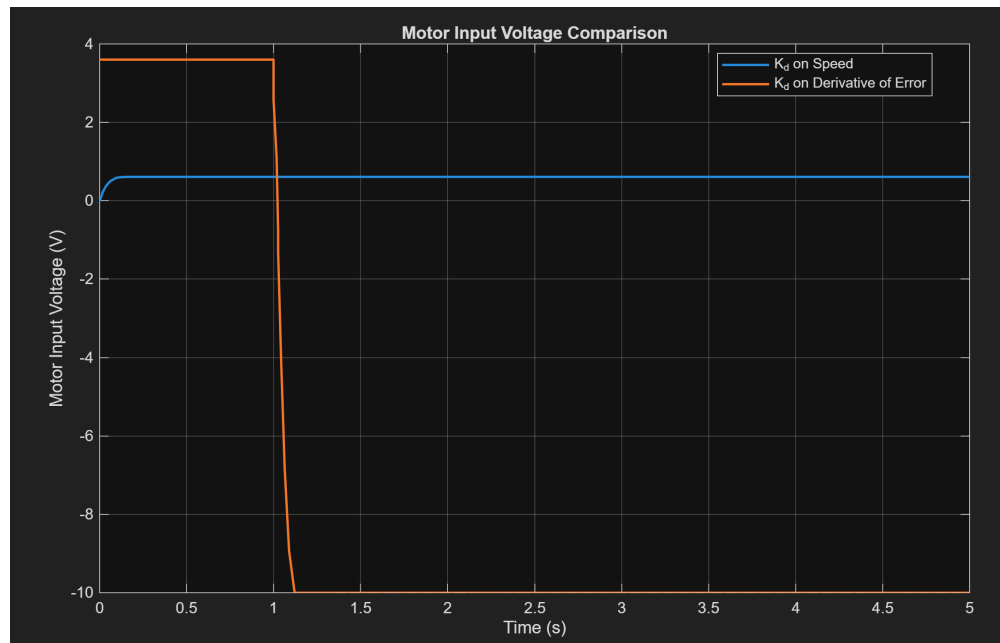


Figure 38: Motor input-voltage comparison for the two derivative control structures.

The voltage comparison shows the same effect. Applying K_d to the derivative of the error causes a larger initial voltage change, while applying K_d to the motor speed gives a smoother voltage response.

5.10 Vivin Sudharsan

5.10.1 Step 5

The proportional controller was first adjusted to obtain a closed-loop bandwidth greater than the required 25 rad/s. The resulting closed-loop bandwidth was approximately 25.1 rad/s, while the phase margin was approximately 66.2°.

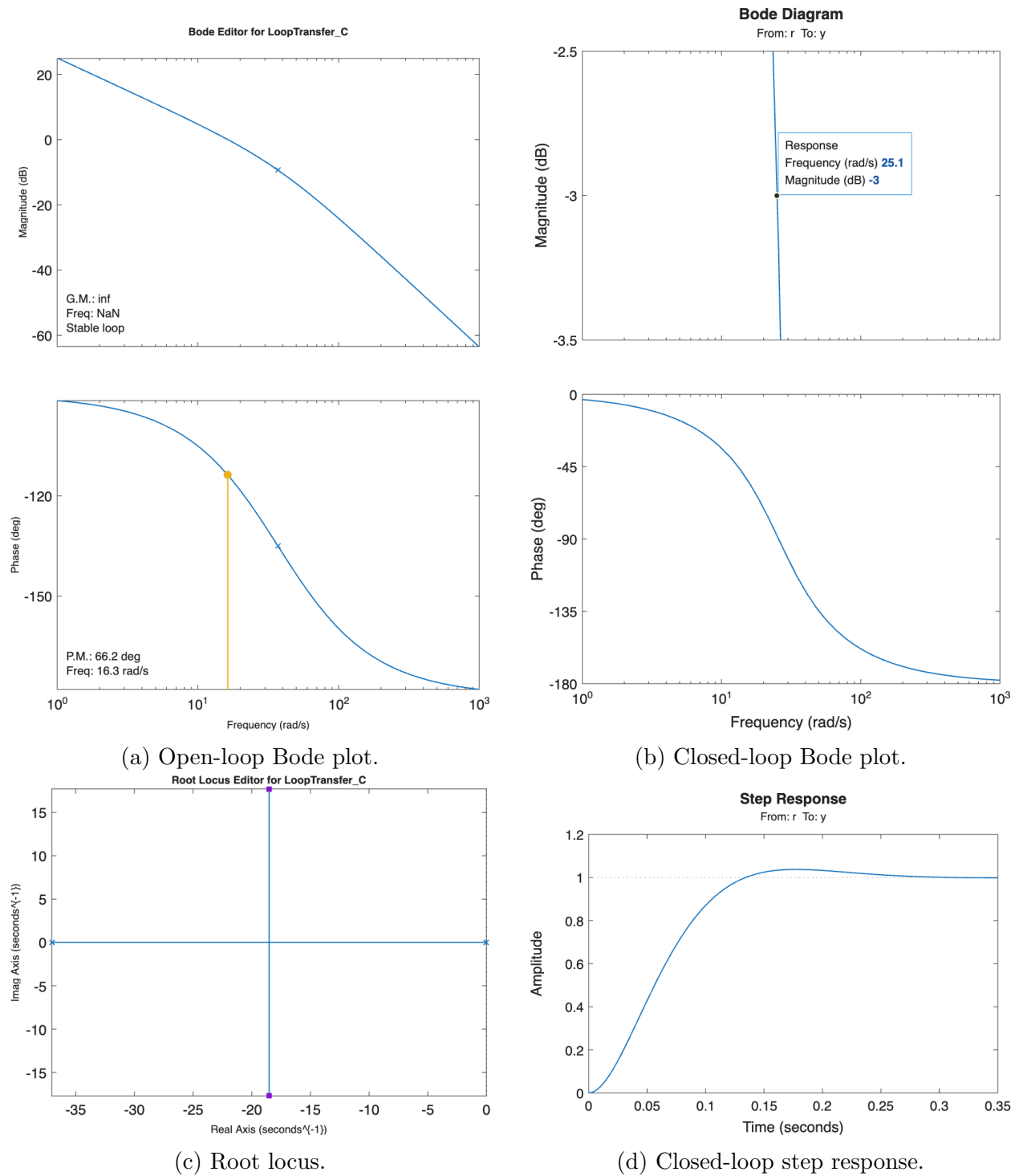


Figure 39: Vivin Sudharsan – Step 5 proportional-controller design results.

The proportional gain used for Step 5 was

$$K_p = \boxed{10.8}. \quad (55)$$

5.10.2 Step 6

A derivative term was then added to form a PD controller. The controller zero was adjusted until both the bandwidth and phase-margin requirements were satisfied. The resulting controller had a phase margin of approximately 70° and a closed-loop bandwidth of approximately 25.3 rad/s.

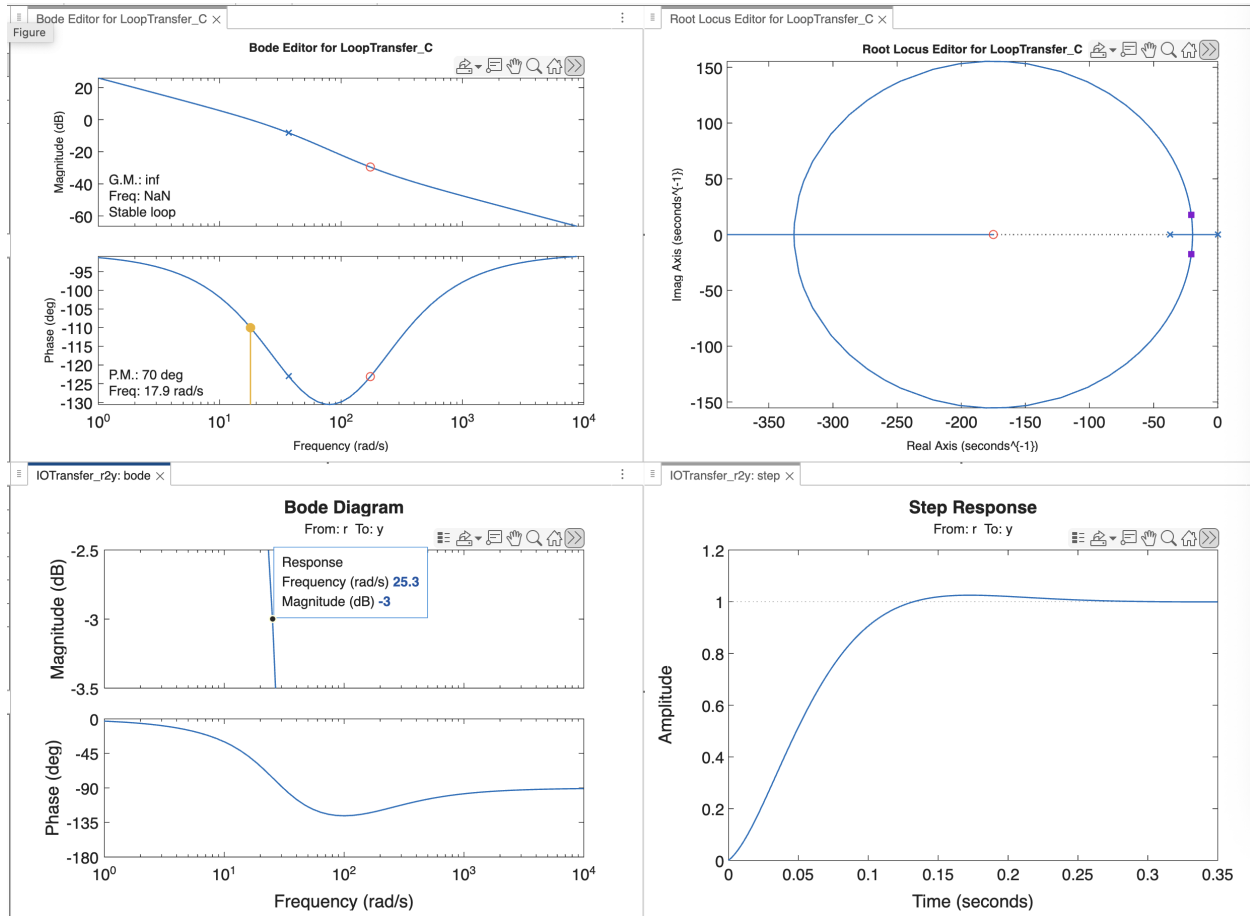


Figure 40: Vivin Sudharsan – Step 6 PD controller design showing the open-loop Bode plot, closed-loop Bode plot, root locus, and step response.

The resulting PD gains were

$$K_p = \boxed{12}, \quad K_d = \boxed{0.0684}. \quad (56)$$

The corresponding PD controller was

$$C_{PD}(s) = K_p + K_d s = 12 + 0.0684s. \quad (57)$$

5.10.3 Step 9

Integral control was added with an initial integral gain equal to one percent of the proportional gain,

$$K_i = 0.01K_p = 0.12. \quad (58)$$

The gains did not require further readjustment in Step 8. Therefore, the final PID gains were

$$K_p = 12, \quad K_d = 0.0684, \quad K_i = 0.12. \quad (59)$$

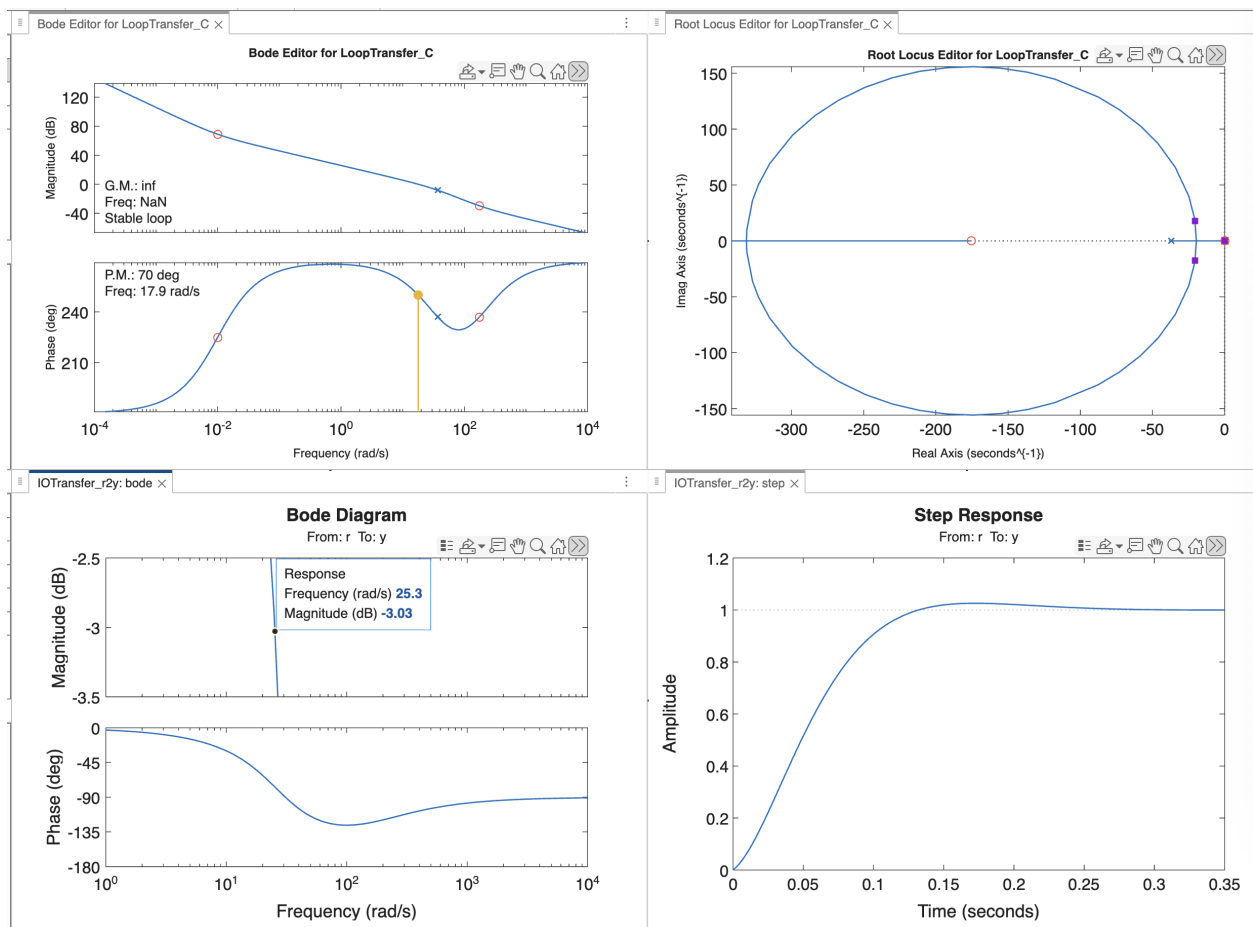


Figure 41: Vivin Sudharsan – Step 9 PID controller design showing the open-loop Bode plot, closed-loop Bode plot, root locus, and step response.

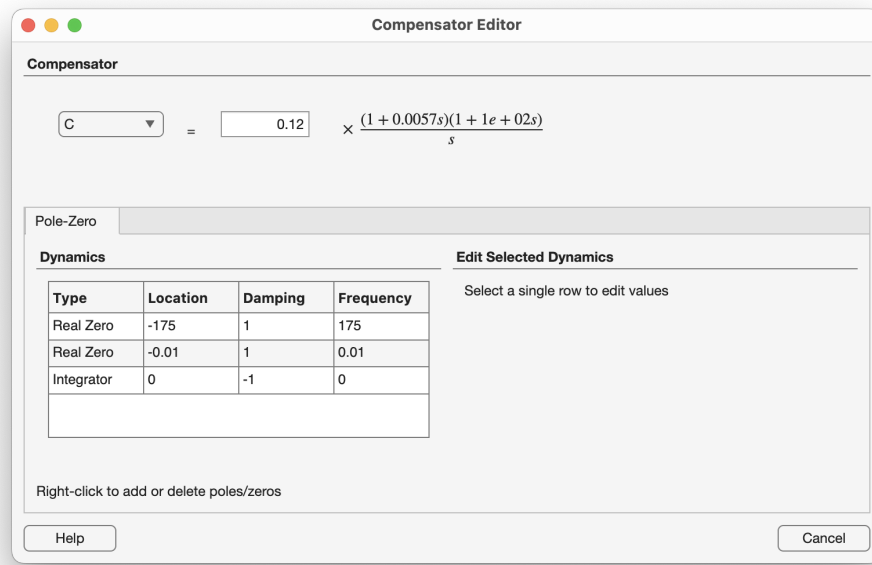


Figure 42: Vivin Sudharsan – Final PID compensator used in Step 9.

5.10.4 Step 10

The closed-loop transfer function was exported from MATLAB Control System Designer. The resulting transfer function was

$$T(s) = \frac{4.188s^2 + 732.9s + 7.329}{s^3 + 41.22s^2 + 732.9s + 7.329}. \quad (60)$$

5.10.5 Step 11

The poles, damping ratios, and natural frequencies of the closed-loop system were obtained using the MATLAB `damp()` function.

Table 4: Vivin Sudharsan – Closed-loop poles, damping ratios, and natural frequencies.

Pole	Damping Ratio, ζ	Natural Frequency, ω_n (rad/s)
-0.0100	1.000	0.0100
$-20.6 + 17.5i$	0.761	27.1
$-20.6 - 17.5i$	0.761	27.1

The complex-conjugate pole pair therefore had

$$\omega_n = \boxed{27.1 \text{ rad/s}}, \quad \zeta = \boxed{0.761}. \quad (61)$$

Texport =

From input "r" to output "y":
 $4.188 s^2 + 732.9 s + 7.329$

 $s^3 + 41.22 s^2 + 732.9 s + 7.329$

Continuous-time transfer function.

[Model Properties](#)

Pole	Damping	Frequency (rad/seconds)	Time Constant (seconds)
-1.00e-02	1.00e+00	1.00e-02	9.99e+01
-2.06e+01 + 1.75e+01i	7.61e-01	2.71e+01	4.85e-02
-2.06e+01 - 1.75e+01i	7.61e-01	2.71e+01	4.85e-02

>>

Figure 43: Vivin Sudharsan – MATLAB output for the closed-loop transfer function, poles, damping ratios, and natural frequencies.

5.10.6 Step 12

The PID controller was implemented in Simulink using individual proportional, integral, and derivative paths. The motor-voltage command was limited using a ± 10 V saturation block, and the validated motor model used $K = 1.649$ and $\tau = 0.0274$ s. The derivative feedback term was applied to the measured angular velocity.

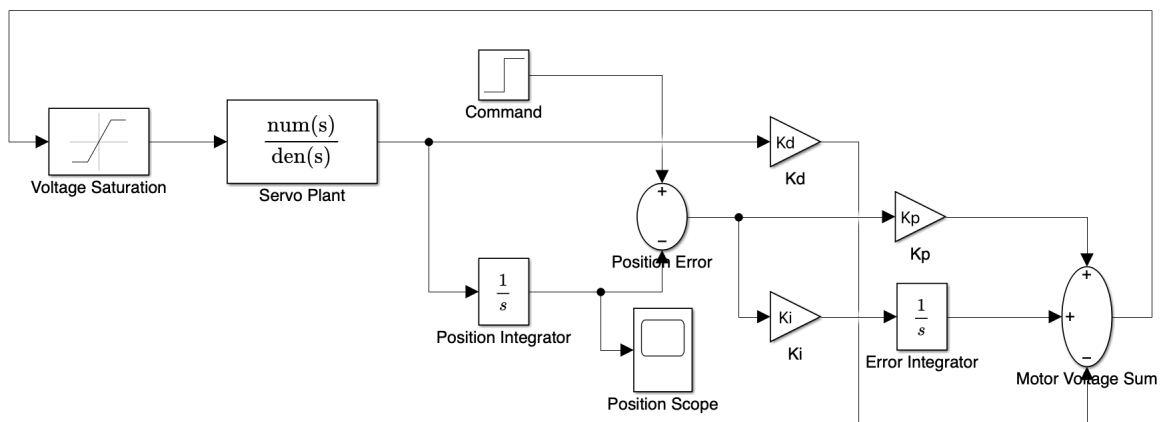


Figure 44: Vivin Sudharsan – Step 12 PID controller Simulink model.

5.10.7 Step 13

The Step 12 Simulink model was modified to use a unit ramp input and was simulated for 30 seconds. Two controller configurations were tested. The PD controller used

$$K_p = 12, \quad K_d = 0.0684, \quad K_i = 0, \quad (62)$$

while the PID controller used

$$K_p = 12, \quad K_d = 0.0684, \quad K_i = 0.12. \quad (63)$$

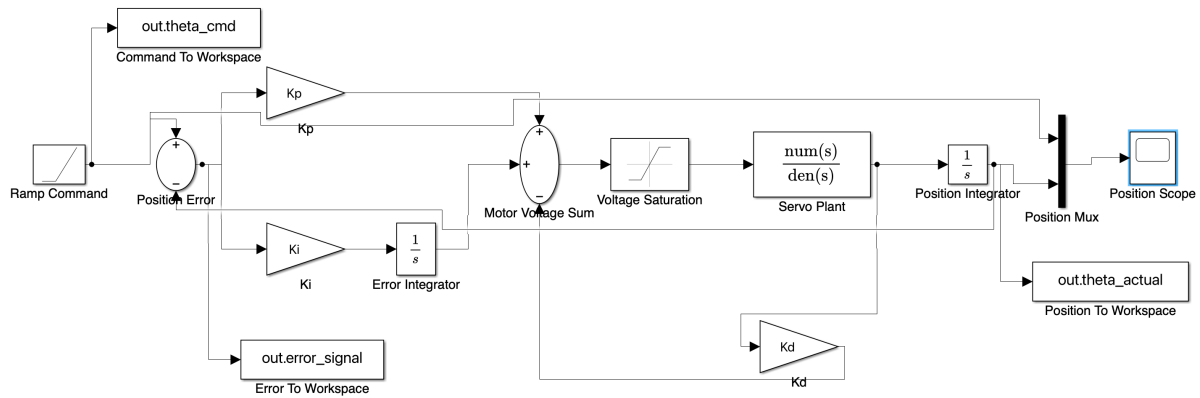


Figure 45: Vivin Sudharsan – Simulink model used for the Step 13 ramp response comparison.

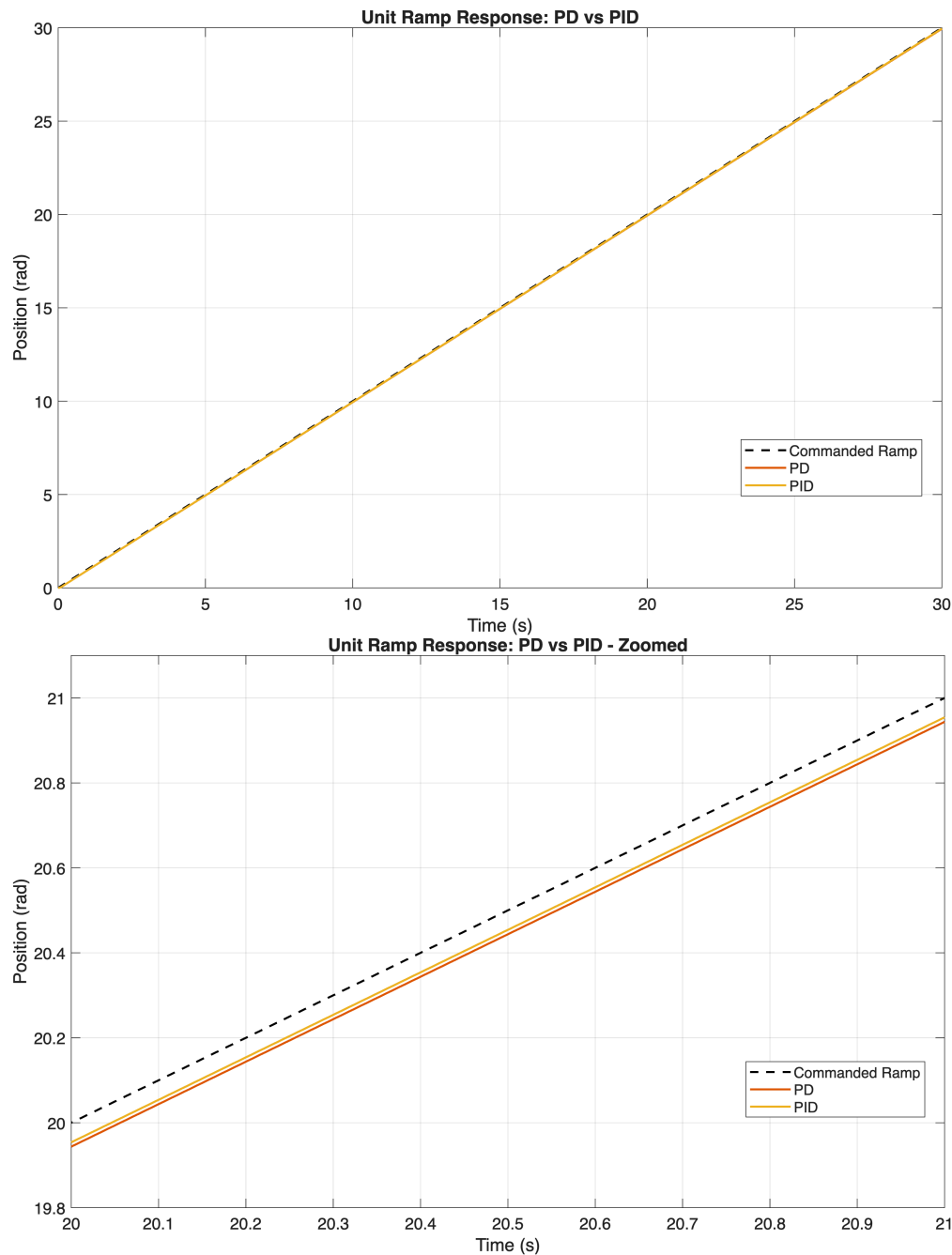


Figure 46: Vivin Sudharsan – PD and PID unit-ramp responses. The lower plot provides a zoomed view of the tracking difference.

The PD and PID controllers both track the unit ramp closely when viewed over the full 30-second interval. However, the zoomed response shows that the PID controller remains slightly closer to the commanded position than the PD controller. The PD response maintains an approximately constant offset from the commanded ramp, while the PID response gradually reduces this offset due to the addition of integral action.

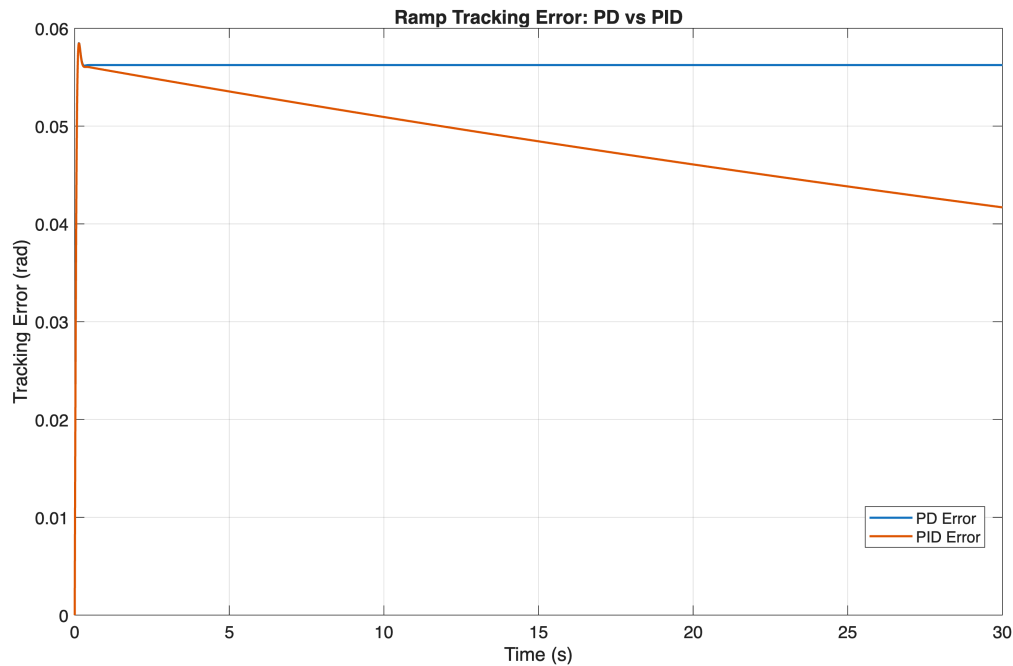


Figure 47: Vivin Sudharsan – PD and PID ramp tracking-error comparison.

The tracking-error plot shows the effect of the integral term more clearly. After the initial transient, the PD controller maintains a nearly constant tracking error of approximately 0.056 rad. In contrast, the PID controller begins at a similar error but gradually decreases to approximately 0.042 rad by the end of the 30-second simulation. Therefore, the integral term reduces the steady ramp-tracking error, although the relatively small value of K_i causes the error to decrease slowly over the simulated interval.

5.10.8 Additional Questions

a. Difference Between Fig. 2 and the Step 6 Control Structure The primary difference between Fig. 2 from the lab manual and the control structure used in Step 6 is the location of the derivative gain and the signal on which the derivative action operates.

In the Fig. 2 architecture used for the Step 12 Simulink model, the derivative gain acts on the measured motor angular velocity, ω , in a feedback path. The controller voltage before saturation is therefore

$$u = K_p e - K_d \omega, \quad (64)$$

where

$$e = \theta_c - \theta. \quad (65)$$

In the Step 6 control structure, the derivative term is instead included in the forward controller and acts on the derivative of the position error. The corresponding PD controller

is

$$u = K_p e + K_d \dot{e}. \quad (66)$$

Therefore, the primary difference is that Fig. 2 uses motor-speed feedback for the derivative term, while the Step 6 architecture applies the derivative gain directly to the derivative of the error.

b. Modified Simulink Model A modified Simulink model was created to reproduce the Step 6 control architecture. The position error is divided into proportional and derivative branches. The derivative of error was calculated using the sampling time of 0.002 s,

$$\dot{e}[k] \approx \frac{e[k] - e[k-1]}{0.002}. \quad (67)$$

The proportional and derivative contributions are then added before the ± 10 V saturation block and motor plant.

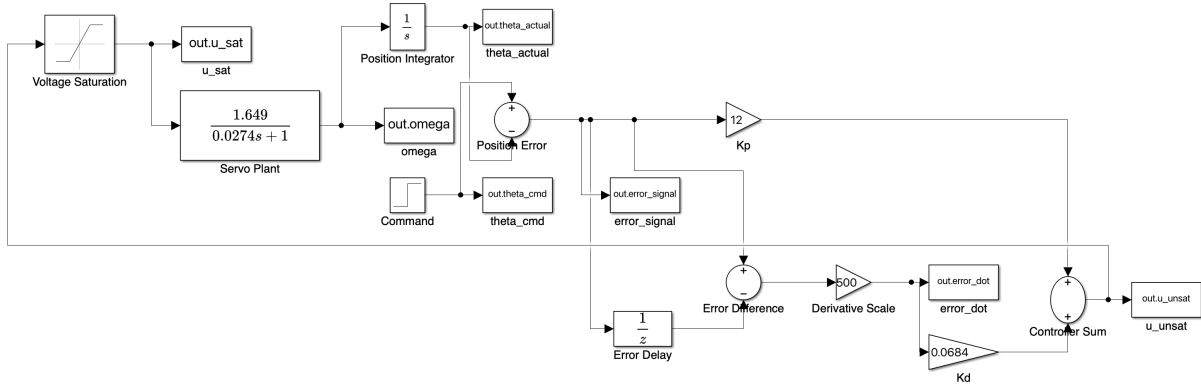


Figure 48: Vivin Sudharsan – Modified Simulink model matching the Step 6 derivative-of-error control architecture.

c. Step Response Comparison The Step 12 and modified models were simulated for five seconds using the required quarter-step input of 0.25 rad. The step was applied at $t = 0.10$ s to allow the response at the command transition to be observed. Both models used the Step 6 PD gains,

$$K_p = 12, \quad K_d = 0.0684, \quad K_i = 0. \quad (68)$$

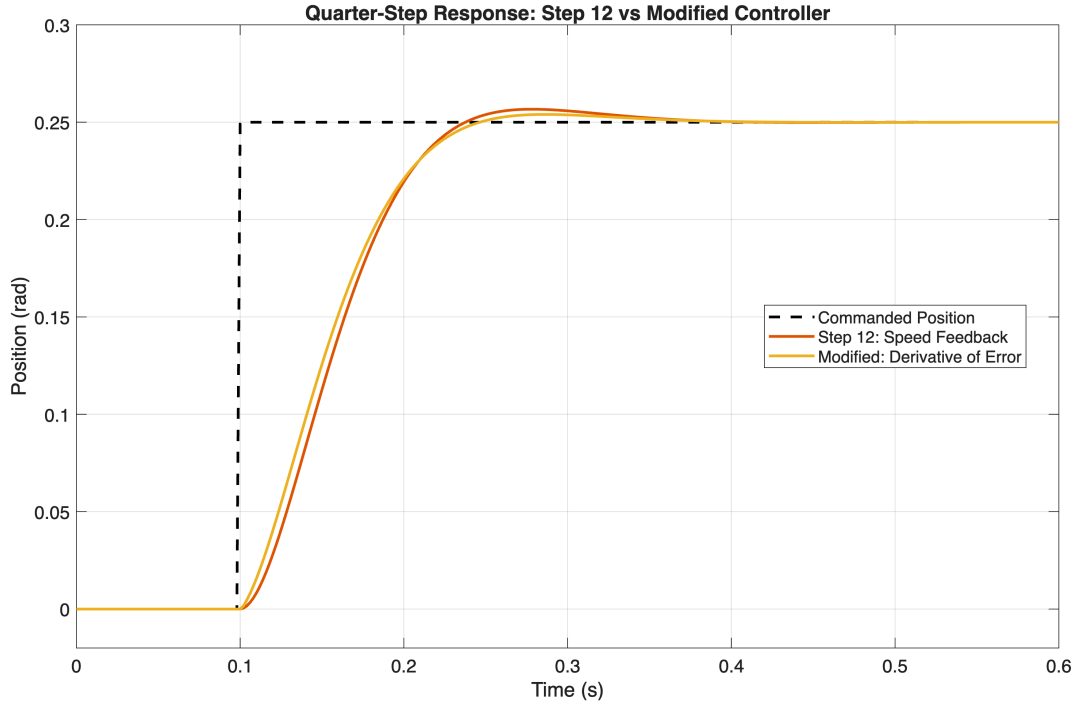


Figure 49: Vivin Sudharsan – Quarter-step response comparison between the Step 12 speed-feedback controller and the modified derivative-of-error controller.

The two controller structures produce similar position responses and both settle to the commanded position of 0.25 rad. A small difference is visible during the initial transient because the derivative-of-error controller responds directly to the sudden change in the reference command.

After the step transition, the commanded position is constant, so

$$\dot{\theta}_c = 0. \quad (69)$$

Since

$$e = \theta_c - \theta, \quad (70)$$

the derivative of the error becomes

$$\dot{e} = \dot{\theta}_c - \dot{\theta} = -\omega. \quad (71)$$

Therefore, after the initial command transition,

$$K_d \dot{e} \approx -K_d \omega. \quad (72)$$

The derivative terms in the two architectures consequently become equivalent after the reference becomes constant. This explains why the two step responses are very similar after

the initial transient and why the difference between the control structures is acceptable for a step input.

d. Derivative-of-Error and Motor-Speed Comparison The derivative of the position error and motor angular velocity were recorded from both controller models during the quarter-step response.

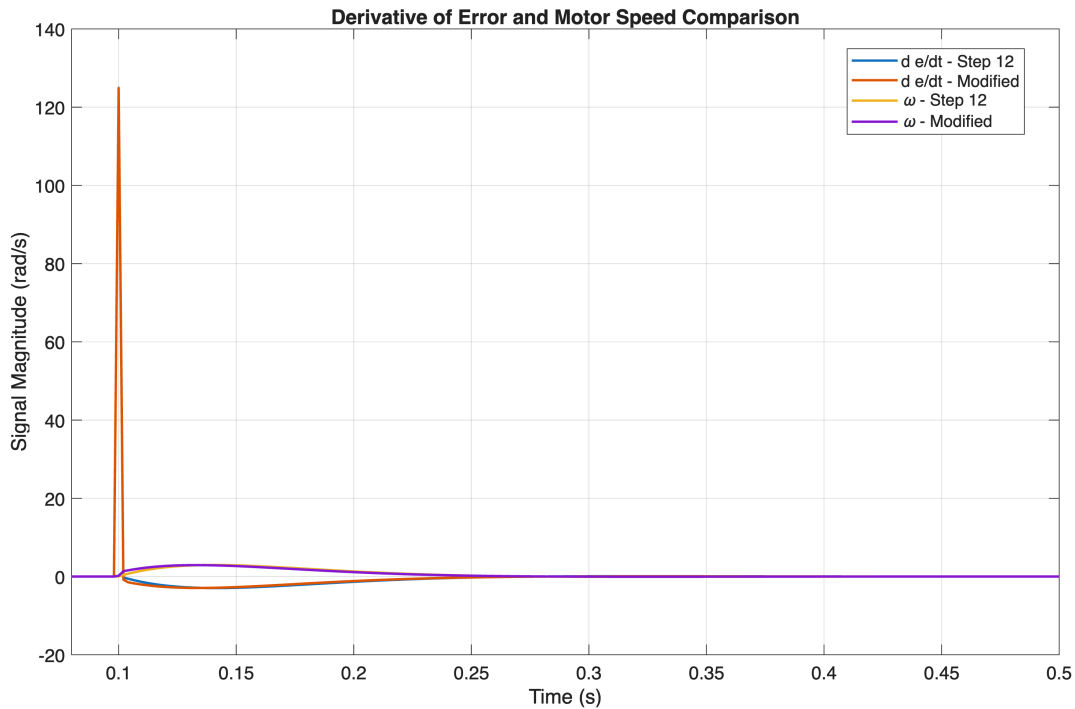


Figure 50: Vivin Sudharsan – Derivative-of-error and motor-speed signals for the Step 12 and modified controller structures.

At the instant of the step, the derivative-of-error signals contain a large positive spike. With a 0.25 rad change occurring over the 0.002s sampling interval, the discrete derivative is approximately

$$\dot{e} \approx \frac{0.25}{0.002} = 125 \text{ rad/s}, \quad (73)$$

which agrees with the initial peak shown in the figure. The motor speed does not contain an equivalent instantaneous spike because the physical plant cannot change angular velocity instantaneously.

To better observe the behavior following the initial command transition, the signals are shown again with the initial spike excluded.

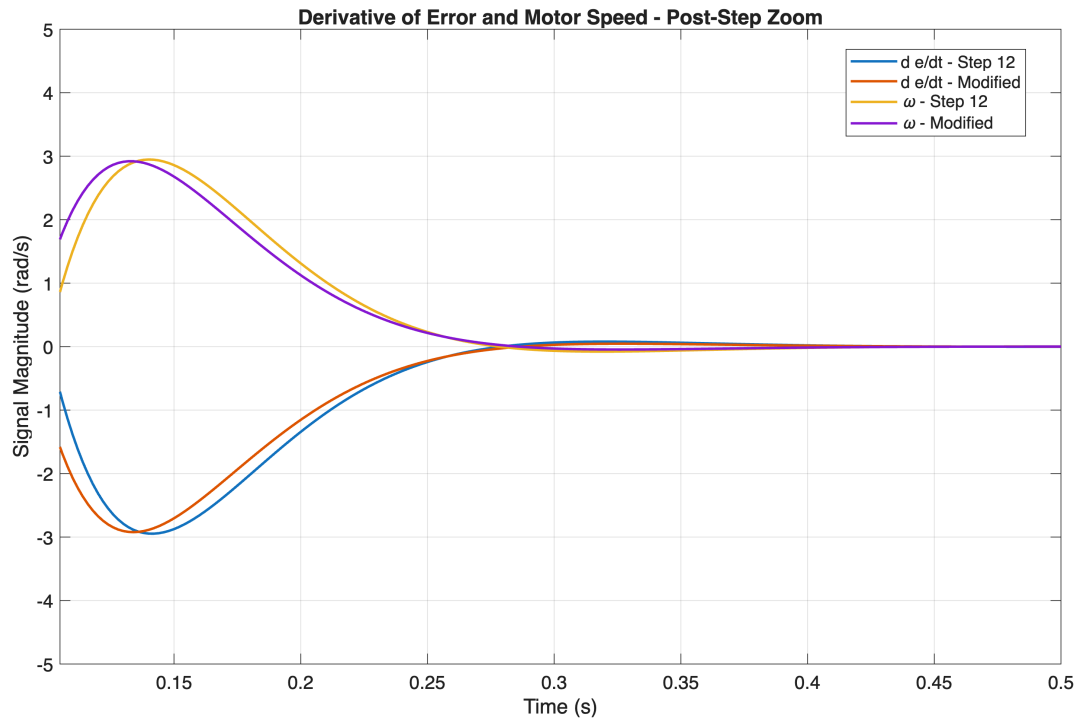


Figure 51: Vivin Sudharsan – Post-step comparison of the derivative-of-error and motor-speed signals.

After the initial step transition, the derivative-of-error signals have approximately the opposite sign of the corresponding motor-speed signals. This agrees with the relationship

$$\dot{e} = -\omega \quad (74)$$

for a constant commanded position. The remaining small differences during the transient are caused by the different controller architectures and the voltage saturation applied to the motor command.

e. Effect of Forward-Path and Feedback Derivative Gain The effect of the two derivative implementations can be seen most clearly in the controller voltage immediately after the quarter-step command.

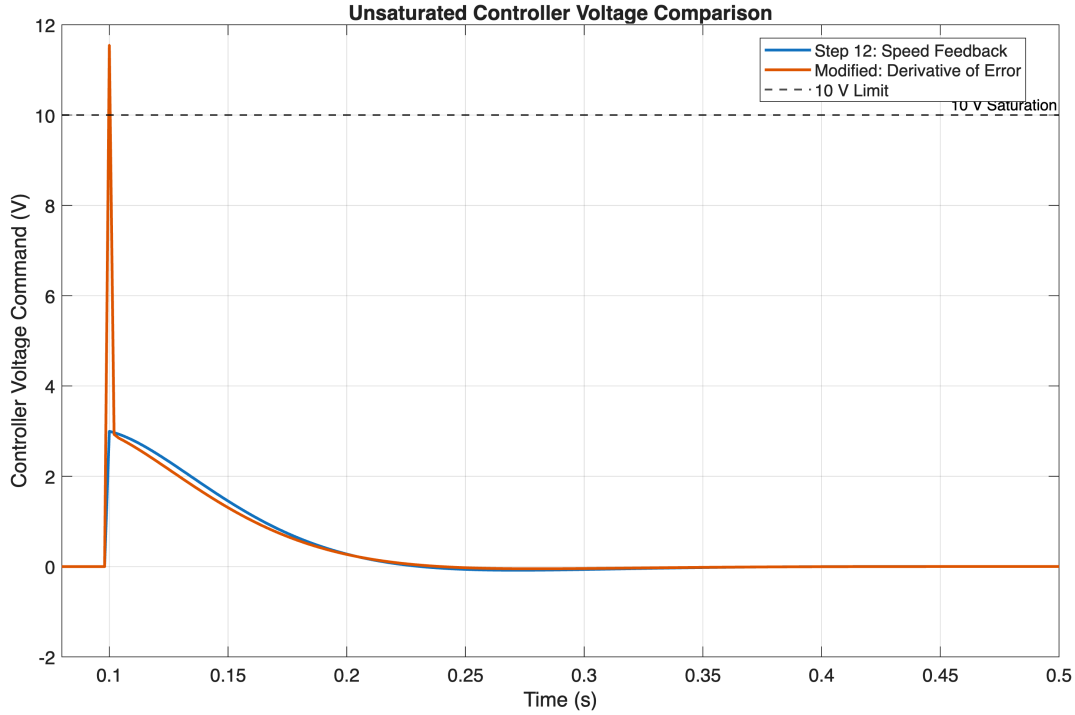


Figure 52: Vivin Sudharsan – Unsaturated controller-voltage commands for the speed-feedback and derivative-of-error controller architectures.

For the Step 12 speed-feedback controller, the motor is initially stationary, so $\omega = 0$ immediately when the command changes. The initial controller command is therefore primarily the proportional term,

$$u \approx K_p e = 12(0.25) = 3.0 \text{ V}. \quad (75)$$

In the modified controller, the derivative gain acts directly on the error. The sudden 0.25 rad change in commanded position produces an initial derivative contribution of approximately

$$K_d \dot{e} = 0.0684(125) = 8.55 \text{ V}. \quad (76)$$

Together with the proportional contribution, the modified controller therefore requests approximately

$$u \approx 3.0 + 8.55 = 11.55 \text{ V}. \quad (77)$$

This large initial command is the derivative kick produced when derivative action is applied directly to the error signal. The Step 12 speed-feedback structure avoids this reference-induced derivative kick because the derivative term depends on the actual motor motion rather than directly on the changing reference.

The controller output is limited to $\pm 10 \text{ V}$ before being applied to the motor. The resulting applied motor voltages are shown below.

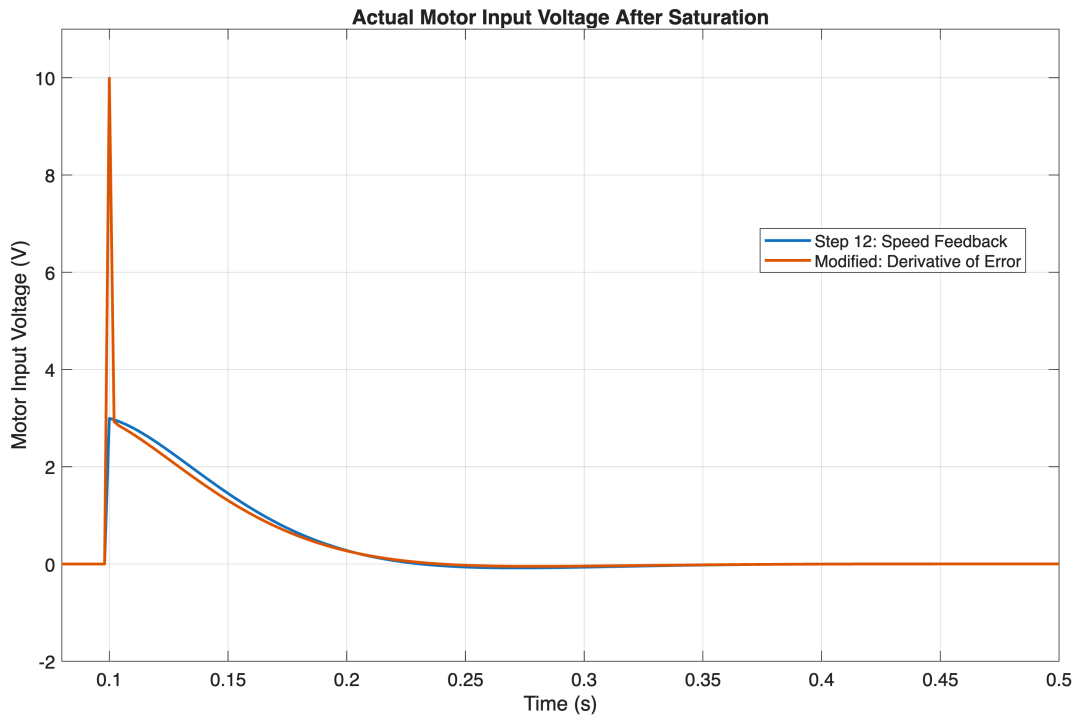


Figure 53: Vivin Sudharsan – Actual motor input voltages after the ± 10 V saturation block for the two derivative-control architectures.

The modified controller's approximately 11.55 V initial request is clipped to 10 V, while the speed-feedback controller initially applies approximately 3 V. After the initial transition, the two voltage responses become increasingly similar because $\dot{e} \approx -\omega$. The saturation therefore reduces the effect of the derivative kick on the resulting position response, which is consistent with the similar step responses observed in Fig. 49.

5.11 Owen Winn

5.11.1 Step 5

A proportional controller was first considered, with

$$C_P(s) = K_p = \boxed{10.75}. \quad (78)$$

The saved design plots show a closed-loop bandwidth of approximately 25 rad/s and a phase margin of 66.1°. Although the bandwidth met the target, the phase margin was below 70°. Proportional control alone therefore did not meet all of the design requirements. The stable position-control loop has zero steady-state error for a unit step.

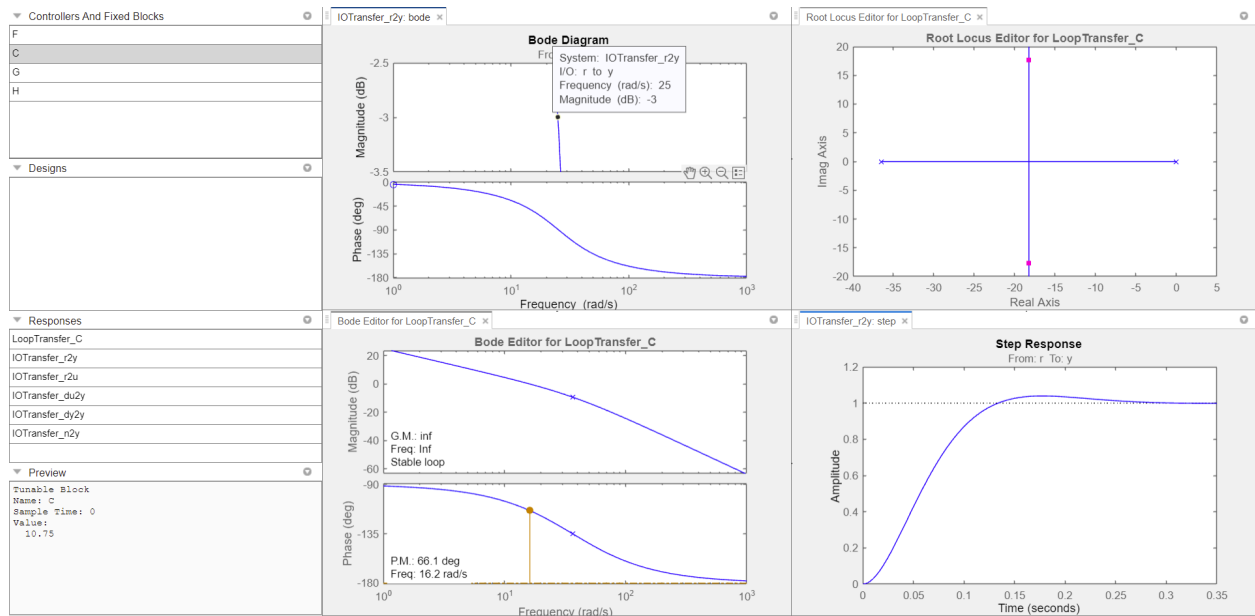


Figure 54: Owen Winn – Step 5 proportional-controller design with closed-loop Bode plot, root locus, open-loop Bode plot, and unit-step response.

5.11.2 Step 6

The saved compensator was approximately

$$C_{PD}(s) = 0.072727(s + 165) \approx 12 + 0.072727s. \quad (79)$$

Thus, the proportional and derivative gains were

$$K_p = \boxed{12}, \quad K_d \approx \boxed{0.072727}. \quad (80)$$

The saved plots indicate a phase margin of approximately 70.1° and a closed-loop bandwidth of approximately 25 rad/s. Derivative action increased the phase margin to the required level. However, the PD-controlled introduced a nonzero steady-state error.

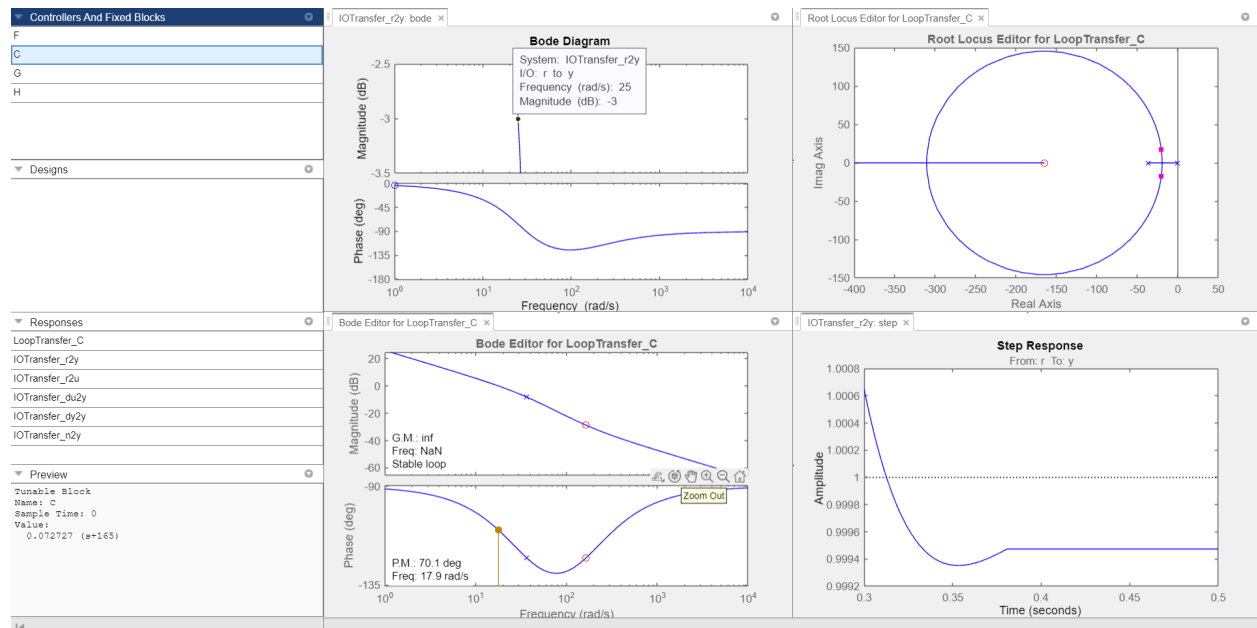


Figure 55: Owen Winn – Step 6 PD-controller design plots. The time-response panel shows a magnified portion of the response near the final value.

5.11.3 Step 9

Integral action was introduced with an integral gain equal to one percent of the proportional gain,

$$K_i = 0.01K_p = 0.12. \quad (81)$$

The recorded gains subsequently used were

$$K_p = \boxed{12}, \quad K_d = \boxed{0.0727}, \quad K_i = \boxed{0.12}. \quad (82)$$

These values give the controller

$$C_{PID}(s) = 12 + 0.0727s + \frac{0.12}{s}. \quad (83)$$

For these gains, the calculated open-loop phase margin is 70.056° and the closed-loop bandwidth is 25.158 rad/s. Both frequency-domain requirements are satisfied. The integral action also brings zero steady-state error for step and ramp commands in the model.

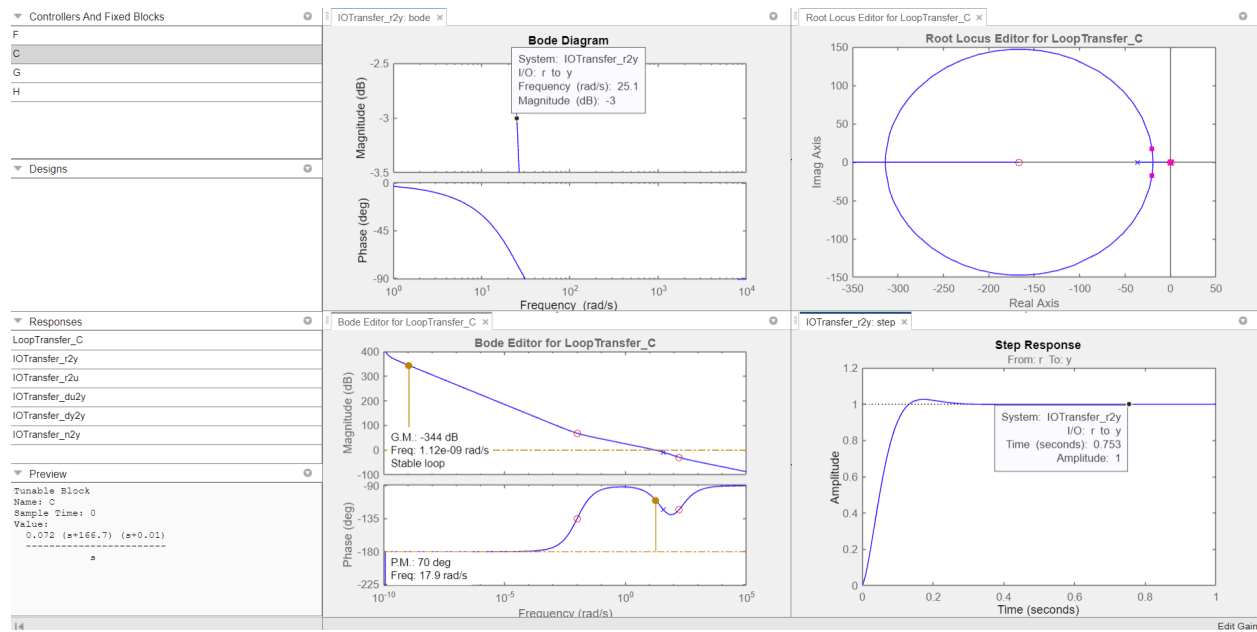


Figure 56: Owen Winn – Saved PID design plots from the integral tuning step.

5.11.4 Step 10

From the Control System Designer, the full PID controller transfer function is

$$T(s) = \frac{C_{PID}(s)G(s)}{1 + C_{PID}(s)G(s)} = \frac{0.1198823s^2 + 19.788s + 0.19788}{0.0274s^3 + 1.1198823s^2 + 19.788s + 0.19788}. \quad (84)$$

5.11.5 Step 11

The closed-loop poles, natural frequencies, and damping ratios are listed in Table 5. All three poles have negative real parts, so the linear closed-loop system is stable.

Table 5: Owen Winn – Closed-loop poles and characteristics.

Pole (s^{-1})	Natural frequency (rad/s)	Damping ratio
-0.0100057	0.0100057	1.0000
$-20.4308 + 17.4460i$	26.8660	0.7605
$-20.4308 - 17.4460i$	26.8660	0.7605

The real pole near -0.0100 s^{-1} is the slowest mode, with a decay time of approximately 99.94 s. It lies close to a zero, so its contribution to the short step response is small. The complex pair describes the rapid oscillating transient and has

$$\omega_n = 26.8660 \text{ rad/s}, \quad \zeta = 0.7605. \quad (85)$$

5.11.6 Step 12

The PID controller was implemented in Simulink. Separate proportional and integral branches act on position error, while the derivative gain acts on measured motor speed. A saturation block limits the applied motor voltage to ± 10 V. The motor-speed plant is $K/(\tau s + 1)$, followed by an integrator to obtain angular position. The model uses a step size of 0.002 s and the gains listed in Step 9.

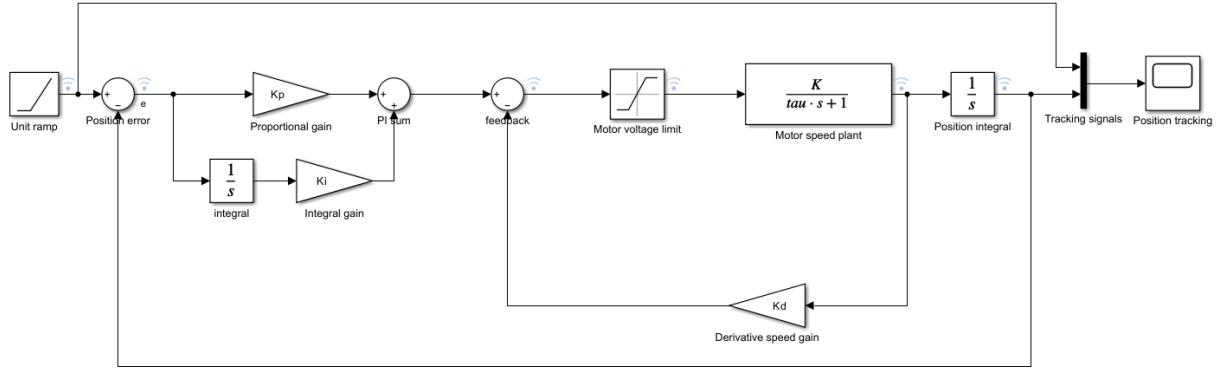


Figure 57: Owen Winn – Step 12 PID Simulink model with individual gain blocks, motor-speed feedback, and motor-voltage saturation.

5.11.7 Step 13

The Step 12 model was simulated for 30 s with a unit ramp command, $\theta_c(t) = t$, using a fixed step size of 0.002 s. Both simulations used $K_p = 12$ and $K_d = 0.0727$. The PD case used $K_i = 0$, while the PID case used $K_i = 0.12$. The ± 10 V motor-voltage limit remained in place.

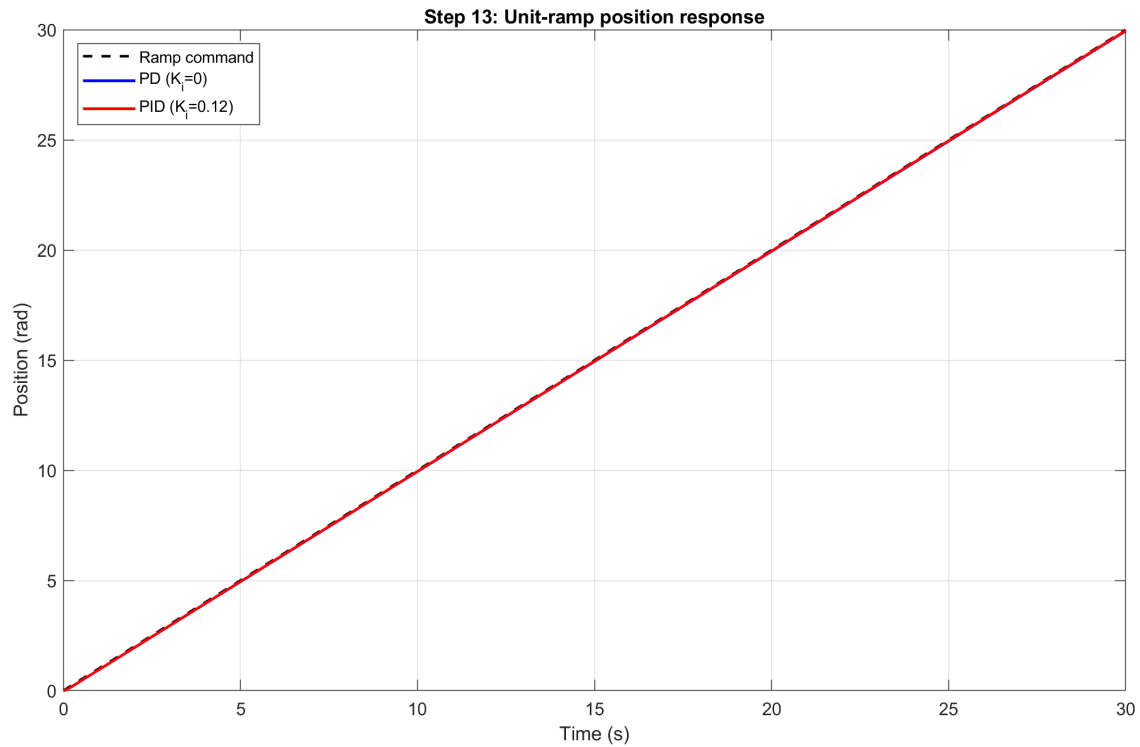


Figure 58: Owen Winn – Unit-ramp position responses with PD and PID control.

The two position responses closely follow the ramp and nearly overlap at the scale of the full simulation. The enlarged view near 30 s shows that the PID response is closer to the commanded position.

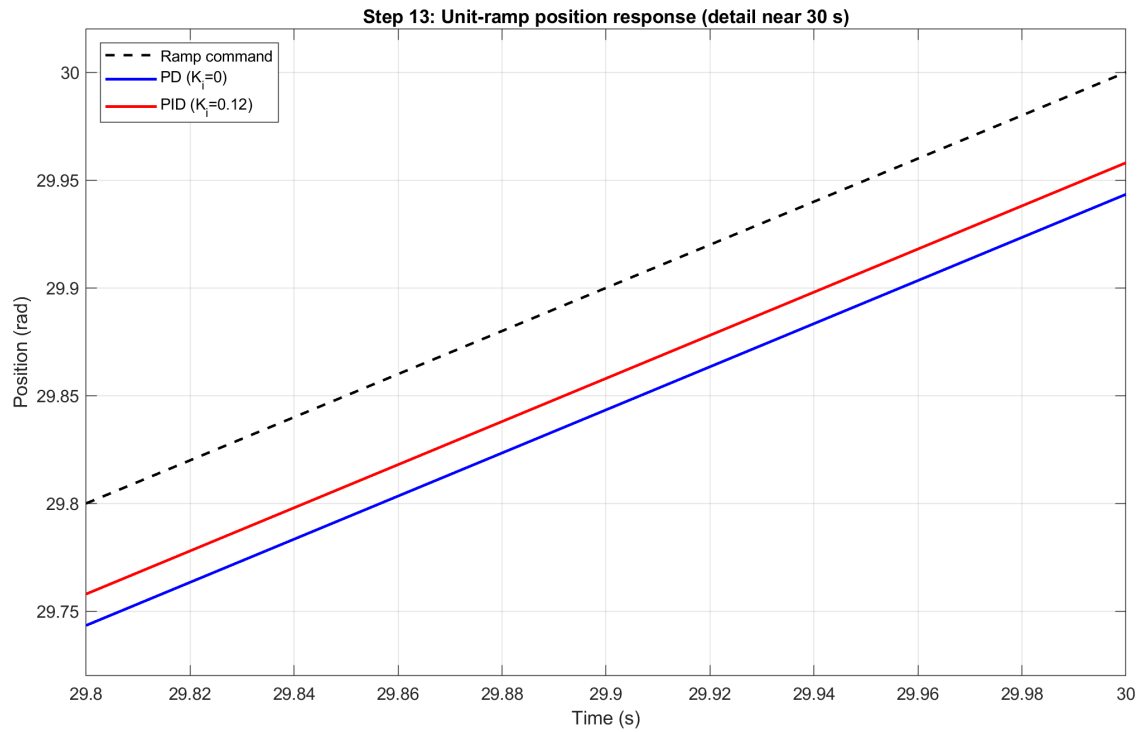


Figure 59: Owen Winn – Enlarged ramp-response comparison near 30 s, showing the smaller tracking lag with PID control.

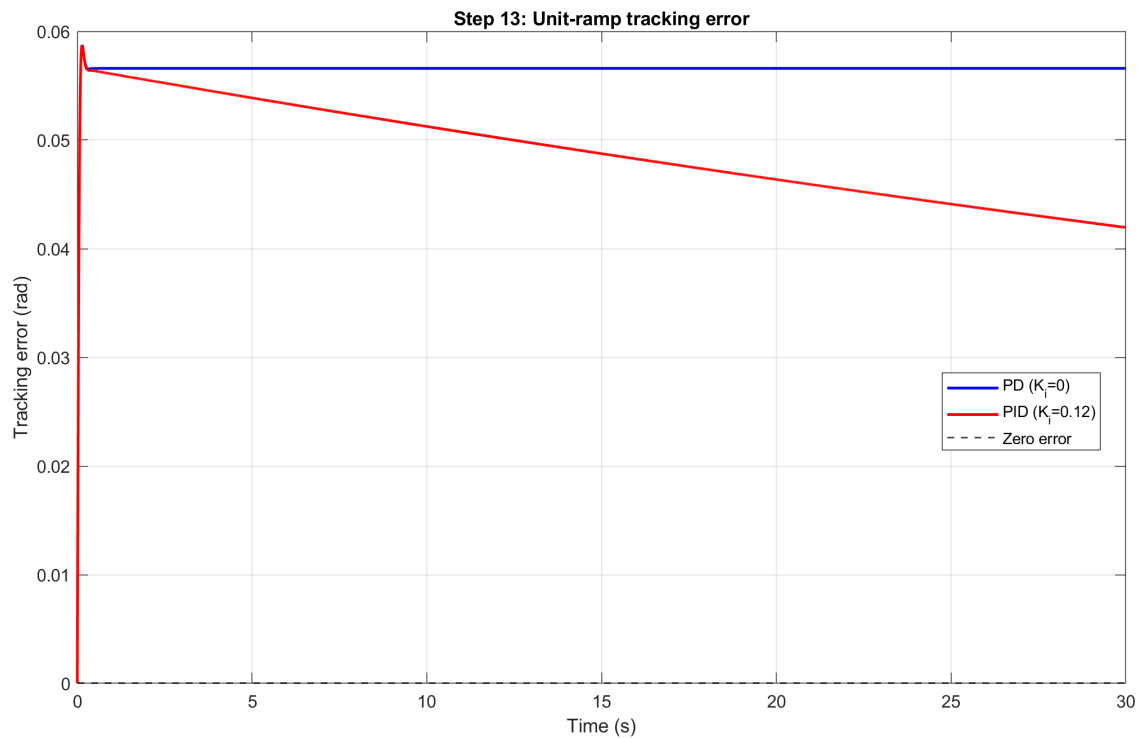


Figure 60: Owen Winn – Ramp tracking error, $e = \theta_c - \theta$, for the PD and PID controllers.

The PD tracking error approaches a constant value of approximately 0.05659 rad. The PID tracking error decreases to approximately 0.04196 rad at $t = 30$ s. This is a finite-time error, not the PID steady-state error. The small integral gain produces a slow correction, consistent with the approximately 100 s decay time of the slow pole identified in Step 11.

The peak applied motor voltage was approximately 0.633 V in both simulations, so the voltage limiter did not activate during these tests.

5.11.8 Additional Questions

a. Difference between Fig. 2 and the Step 6 architecture In Fig. 2 of the lab manual, the derivative gain acts on measured motor speed in a negative-feedback path. In the saved Step 6 Control System Designer architecture, the complete controller acts on position error in the forward path. In other words, Fig. 2 applies the derivative gain to motor speed, while the Step 6 controller applies it to the derivative of position error.

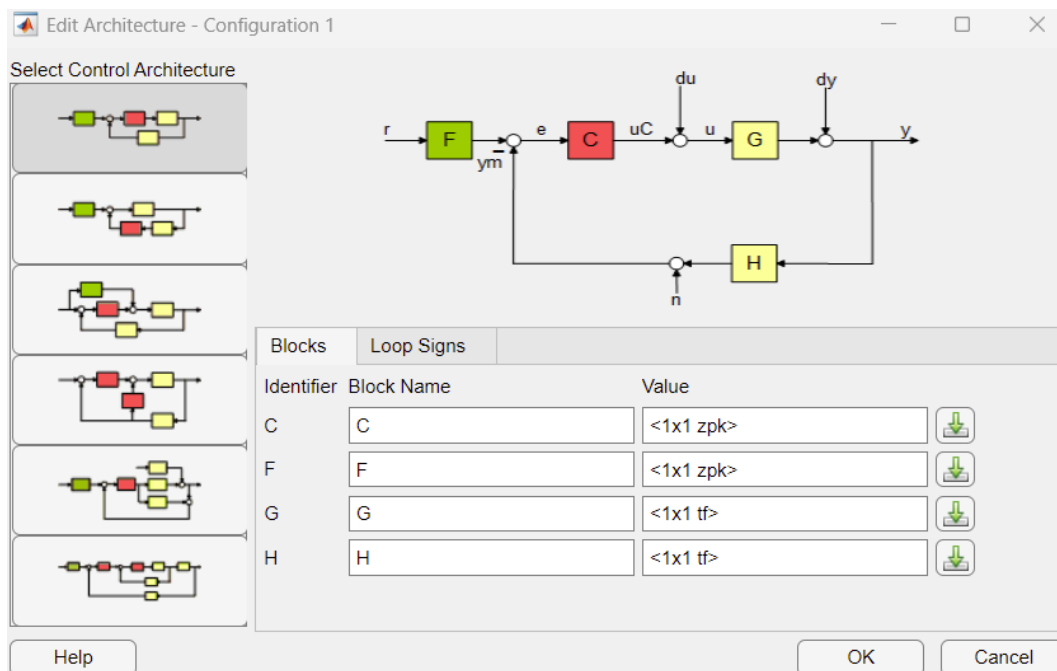


Figure 61: Owen Winn – Saved Step 6 Control System Designer architecture, with the PD controller in the forward error path.

b. Modified Simulink model Two comparison models were prepared from the Step 12 model. Both used the Step 6 gains $K_p = 12$, $K_d = 12/165 \approx 0.072727$, and $K_i = 0$. The plant parameters were $K = 1.649$ and $\tau = 0.0274$ s, and the motor voltage was limited to ± 10 V. Both models used a fixed solver step of 0.002 s and a five-second simulation.

The original model subtracts $K_d\omega$ from the proportional term. The modified model instead

adds $K_d \dot{e}$. The error derivative was calculated from the change in error between samples:

$$\dot{e}[k] \approx \frac{e[k] - e[k-1]}{T_s}, \quad T_s = 0.002 \text{ s.} \quad (86)$$

This calculation was used to plot the error derivative for both models, but only the modified model used it to control the motor.

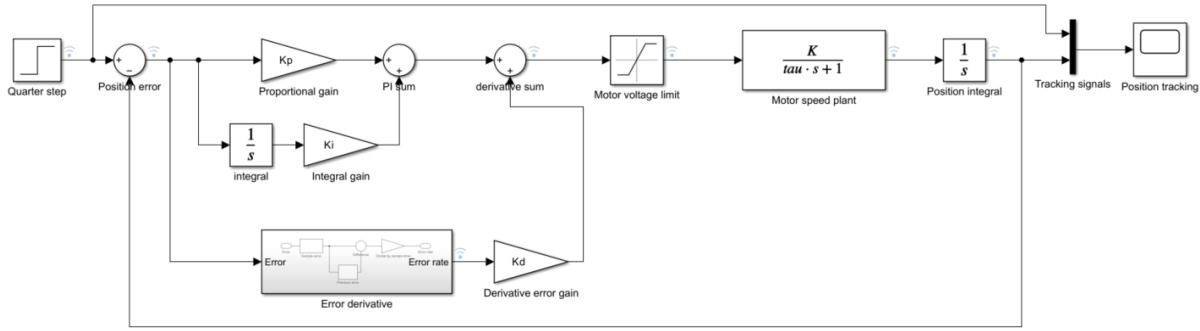


Figure 62: Owen Winn – Modified Simulink model matching the Step 6 control architecture.

c. Step responses and acceptability of the two structures A quarter-step command of 0.25 rad was applied at $t = 0.10$ s in both models. Figure 63 compares the position responses over the full five-second test and during the initial transient.

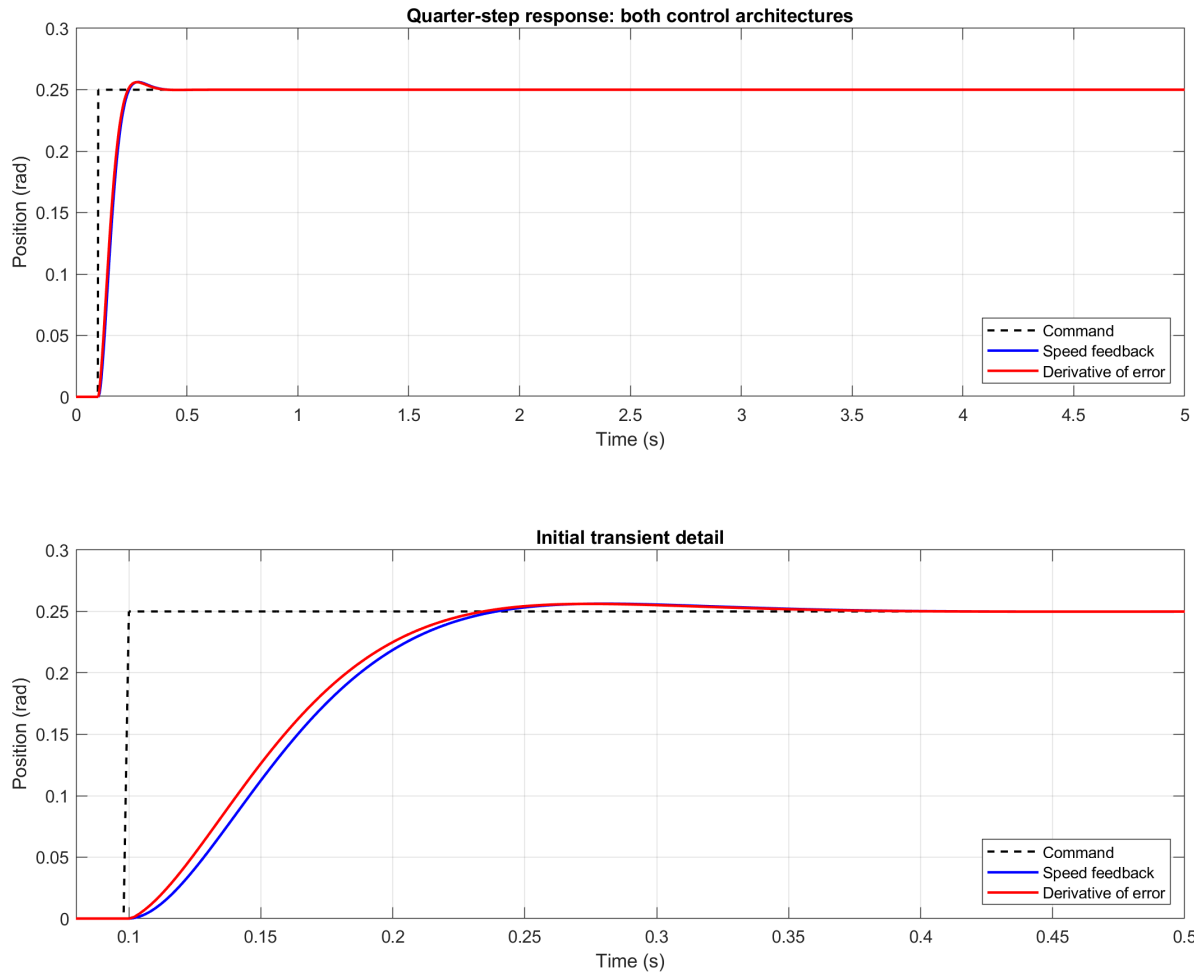


Figure 63: Owen Winn – Quarter-step position responses for the speed-feedback and derivative-of-error controllers, with a transient detail below.

After the command transition, $\dot{\theta}_c = 0$, so

$$\dot{e} = \dot{\theta}_c - \dot{\theta} = -\omega. \quad (87)$$

After the step, the command stays constant, so the two derivative terms have the same effect. The plots show a small difference at the beginning because the modified controller also responds to the sudden change in command. Both responses settle to 0.25 rad, which explains why either structure works for this step input.

d. Error-derivative and motor-speed signals Figure 64 shows all four requested signals: the error derivative and motor speed from each model. At the command transition, the error changes by 0.25 rad and the backward difference produces a positive pulse of

$$\dot{e} \approx \frac{0.25}{0.002} = 125 \text{ rad/s}. \quad (88)$$

Motor speed does not jump in the same way because the motor takes time to respond. The lower plot zooms in after the spike and shows that the error derivative is approximately the negative of motor speed for each model.

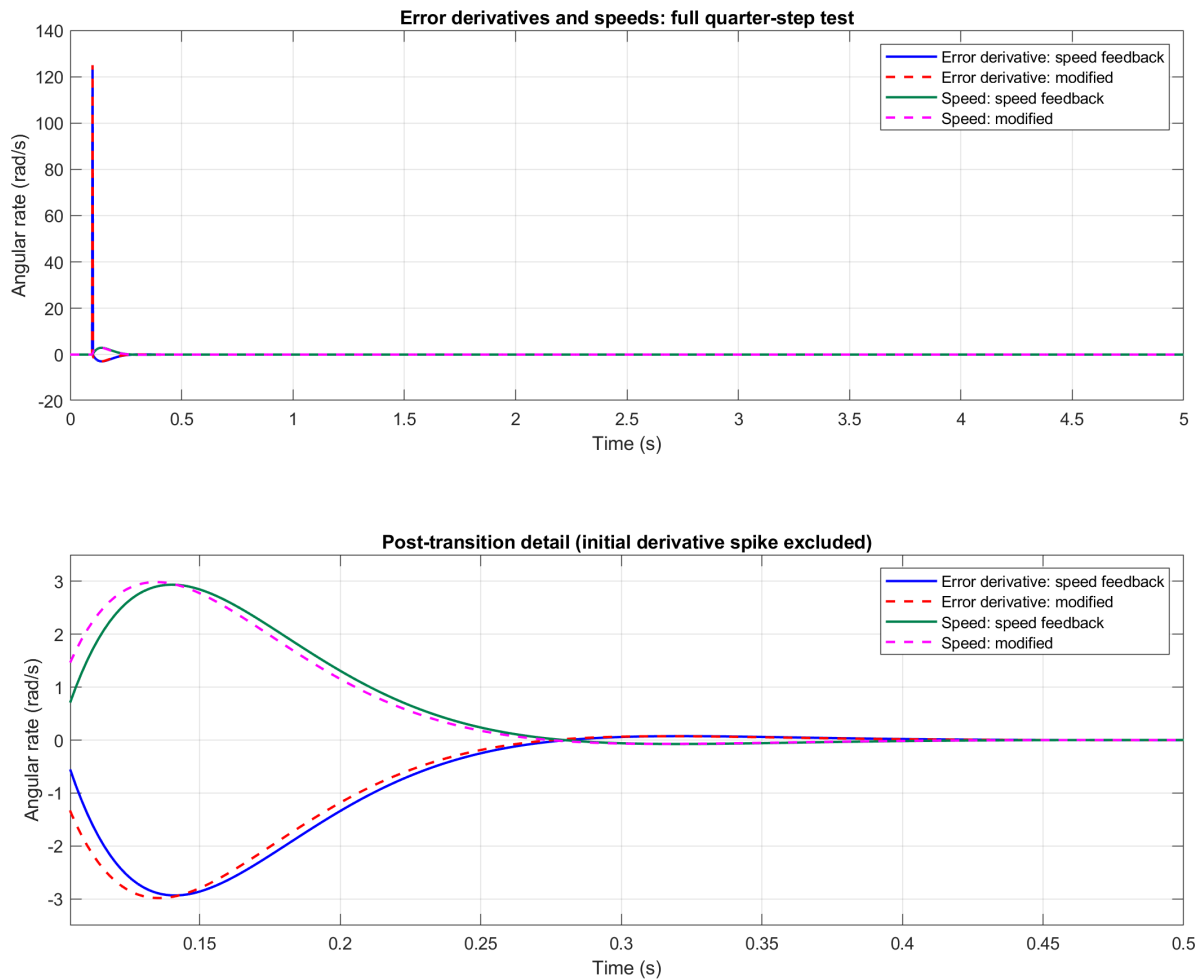


Figure 64: Owen Winn – Error derivatives and motor speeds for both architectures. All four signals appear in each panel; the lower panel resolves the motion after the initial derivative pulse.

e. Influence of derivative placement on motor input voltage The speed-feedback controller initially requests about 3 V. The modified controller also reacts to the sudden change in error, producing a derivative kick and a total voltage request of about 12.09 V in the sampled model. The saturation block limits this to 10 V. This larger initial voltage explains the brief difference between the position responses.

6 Controller Implementation

The controller designed in Part B was implemented using both the Quanser Digital Twin and the physical Rotary Servo Base. The proportional and derivative gains obtained during the control-design portion,

$$K_p = 12, \quad K_d = 0.0684, \quad (89)$$

satisfied the required transient-response criteria without additional tuning. Therefore, these gains were retained for the Part C experiments. For PID testing, the integral gain was set to

$$K_i = 0.01K_p = 0.12. \quad (90)$$

The Digital Twin and hardware were subjected to the same reference inputs so that their closed-loop responses could be compared directly. Step response tests were conducted using sampling times of 0.002 s and 0.04 s, while ramp response tests were conducted at 0.002 s using both PD and PID control.

6.1 Step Response

6.1.1 PD Controller, 0.002 s Sampling Time

With a sampling time of 0.002 s, both the Digital Twin and the physical Rotary Servo Base closely followed the commanded step input. The simulated and experimental responses exhibited similar transient behavior and reached approximately the commanded steady-state value. The agreement between the two responses indicates that the Digital Twin provided a reasonable representation of the physical servo system.

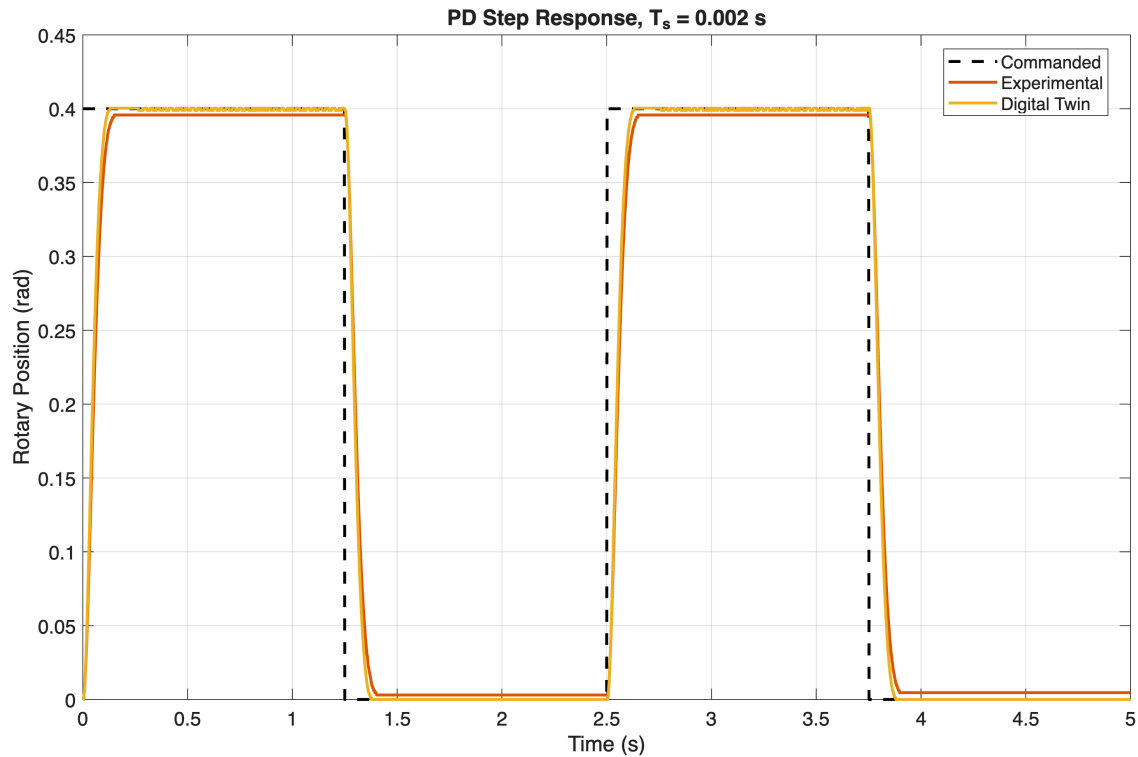


Figure 65: Step response of the commanded position, physical Rotary Servo Base, and Quanser Digital Twin using PD control with $K_p = 12$, $K_d = 0.0684$, and a sampling time of 0.002 s.

6.1.2 PD Controller, 0.04 s Sampling Time

Increasing the sampling time from 0.002 s to 0.04 s degraded the closed-loop response. Both the Digital Twin and experimental hardware exhibited increased oscillation and overshoot compared with the 0.002 s case. The larger sampling interval reduces how frequently the controller measures the system state and updates the motor command, which introduces additional effective delay into the feedback loop.

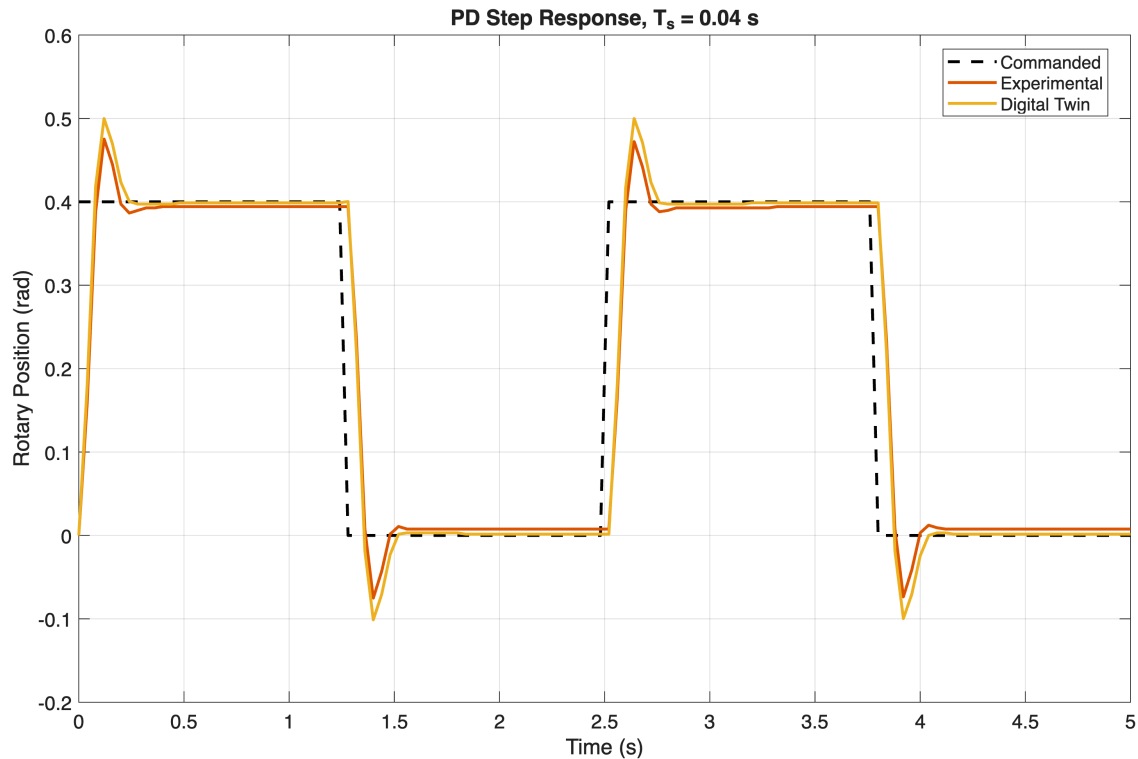


Figure 66: Step response of the commanded position, physical Rotary Servo Base, and Quanser Digital Twin using PD control with $K_p = 12$, $K_d = 0.0684$, and a sampling time of 0.04 s.

6.1.3 Step Response Discussion

Why is there no steady-state error in a PD controller with a step command?

The position-control plant contains an integrator because angular position is the integral of angular velocity. As a result, the open-loop system is Type 1 even when the controller contains only proportional and derivative terms. A Type 1 system has zero steady-state error for a constant step input. The derivative term primarily affects the transient response and damping and does not change the system type.

What is the effect of sampling time on closed-loop system behavior?

Increasing the sampling time decreases the frequency at which the controller receives measurements and updates the actuator command. The 0.04 s case therefore introduces more delay than the 0.002 s case. This reduces the effective stability margin and produces greater overshoot and oscillation. The experimental results show that the 0.002 s sampling time provides smoother and more accurate command tracking.

6.2 Ramp Response

6.2.1 PD Controller, 0.002 s

With PD control, both the Digital Twin and the physical servo followed the triangular ramp command closely. However, a finite tracking error remained while the command was changing. This behavior is expected because the PD-controlled position system is Type 1. A Type 1 system has zero steady-state error for a step input but a nonzero steady-state error for a ramp input.

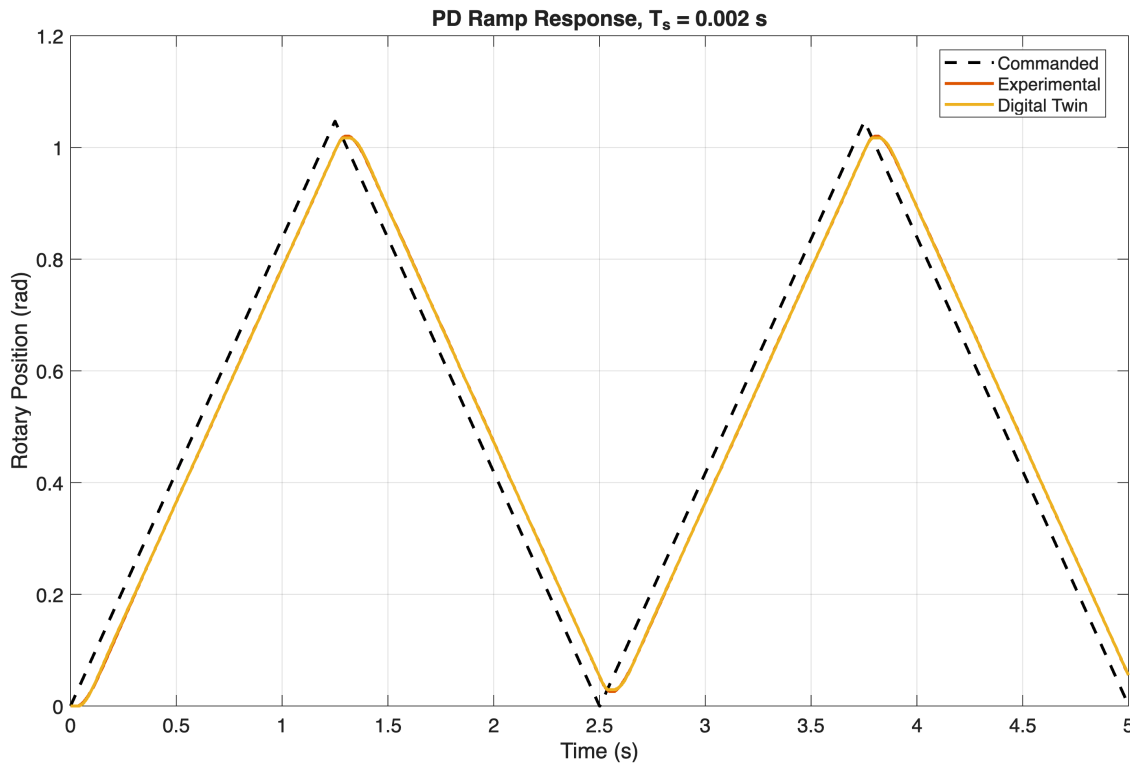


Figure 67: Ramp response of the commanded position, physical Rotary Servo Base, and Quanser Digital Twin using PD control with $K_p = 12$, $K_d = 0.0684$, and a sampling time of 0.002 s.

6.2.2 PID Controller, 0.002 s

Integral control was added using $K_i = 0.12$. The integral term accumulates position error and provides additional corrective input, reducing the persistent tracking error associated with a ramp command. Because the integral gain was relatively small compared with the proportional gain, its effect on the overall transient response was modest.

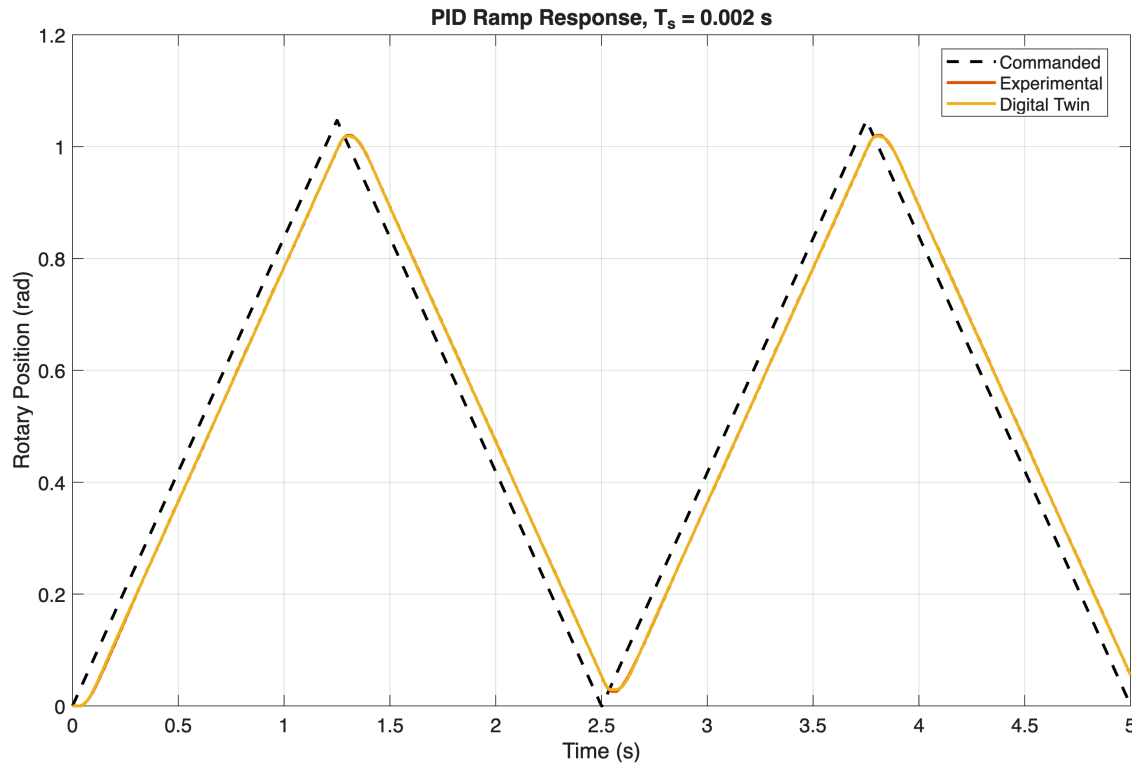


Figure 68: Ramp response of the commanded position, physical Rotary Servo Base, and Quanser Digital Twin using PID control with $K_p = 12$, $K_d = 0.0684$, $K_i = 0.12$, and a sampling time of 0.002 s.

6.2.3 Ramp Response Discussion

What is the effect of I-gain in ramp response?

The integral gain responds to accumulated position error. Under a ramp input, PD control produces a finite tracking error, while the addition of integral action increases the system type and reduces the steady-state ramp error. Since the integral gain used in this experiment was small, $K_i = 0.12$, the improvement in tracking was gradual and the overall transient response remained similar to the PD case.

Explain differences in ramp responses with PD gains between Digital Twin and experimental results.

The Digital Twin and physical system produced similar ramp responses, but small differences were present because the physical system contains effects that are not represented perfectly in the model. These include friction, motor dead zone, sensor noise, component variation, and other nonlinear behavior. These effects can cause small differences in tracking error and transient response between the simulated and experimental systems.

Explain differences in ramp responses with PID gains between Digital Twin and experimental results.

The PID responses also showed close agreement between the Digital Twin and the physical

servo. The experimental system exhibited small differences because the real motor is affected by friction, dead zone, sensor noise, and other nonlinearities that are only approximated in the Digital Twin. Integral action helps reduce persistent tracking error, but the physical nonlinearities can still produce small deviations between the simulated and measured responses.

6.3 Closed-Loop System

The gains obtained during Part B satisfied the Part C response requirements without additional tuning. Therefore, the same controller gains were used for the closed-loop system analysis:

$$K_p = 12, \quad K_d = 0.0684. \quad (91)$$

Using the validated plant model,

$$G(s) = \frac{1.649}{s(0.0274s + 1)}, \quad (92)$$

and the PD controller,

$$C(s) = 12 + 0.0684s, \quad (93)$$

the closed-loop bandwidth and open-loop phase margin were evaluated.

6.3.1 Bandwidth

The bandwidth of the closed-loop system with the retained Part B gains was approximately

$$\omega_{BW} = \boxed{25.33 \text{ rad/s}}. \quad (94)$$

This value is consistent with the original controller-design target of approximately 25 rad/s.

6.3.2 Phase Margin

The phase margin of the open-loop system with the retained Part B gains was approximately

$$PM = \boxed{69.73^\circ}. \quad (95)$$

This value is approximately equal to the original design target of 70°.

6.3.3 Dead Zone

A DC motor dead zone is a range of small input voltages for which the motor does not produce enough torque to overcome effects such as static friction. Within this range, the controller

may command a nonzero voltage without producing significant shaft motion. This can create residual tracking error, especially near small command changes or direction reversals.

Integral control can reduce this effect by accumulating position error until the commanded voltage becomes large enough to overcome the dead zone. However, excessive integral action may also increase overshoot or oscillation once the motor begins moving.

7 Conclusion

This lab covered the process of modeling the Rotary Servo Base, designing a position controller, and testing it in simulation and on the hardware. We first used frequency-response and step-response tests to estimate the motor's gain and time constant, then compared these results with the nominal model. The selected model was used to design P, PD, and PID controllers in MATLAB and Simulink. Finally, the controllers were tested using the Digital Twin and physical servo with step and ramp commands.

In open-loop operation, a constant voltage caused the motor speed to approach a steady value. The frequency-response test also showed that the speed response became smaller as the input frequency increased. These tests helped describe the motor's behavior, but without position feedback, the system could not correct its motion to reach a commanded position. Closing the loop allowed the controller to adjust the motor voltage based on the position error. The closed-loop step responses reached approximately the commanded position, while the ramp tests showed a remaining tracking error with PD control.

PID control was selected because the design needed both a fast, well-damped response and accurate tracking of step and ramp commands. The design targets were a bandwidth of at least 25 rad/s and a phase margin of at least 70 degrees. Proportional control alone did not meet both targets, so derivative action was added to improve damping and phase margin. PD control provided good step tracking, but it did not eliminate ramp tracking error. Adding integral action reduced this error over time. The improvement was gradual because the integral gain was small, and the error had not reached zero by the end of the 30-second simulations.

The Digital Twin and hardware showed similar overall behavior, with some differences that could be caused by friction, motor dead zone, and sensor noise. Increasing the sampling time from 0.002 s to 0.04 s also increased oscillation and overshoot. These results showed that controller gains, sampling time, and the motor's voltage limits all need to be considered when moving from a model to the physical system.